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Valid F-screening in linear regression

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Abstract

Suppose that a data analyst wishes to report the results of a least squares linear regression *only if the overall null hypothesis*—$H_0^{1:p} : \beta_1 = \beta_2 = \ldots = \beta_p = 0$—*is rejected*. This practice, which we refer to as *F-screening* (since the overall null hypothesis is typically tested using an F-statistic), is in fact common across a number of applied fields. Unfortunately, it poses a problem: standard guarantees for the inferential outputs of linear regression, such as Type 1 error control of hypothesis tests and nominal coverage of confidence intervals, hold *unconditionally*, but fail to hold conditional on rejection of the overall null hypothesis. In this paper, we develop an inferential toolbox for the coefficients in a least squares model that are valid conditional on rejection of the overall null hypothesis. We develop selective p-values that lead to tests that are consistent and control the selective Type 1 error, i.e., the Type 1 error conditional on having rejected the overall null hypothesis. Furthermore, they can be computed without access to the raw data, i.e., using only the standard outputs of a least squares linear regression, and therefore are suitable for use in a retrospective analysis of a published study. We also develop confidence intervals that attain the nominal selective coverage, and point estimates that account for having rejected the overall null hypothesis. We derive an expression for the Fisher information about the coefficients resulting from the proposed approach, and compare this to the Fisher information that results from an alternative approach that relies on sample splitting. We investigate the proposed approach in simulation and via re-analysis of two datasets from the biomedical literature.

1 Introduction

Not all data analyses yield interesting results, and many uninteresting findings are left in the "file drawer" (Rosenthal 1979) rather than published. Furthermore, when papers are

published, analyses yielding null results are often omitted (Benjamini & Yekutieli 2005). Such *selective reporting* has been empirically observed in the biomedical literature (Chan et al. 2004, Ioannidis 2005).

Why is selective reporting a problem? Standard statistical guarantees, such as Type 1 error control for hypothesis tests and nominal coverage of confidence intervals, only hold when results are reported regardless of whether they are "interesting."

In this work, we consider a particular and pervasive form of selective reporting that arises in the context of a regression model of the form

$$Y = \beta_0 + X_1\beta_1 + \ldots + X_p\beta_p + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2). \tag{1}$$

Given n independent observations from (1), suppose that the data analyst first tests

$$H_0^{1:p} : \beta_1 = \beta_2 = \ldots = \beta_p = 0 \tag{2}$$

at level $\alpha_0 \in (0, 1)$ in order to determine whether the model in (1) merits further study. If they fail to reject $H_0^{1:p}$, then (1) receives no further consideration, and the fact that (1) was considered may not even be reported. On the other hand, if (and only if) they reject $H_0^{1:p}$, they then proceed to test $H_0^M : \beta_M = 0$ for a fixed (and typically strict) subset M of $\{1, \ldots, p\}$, where $\beta_M := \{\beta_j : j \in M\}$. Because $H_0^{1:p}$ is typically tested by means of an F-statistic, we refer to this procedure—summarized in Box 1—as *F-screening*. If conducted, the test in Step 2 of Box 1 is sometimes referred to as a *post hoc* test.

Box 1: *F-screening in the model* (1).

Step 1: Test $H_0^{1:p}$ as given in (2) at level α_0, using an F-test.

Step 2: Only if $H_0^{1:p}$ is rejected, conduct inference on β_M for fixed $M \subset \{1, \ldots, p\}$.

To see that conducting "standard" inference in Step 2 of F-screening will not provide appropriate statistical guarantees, consider the extreme setting where $p = 1$ and $\beta_1 = 0$. In this case, $H_0^{1:p} \equiv H_0^1$. Suppose we carry out F-screening in Box 1 using the "standard" t-test for H_0^1 in Step 2. Then for fixed α_0, the tests in Steps 1 and 2 are identical, and so we will report a result in Step 2 only if H_0^1 is rejected. Thus the Type 1 error rate among reported results equals one! The left plot in Fig. 5 illustrates the inflation of Type 1 error that results from conducting "standard" inference in Step 2 in a setting with $p = 10$ and $n = 100$.

As pointed out by Olshen (1973), conducting a test of H_0^M that accounts for the fact that H_0^M is tested if and only if $H_0^{1:p}$ is rejected is distinct from the task of correcting multiple tests of the form $H_0^M : \beta_M = 0$ for multiple comparisons. The latter has been very well-explored in the literature: see Dunn (1961), Šidák (1967), Tukey (1949), Scheffé (1953). For instance, if our interest in Step 2 of Box 1 lies in $H_0^j : \beta_j = 0$ for $j \in \{1, \ldots, p\}$, then we could consider testing each null hypothesis using a standard test, and applying a Bonferroni correction to account for the fact that p tests were conducted. However, this will not address the fact that F-screening was performed: namely, that we test $\{H_0^j\}_{j=1}^p$ *if and only if we have rejected* $H_0^{1:p}$. In Appendix A, we elaborate on how the guarantees provided by multiple

testing corrections—such as family-wise error rate control—differ from those achieved by our proposed method. Further, we describe a closed-testing procedure that parallels F-screening and show that its guarantees are quite different. Fig. A.1 demonstrates empirically that muliple testing approaches fail to provide valid inference conditional on the rejection of $H_0^{1:p}$.

Our goal of conducting inference on β_M that is valid *conditional on the fact that we rejected* $H_0^{1:p}$ falls under the umbrella of *selective inference* or *post-selection inference*; we will view rejection of $H_0^{1:p}$ as the "selection event." The selective inference framework was laid out in Fithian et al. (2017) and further explored in a variety of settings (Taylor & Tibshirani 2015, Lee et al. 2016, Zhang et al. 2019, Chen & Bien 2020, Jewell et al. 2022, Chen & Gao 2024, Chen et al. 2023, Chen & Witten 2023, Gao et al. 2024, Neufeld et al. 2022, among others). We with to control the selective Type 1 error for the Step 2 test, conditional on rejection in Step 1; that is,

$$\Pr_{H_0^M}\left(H_0^M \text{ is rejected} \mid H_0^M \text{ is tested}\right) \leq \alpha.$$

Our work is related to Heller et al. (2019), who considered inference following aggregate testing for a coefficient that follows a multivariate normal distribution with known covariance. However, in Section B, we show that their proposal does not control the selective Type 1 error in the setting of Box 1.

Our contributions are as follows:

1. We provide a valid end-to-end approach for conducting F-screening in the sense of Box 1. That is, we enable reporting of all of the standard multiple linear regression outputs—such as p-values, point estimates, and confidence intervals—conditional on rejection of $H_0^{1:p}$ in Step 1; we will refer to them as *selective* p-values, point estimates, and confidence intervals.

2. We establish consistency and selective Type 1 error control of the test resulting from our selective p-value.

3. We characterize the Fisher information about β_M resulting from our procedure, and we show that it compares favorably to that of a procedure that relies on sample splitting.

4. We show that the selective p-values can be computed using only the standard multiple linear regression outputs *without access to the raw data*. This is particularly critical because, while it is firmly established that selective reporting is common practice in the published literature (Chan et al. 2004), *the raw data is often not available to the reader who wishes to correct for it.*

5. We specialize our approach to the setting of analysis of variance (ANOVA), in which the F-screening procedure in Box 1, with Step 2 conducted using a "standard" approach that does not account for Step 1, is particularly pervasive (see, e.g., Fig. 1).

Addressing the inferential problems posed by F-screening is especially important from an applied perspective. A theoretical statistician unaccustomed to scientific collaboration might believe that F-screening is an unusual procedure unlikely to be conducted in practice, or that conducting Step 2 of Box 1 without accounting for Step 1 is "obviously wrong" and

Characteristic	Participants, No. (%)			Statistical test	*P* value	*P* value for post hoc test[a]
	Aged <65 y (n = 150)	Aged 65-84 y (n = 449)	Aged ≥85 y (n = 227)			
Sample 1 (n = 826)						
Depression measure, CES-D, total score, mean (SD) [range]	9.33 (7.16) [0-33]	7.60 (5.84) [0-38]	8.79 (5.68) [0-35]	1-Way ANOVA	.003	.05,[c] .008[d]
Self-reported sleep duration, mean (SD), h	6.65 (0.87)	6.93 (0.87)	7.07 (1.02)	1-Way ANOVA	<.001	.004,[c] <.001[e]
Mini-Mental State Examination Score	29.31 (1.19) [23-30]	28.12 (2.42) [14-30]	26.14 (4.15) [1-30]	1-Way ANOVA	<.001	<.001[c,d,e]
Mattis Dementia Rating Scale	141.5 (2.57) [130-144]	139.0 (5.85) [101-144]	131.6 (14.89) [16-144]	1-Way ANOVA	<.001	.01,[c] <.001[d,e]

Abbreviations: ANOVA, analysis of variance; CES-D, Center for Epidemiological Studies Depression Scale; NA, not applicable.

[a] The Šídák multiple comparison test was used for all post hoc analyses.
[b] Other race and ethnicity includes Asian or Pacific Islander, Black, Hispanic, Native American, and other participant responses.
[c] Participants aged 65-84 y vs aged <65 y.
[d] Participants aged ≥85 y vs aged 65-84 y.
[e] Participants aged ≥85 y vs aged <65 y.

Figure 1: A subset of Table 1 from Keil et al. (2023). The $n = 826$ participants in the dataset are split into three age groups: under 65 years old, between 65 and 85 years old, and over 85 years old. Each row in the table corresponds to a continuous variable of interest, with means and standard deviations reported for each age group. For each of these continuous variables, ANOVA is conducted to test for a difference in means between the three age groups (corresponding to Step 1 of Box 1). The associated p-value is reported in the "P Value" column. The last column holds p-values for "post hoc tests" which are typically conducted *only* if the p-value associated with the ANOVA test is small (corresponding to Step 2 of Box 1). In this case, the p-values in the last column would be invalid since they do not account for the fact that they were only computed because $H_0^{1:p}$ (2) was rejected.

no data analyst would consider it. The theoretical statistician would be mistaken on both counts. Statistical references that encourage F-screening with "standard" inference in Step 2 as part of a routine data analysis abound, and some are authored by luminaries in the field.[1] Thus, it is hardly surprising that in the scientific literature, conducting Step 2 of Box 1 with a "standard" approach that does not account for having conducted Step 1 is often viewed as the correct analysis.[2] In clinical journals, F-screening is a routine part of "Table 1," in which

[1] Fisher (1935, §24) writes: "If the yields of the different varieties in the experiment fail to satisfy the test of significance they will not often need to be considered further . . . If, however . . . the null hypothesis has been falsified . . . we shall thereafter proceed to interpret the differences between the varietal yields . . . and shall be concerned to know with what precision these different yields have been evaluated." A few years later, Lindquist (1940, page 98) writes "The student may well ask why it was necessary to apply the F-test at all, if we were subsequently going to test the individual differences anyway. The answer is that the F-test tells us whether it is worthwhile to test the individual difference at all. If the F-test had not proven significant, we would have known at once that *all* observed differences in . . . means could be due to chance alone. In that case it would not only have been unnecessary, but decidedly *improper* to apply the t-test to individual differences." The abstract of Scheffé (1953) poses the following question: "Under the usual assumptions, if the conventional F-test of $H : \mu_1 = \mu$ . . . rejects H, what further inference are valid about the contrasts H_i?" Scheffé (1953) later remarks that "The problem of making further inferences about the contrasts, arising when the F-test rejects H, has been considered by various writers, including Fisher (1935), Newman (1939), Tukey (1949), Nandi (1951)." As pointed out in Olshen (1973), to the extent that these references consider selection, they focus on the multiplicity issues that arise from testing $H_0^j : \beta_j = 0$, $j = 1, \ldots, p$, rather than on the challenge that arises from inference conditional on rejecting $H_0^{1:p}$. See Appendix A for simulation results demonstrating that multiple testing corrections do not correct for selection in the F-screening procedure.

[2] As one of many possible examples, Webb et al. (2024) describe their analysis as follows: "A one-way ANOVA revealed that amygdala reactivity was not significantly different by trajectory groups . . . Therefore, additional analyses testing whether greenspace was associated with trajectory assignment via amygdala

patient characteristics are summarized; an example is shown in Fig. 1.

In Fig. 1, Keil et al. (2023) report the results of multiple ANOVAs for several outcomes of interest, along with the corresponding post hoc tests. We note that if these post hoc results are reported for all outcomes, regardless of the result in Step 1 of Box 1, then no selective reporting problem arises and standard marginal guarantees remain intact. However, if subsequent inference is only conducted or reported when $H_0^{1:p}$ (2) is rejected, then this inference is implicitly conditional on the rejection event, and standard procedures no longer provide valid guarantees. Further, we emphasize that this issue is not limited to studies that consider multiple ANOVAs or linear models: *even when Step 1 of Box 1 is conducted for a single model, Step 2 must account for the fact that we would not have conducted it had* $H_0^{1:p}$ *not been rejected in Step 1.*

Finally, we note that F-screening is likely conducted much more frequently than the published literature suggests, both because (i) it is typically unknown whether an analysis would have been included in a published paper if $H_0^{1:p}$ had not been rejected, and (ii) an analysis for which $H_0^{1:p}$ is not rejected may not be published at all. Given that F-screening is both commonplace and problematic if inference in Step 2 does not account for Step 1, it came as a surprise to us that—to the best of our knowledge—no statistical solution has yet been proposed for valid inference in Step 2. The rest of this paper fills this gap.

The paper is organized as follows. In Section 2, we propose a conditional selective inference approach for valid inference in Step 2 of Box 1. We establish its theoretical properties and derive an expression for the leftover Fisher information after rejection of $H_0^{1:p}$ (2) in Step 1. Section 3 shows that it is possible to compute selective p-values using the results of a "standard" (non-selective) analysis in Step 2, *without having access to the raw data*. Section 4 specializes the results in Sections 2 and 3 to the case of ANOVA. Simulation results are in Section 5 (and Section D), and re-analyses of published findings in Section 6. The discussion is in Section 7. The main text focuses on developing valid selective p-values; confidence intervals and point estimates are deferred to Appendix C. Section E contains mathematical details for Figures 2 and 3. Proofs of all results are given in Appendix F.

2 Developing valid p-values in Step 2 of F-screening

2.1 Preliminaries

Let $P_A := A(A^\top A)^{-1}A^\top$ denote the projection onto the column space of a matrix A with full column rank, and let A_{-S} denote the matrix A with the columns indexed by $S \subset \{1, \ldots, p\}$ removed. Furthermore, let $\mathbb{1}_n$ denote the n-dimensional column vector of 1's, and I_n the $n \times n$ identity matrix. This section restates some standard results in the theory of linear models (a more thorough treatment can be found elsewhere, e.g., Seber & Lee 2003).

Consider n independent observations from the model in (1). We assume that the intercept is not of scientific interest, and so both $H_0^{1:p}$ and H_0^M will operate on centered data. To do this, we let $U \in \mathbb{R}^{n\times(n-1)}$ be any matrix satisfying both $U^\top U = I_{n-1}$ and $UU^\top = I_n - P_{\mathbb{1}_n}$, and then with slight abuse of notation we redefine $Y := U^\top Y$, $X_1 := U^\top X_1$, and so on. Thus, in what follows, we will treat $\mathbb{R}^{n-1}$ as the ambient space and dispense with the intercept.

reactivity were not conducted."

First, consider the null hypothesis $H_0^{1:p}$ given in (2). Under $H_0^{1:p}$, the F-statistic defined as

$$F_{H_0^{1:p}} := \frac{n-p-1}{p} \cdot \frac{Y^\top P_X Y}{Y^\top (I_{n-1} - P_X) Y} \tag{3}$$

follows an $F_{p,n-p-1}$ distribution (that is, an F distribution with p and $n-p-1$ degrees of freedom). Next, fix H_0^M by letting $M \subset \{1, \ldots, p\}$ denote a set of indices with cardinality $|M| = m < p$, and let β_M denote the subset of coefficients indexed by M (in the simplest case, $M = \{j\}$ for fixed $j \in \{1, \ldots, p\}$). Under the null hypothesis H_0^M, the F-statistic

$$F_{H_0^M} := \frac{n-p-1}{m} \cdot \frac{Y^\top (P_X - P_{X_{-M}}) Y}{Y^\top (I_{n-1} - P_X) Y} \tag{4}$$

follows an $F_{m,n-p-1}$ distribution. Thus, for a realization y of the random variable Y, we can compute the p-value for $H_0^M : \beta_M = 0$ as

$$\begin{aligned} p_{H_0^M}(y) &:= \Pr_{\beta_M = 0} \left(\frac{Y^\top (P_X - P_{X_{-M}}) Y}{Y^\top (I_{n-1} - P_X) Y} \geq \frac{y^\top (P_X - P_{X_{-M}}) y}{y^\top (I_{n-1} - P_X) y} \right) \\ &= 1 - F_{m,n-p-1} \left(\frac{y^\top (P_X - P_{X_{-M}}) y}{y^\top (I_{n-1} - P_X) y} \cdot \frac{(n-p-1)}{m} \right), \end{aligned} \tag{5}$$

where $F_{m,n-p-1}(\cdot)$ is the cumulative distribution function of a random variable with an $F_{m,n-p-1}$ distribution. This is the "standard" test of $H_0^M : \beta_M = 0$, i.e., it does not account for Step 1 of Box 1.

Fig. 2 displays the rejection regions corresponding to a test of $H_0^{1:p}$ using (3) and a test of H_0^M using (4) in a simple setting with $n = 3$, $p = 2$, and X orthogonal. The geometry of these two events interplays in non-trivial ways, and failing to account for the first when evaluating the second can yield invalid inference. By contrast, the selective test that we will develop in the next section will take the sequential nature of the analysis into account, and it will adjust the geometry of the second test accordingly.

2.2 A conditional selective inference approach

We test H_0^M in Step 2 of Box 1 if and only if we reject $H_0^{1:p}$ (2) at level $\alpha_0 \in (0, 1)$: that is, if and only if $Y \in E_1$, where

$$E_1 := \left\{ Y : \frac{n-p-1}{p} \cdot \frac{Y^\top P_X Y}{Y^\top (I_{n-1} - P_X) Y} \geq F^{-1}_{p,n-p-1}(1-\alpha_0) \right\}. \tag{6}$$

The left panel of Fig. 5 shows that conditional on $Y \in E_1$ (6), the p-values obtained using $F_{H_0^M}$ are far from uniform and indeed anti-conservative—i.e., a test based on $p_{H_0^M}$ (5) does not control the selective Type 1 error in Step 2 of Box 1. This is because $F_{H_0^M}$ (4) does not follow an $F_{m,n-p-1}$ distribution under H_0^M when we condition on $Y \in E_1$ (6). Our goal is to develop an alternative to $p_{H_0^M}$ (5) that *does* lead to selective Type 1 error control.

To illustrate this idea, we consider in Fig. 3 a simple setting with $p = 2$ and σ^2 known. The left-hand panel displays the rejection regions for χ^2-tests of $H_0^{1:2}$ and H_0^1, respectively;

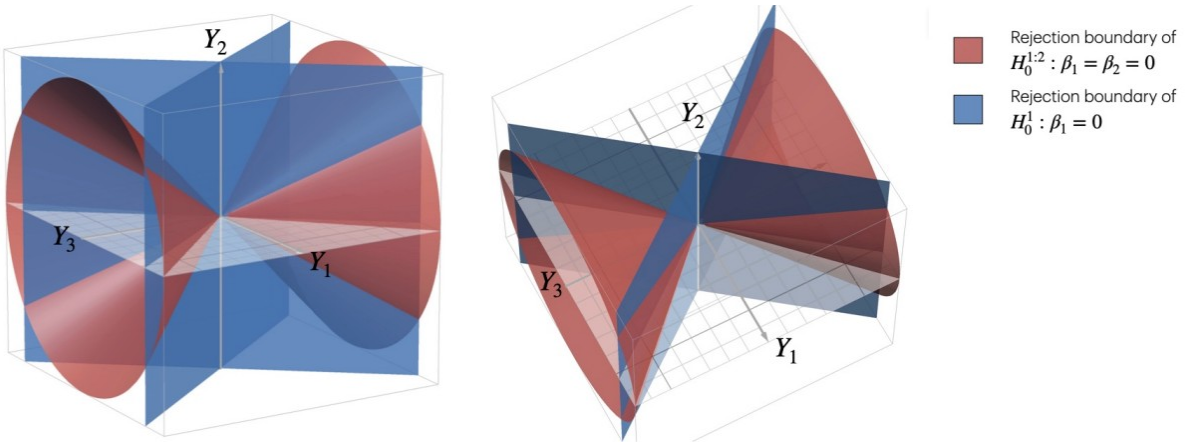


Figure 2: We consider testing $H_0^{1:2} : \beta_1 = \beta_2 = 0$ using $F_{H_0^{1:2}}$ (3) and $H_0^1 : \beta_1 = 0$ using $F_{H_0^1}$ (4) in the model $Y = X\beta + \epsilon$, where $X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. The red double cone corresponds to the rejection boundary of the test $H_0^{1:2}$, and the blue planes correspond to the rejection boundary of H_0^1. Thus data points that lie farther from the Y_3-axis than the double cone lead to rejection of $H_0^{1:2}$, and data points that lie farther from the (Y_2, Y_3)-plane than the two blue planes lead to rejection of H_0^1. See Section E for mathematical details.

this does not account for the sequential nature of the two tests in the context of F-screening. The right-hand panel displays the rejection region of a χ^2-test of $H_0^{1:2}$, alongside the rejection region of a test of H_0^1 that conditions upon rejection of $H_0^{1:2}$.

To extend this idea to the setting of the paper, we make use of F-tests rather than χ^2-tests. We modify $p_{H_0^M}$ in (5) to condition on the selection event in (6), leading to

$$\Pr_{\beta_M=0}\left(\frac{Y^\top(P_X - P_{X_{-M}})Y}{Y^\top(I_{n-1} - P_X)Y} \geq \frac{y^\top(P_X - P_{X_{-M}})y}{y^\top(I_{n-1} - P_X)y}\middle| Y \in E_1\right). \tag{7}$$

Unfortunately, (7) is not a well-defined probability: it involves nuisance parameters $\{\beta_j\}_{j\notin M}$ and σ^2 which are not fixed under H_0^M, and thus it cannot be computed. To remedy this, we additionally condition on the events

$$E_2(y) := \left\{Y : \|P_{X_{-M}}Y\| = \|P_{X_{-M}}y\|\right\} \quad \text{and} \quad E_3(y) = \{Y : \|Y\| = \|y\|\}. \tag{8}$$

We then define

$$E(y) := E_1 \cap E_2(y) \cap E_3(y). \tag{9}$$

This leads to the p-value

$$p_{H_0^M|E}(y) := \Pr_{\beta_M=0}\left(\frac{Y^\top(P_X - P_{X_{-M}})Y}{Y^\top(I_{n-1} - P_X)Y} \geq \frac{y^\top(P_X - P_{X_{-M}})y}{y^\top(I_{n-1} - P_X)y}\middle| Y \in E(y)\right). \tag{10}$$

Remark 1. *In $p_{H_0^M|E}$, we condition on the realizations of $\|Y\|$ and $\|P_{X_{-M}}Y\|$. As discussed in Section 4.1 of Fithian et al. (2017), the natural parameters for the model $Y \sim$*

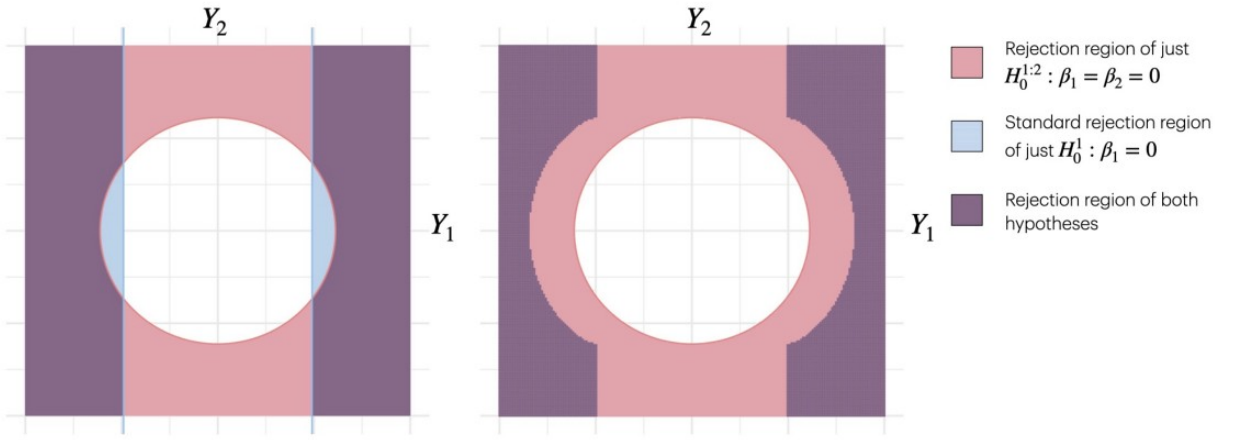


Figure 3: We consider testing $H_0^{1:2} : \beta_1 = \beta_2 = 0$ and $H_0^1 : \beta_1 = 0$ in the model $Y = X\beta + \epsilon$ where $X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and we take the variance of ϵ to be known. In the left plot we use χ^2-tests to test $H_0^{1:2}$ and H_0^1; this is in analogy to F-screening in Box 1 where Step 2 is conducted using a "standard" test. On the right, we use a χ^2-test to test $H_0^{1:2}$, and then we use a test of H_0^1 in Step 2 that accounts for the rejection of $H_0^{1:2}$ in Step 1. This leads to a substantial change in the geometry of the rejection region of H_0^1. Mathematical details are in Section E.

$\mathcal{N}_{n-1}(X\beta, \sigma^2 I_{n-1})$ are β/σ^2 and $1/(2\sigma^2)$, with sufficient statistics $X_k^\top Y$ for each β_k/σ^2 and $\|Y\|^2$ for $1/(2\sigma^2)$. Hence, conditioning on $\|Y\|$ and $P_{X_{-M}}Y$ removes dependence on β_{-M} and σ^2, and we are left with a distribution that depends on β_M/σ^2. In fact, it suffices to condition on $\|P_{X_{-M}}Y\|$ in place of the full vector $P_{X_{-M}}Y$. Thus, we are able to test $\beta_M/\sigma^2 = 0$, or equivalently $H_0^M : \beta_M = 0$, as we show in Proposition 2.1.

The next result establishes that a test of H_0^M based on $p_{H_0^M|E}(y)$ controls the selective Type 1 error, conditional on the event that $H_0^{1:p}$ was rejected.

Theorem 2.1. *A test of H_0^M based on the p-value $p_{H_0^M|E}(y)$ (10) controls the selective Type 1 error, conditional on the event that $H_0^{1:p}$ was rejected. That is, for any $\alpha' \in (0, 1)$,*

$$\Pr_{\beta_M=0}\left(p_{H_0^M|E}(Y) \le \alpha' \mid Y \in E_1\right) = \alpha'.$$

The left panel of Fig. 5 empirically corroborates this result. The next result shows that $p_{H_0^M|E}(y)$ can be easily approximated via Monte Carlo.

Proposition 2.1. *Let $Z \sim \chi^2_{n-p-1}$ and $W \sim \chi^2_m$, where $Z \perp W$ (i.e., Z is independent of W). Define*

$$a(y) := \|y\|^2, \quad c := F^{-1}_{p,n-p-1}(1-\alpha_0) \cdot \tfrac{p}{n-p-1}, \quad d(y) := \|P_{X_{-M}} y\|^2, \quad r(y) := \tfrac{y^\top (P_X - P_{X_{-M}})y}{y^\top (I_{n-1} - P_X)y}. \tag{11}$$

Then,

$$p_{H_0^M|E}(y) = \Pr\left(\frac{W}{Z} \ge r(y) \,\middle|\, \frac{a(y)\cdot W + d(y)\cdot Z}{(a(y) - d(y))\cdot Z} \ge c\right). \tag{12}$$

To summarize, a test based on $p_{H_0^M|E}(y)$ in (10) controls the selective Type 1 error for H_0^M in the context of F-screening as described in Box 1 (Theorem 2.1). Furthermore, $p_{H_0^M|E}(y)$ can be easily approximated via Monte Carlo (Proposition 2.1). Obtaining confidence intervals for $\beta_1, \ldots, \beta_p$ that attain the nominal selective coverage (conditional on rejecting $H_0^{1:p}$), and correcting the corresponding point estimates for selection, are straightforward extensions of the ideas in this section. Details are provided in Appendix C.

We close this subsection with a result concerning the power of the test that results from using the p-value in (12).

Proposition 2.2. *Consider a fixed sequence of design matrices $X_n \in \mathbb{R}^{n\times p}$ and corresponding realizations of the linear model $Y_n = X_n\beta + \epsilon_n$ from (1). Suppose that the design matrices satisfy $\frac{1}{n}X_n^\top X_n \to Q$ for some positive definitive Q. Then the test of $H_0^M : \beta_M = 0$ that uses the selective p-value in (12) is consistent under every alternative, conditional on $Y_n \in E_1$. That is, for any β satisfying $\beta_M \neq 0$ and arbitrary $\alpha' \in (0,1)$, $\lim_{n\to\infty} \Pr_\beta\left(p_{H_0^M|E}(Y_n) \leq \alpha' | Y_n \in E_1\right) = 1$.*

Remark 2. *In Proposition 2.2, the following quantities are fixed with respect to n: $p \geq 1$, $\beta \in \mathbb{R}^p$, $\sigma^2 > 0$, $M \subset \{1, \ldots, p\}$. In contrast, X_n and Y_n grow in length with n, and the p-value in (12) depends on n through $Z \sim \chi^2_{n-p-1}$, as well as c, d, and r in (11). In the remainder of the paper, for conciseness, we only include the subscript n when we wish to emphasize dependence on n. We may also suppress the dependence of some quantities on y (e.g., $p_{H_0^M|E}(y)$ in (12) may be written $p_{H_0^M|E}$).*

2.3 F-Screening and (Conditional) Fisher Information

We now consider the test of H_0^M in Step 2 of Box 1 through the lens of Fisher information in a simplified setting in which σ^2 is known and $\beta_0 = 0$. Hence, we consider the model (1) without the intercept, and the F-tests in Steps 1 and 2 of Box 1 are replaced with χ^2-tests. The hypothesis of interest in Step 2 is $H_0^1 : \beta_1 = 0$. We compare our proposed approach (specialized to this setting) to a sample splitting procedure.

In the sample splitting procedure, we partition the data into $(X^{\text{tr}}, Y^{\text{tr}})$ and $(X^{\text{te}}, Y^{\text{te}})$, where a proportion $\rho \in (0,1)$ of the observations are assigned to the training set and the remaining observations are assigned to the test set such that ρn and $(1-\rho)n$ are integers. We test $H_0^{1:p}$ with $(X^{\text{tr}}, Y^{\text{tr}})$, and then $(X^{\text{te}}, Y^{\text{te}})$ is used to test H_0^1 if and only if $H_0^{1:p}$ is rejected.

In the spirit of Fithian et al. (2017, Section 2.5), we derive expressions for: (i) the leftover Fisher information in Y^{te} about β_1 conditional on the event that $H_0^{1:p}$ is rejected with $(X^{\text{tr}}, Y^{\text{tr}})$; and (ii) the leftover Fisher information in Y about β_1 conditional on the event that both the overall null $H_0^{1:p}$ is rejected using the full data (X, Y) and $P_{X_{-1}}Y = P_{X_{-1}}y$ for some realization $y \in \mathbb{R}^n$ (note that we do not additionally condition on the realization of $\|Y\|$ because σ^2 is known).

Definition 1 (adapted from Fithian et al. 2017)**.** *Let $\ell(\theta; Y \mid Y \in A)$ denote the conditional log-likelihood of Y given $Y \in A$. The leftover Fisher information about the parameter θ in data Y conditional on $Y \in A$ is $\mathcal{I}_{Y|Y\in A}(\theta \mid Y \in A) := -\mathbb{E}\left[\nabla^2 \ell(\theta; Y \mid Y \in A) \mid Y \in A\right]$.*

Proposition 2.3.

i. Define $c_\chi = \sigma^2(\chi^2_p)^{-1}(1-\alpha_0)$ *and the set*

$$R_\rho(X) := \{Y \in \mathbb{R}^{(\rho n)} : Y^\top P_X Y \geq c_\chi\}. \tag{13}$$

Treating $\beta_2, \ldots, \beta_p$ *as nuisance parameters, the leftover Fisher information in* Y^{te} *about* β_1 *conditional on* $Y^{\text{tr}} \in R_\rho(X^{\text{tr}})$ *equals the (unconditional) Fisher information about* β_1 *in* Y^{te}*. This quantity is* $\frac{1}{\sigma^2}\left((X^{\text{te}})_1^\top (I - P_{(X^{\text{te}})_{-1}})(X^{\text{te}})_1\right)$*.*

ii. For any $\tilde{y} \in \mathbb{R}^n$*, define*

$$R(\tilde{y}; X) := \{Y \in \mathbb{R}^n : Y^\top P_X Y \geq c_\chi,\ P_{X_{-1}} Y = P_{X_{-1}} \tilde{y}\}. \tag{14}$$

Treating $\beta_2, \ldots, \beta_p$ *as nuisance parameters, the leftover Fisher information in* Y *about* β_1 *conditional on* $Y \in R(\tilde{y}; X)$ *is given by*

$$\begin{aligned}
&\mathcal{I}_{Y|Y\in R(\tilde{y};X)}(\beta_1; Y \in R(\tilde{y}; X)) \\
&= \begin{cases} \frac{X_1^\top (I - P_{X_{-1}}) X_1}{\sigma^2} & \text{if } \ \tilde{y}^\top P_{X_{-1}} \tilde{y} \geq c_\chi \\ \frac{X_1^\top (I - P_{X_{-1}}) X_1}{\sigma^2} \left[1 - \left(\frac{[\phi(a) - \phi(b)]^2}{[1 - \Phi(b) + \Phi(a)]^2} + \frac{\phi(a)\cdot a - \phi(b)\cdot b}{1 - \Phi(b) + \Phi(a)}\right)\right] & \text{if } \ \tilde{y}^\top P_{X_{-1}} \tilde{y} < c_\chi \end{cases}
\end{aligned} \tag{15}$$

where

$$a = \frac{-\sqrt{c_\chi - \tilde{y}^\top P_{X_{-1}} \tilde{y}} - \beta_1 \sqrt{X_1^\top (I_n - P_{X_{-1}}) X_1}}{\sigma}, \quad b = \frac{\sqrt{c_\chi - \tilde{y}^\top P_{X_{-1}} \tilde{y}} - \beta_1 \sqrt{X_1^\top (I_n - P_{X_{-1}}) X_1}}{\sigma}, \tag{16}$$

and ϕ *and* Φ *denote the density and distribution functions, respectively, of the standard normal distribution.*

For insight into the two cases in (15), note that if $\tilde{y}^\top P_{X_{-1}} \tilde{y} \geq c_\chi$, then the set $R(\tilde{y}; X)$ in (14) can more succinctly be characterized as $R(\tilde{y}; X) := \left\{Y \in \mathbb{R}^n : P_{X_{-1}} Y = P_{X_{-1}} \tilde{y}\right\}$, because $\tilde{y}^\top P_{X_{-1}} \tilde{y} \geq c_\chi$ and $P_{X_{-1}} Y = P_{X_{-1}} \tilde{y}$ imply that $Y^\top P_{X_{-1}} Y \geq c_\chi$ and hence $Y^\top P_X Y \geq c_\chi$. Furthermore, $\frac{\partial^2}{\partial \beta_1^2} \ell(\beta_1; Y \mid P_{X_{-1}} Y = P_{X_{-1}} \tilde{y}) = X_1^\top (I - P_{X_{-1}}) X_1 / \sigma^2$; this is simply the (unconditional) Fisher information about β_1 when $\beta_2, \ldots, \beta_p$ are treated as nuisance parameters. Therefore, when $\tilde{y}^\top P_{X_{-1}} \tilde{y} \geq c_\chi$, conditioning on the set $R(\tilde{y}; X)$ does not affect the Fisher information about β_1. On the other hand, when $\tilde{y}^\top P_{X_{-1}} \tilde{y} < c_\chi$, this more succinct characterization of $R(\tilde{y}; X)$ is no longer possible, and $\frac{\partial^2}{\partial \beta_1^2} \ell(\beta_1; Y \mid R(\tilde{y}; X)) \neq X_1^\top (I - P_{X_{-1}}) X_1 / \sigma^2$. Thus, when $\tilde{y}^\top P_{X_{-1}} \tilde{y} < c_\chi$, conditioning on $Y \in R(\tilde{y}; X)$ indeed affects the Fisher information about β_1.

Recall that we test H_0^1 if and only $H_0^{1:p}$ is rejected with a χ^2-test, so the information about β_1 is only relevant when $Y^\top P_X Y \geq c_\chi$. Therefore, the quantity of interest is actually the leftover Fisher information available for inference about β_1 *times an indicator for the rejection of* $H_0^{1:p}$: that is, the random variable $\mathcal{I}_{Y|Y\in R(\tilde{Y};X)}(\beta_1; Y \in R(\tilde{Y}; X)) \cdot \boldsymbol{I}_{R_1(X)}(\tilde{Y})$, where $\tilde{Y} \overset{d}{=} Y$,

$R_1(X)$ arises from setting $\rho = 1$ in (13), and $\boldsymbol{I}_{R_1(X)}(\tilde{Y})$ is the indicator for $\tilde{Y} \in R_1(X)$. The expectation of this quantity (with respect to $\tilde{Y}$) can be expressed as

$$\mathbb{E}\Big[\mathcal{I}_{Y|Y\in R(\tilde{Y};X)}(\beta_1; Y \in R(\tilde{Y};X)) \cdot \boldsymbol{I}_{R_1(X)}(\tilde{Y})\Big] \\ = \mathbb{E}\left[\mathcal{I}_{Y|Y\in R(\tilde{Y};X)}(\beta_1; Y \in R(\tilde{Y};X)) \mid \tilde{Y} \in R_1(X)\right] \cdot \Pr(\tilde{Y} \in R_1(X)). \quad (17)$$

The right-hand side of (17) is the product of two terms: the expected leftover Fisher information conditional on rejecting $H_0^{1:p}$, and the probability of rejecting $H_0^{1:p}$.

To approximate the conditional expectation in (17), in the left-hand panel of Fig. 4 we generate 1000 draws of $\tilde{Y}$ according to (1) (with $\beta_0 = 0$) and display the average of (15) *over all realizations of* $\tilde{Y}$ *such that* $\tilde{Y}^\top P_X \tilde{Y} \geq c_\chi$. The second term on the right-hand side of (17) can be computed explicitly by noting that, for $R_\rho(X)$ defined in (13) and $\lambda = \frac{1}{2\sigma^2}\beta^\top X^\top X\beta$, $\Pr(Y \in R_\rho(X)) = 1 - F_{p,n\cdot\rho-p-1}(\lambda; F^{-1}_{p,n\cdot\rho-p-1}(0; 1-\alpha_0))$, where $F_{df_1,df_2}(\lambda;\cdot)$ denotes the cumulative distribution function of a noncentral F random variable with df_1 and df_2 degrees of freedom and noncentrality parameter λ. This quantity is displayed in the middle panel of Fig. 4. Finally, in the right-hand panel of Fig. 4 we display the product of the two terms in the left-hand and center panels, i.e., the right-hand side of (17).

For the sample splitting procedure, we plot an analogous set of quantities. Recalling from Proposition 2.3(i) that the leftover Fisher information in Y^{te} about β_1 conditional on $Y^{\text{tr}} \in R_\rho(X^{\text{tr}})$ is $\frac{1}{\sigma^2}\left((X^{\text{te}})_1^\top (I - P_{(X^{\text{te}})_{-1}})(X^{\text{te}})_1\right)$, it follows that

$$\mathbb{E}\left[\frac{(X^{\text{te}})_1^\top (I-P_{(X^{\text{te}})_{-1}})(X^{\text{te}})_1}{\sigma^2} \cdot \boldsymbol{I}_{R_\rho(X^{\text{tr}})}(Y^{\text{tr}})\right] = \frac{(X^{\text{te}})_1^\top (I-P_{(X^{\text{te}})_{-1}})(X^{\text{te}})_1}{\sigma^2} \cdot \Pr(Y^{\text{tr}} \in R_\rho(X^{\text{tr}})). \quad (18)$$

In the left-hand and center panels of Fig. 4, we display the two terms on the right-hand side of (18), and in the right-hand panel we display their product.

In the left-hand panel of Fig. 4, we see that the selective approach generally results in higher expected leftover Fisher information about β_1, conditional on rejection of $H_0^{1:p}$, as compared to sample splitting. Furthermore, the center panel shows that the probability of rejecting $H_0^{1:p}$ is always larger using the selective approach than using sample splitting; this is due to the fact that the former uses all n observations whereas the latter uses only $\rho n \leq n$. Finally, the right-hand panel shows that the product of the left and middle panels (representing (17) and (18)) is larger for the selective approach than for sample splitting, for all values of $\beta_1, \ldots, \beta_p$ considered.

Remark 3. *We see from* (15) *that when* $\beta_1 = 0$*, conditioning on* $R(\tilde{Y}; X)$ *in* (14) *can increase—and cannot decrease—the leftover Fisher information about* β_1 *relative to the unconditional Fisher information. This is reflected in the bottom-left panel of Figure 4. However, for larger values of* $|\beta_1|$*, conditioning typically does not increase the leftover Fisher information.*

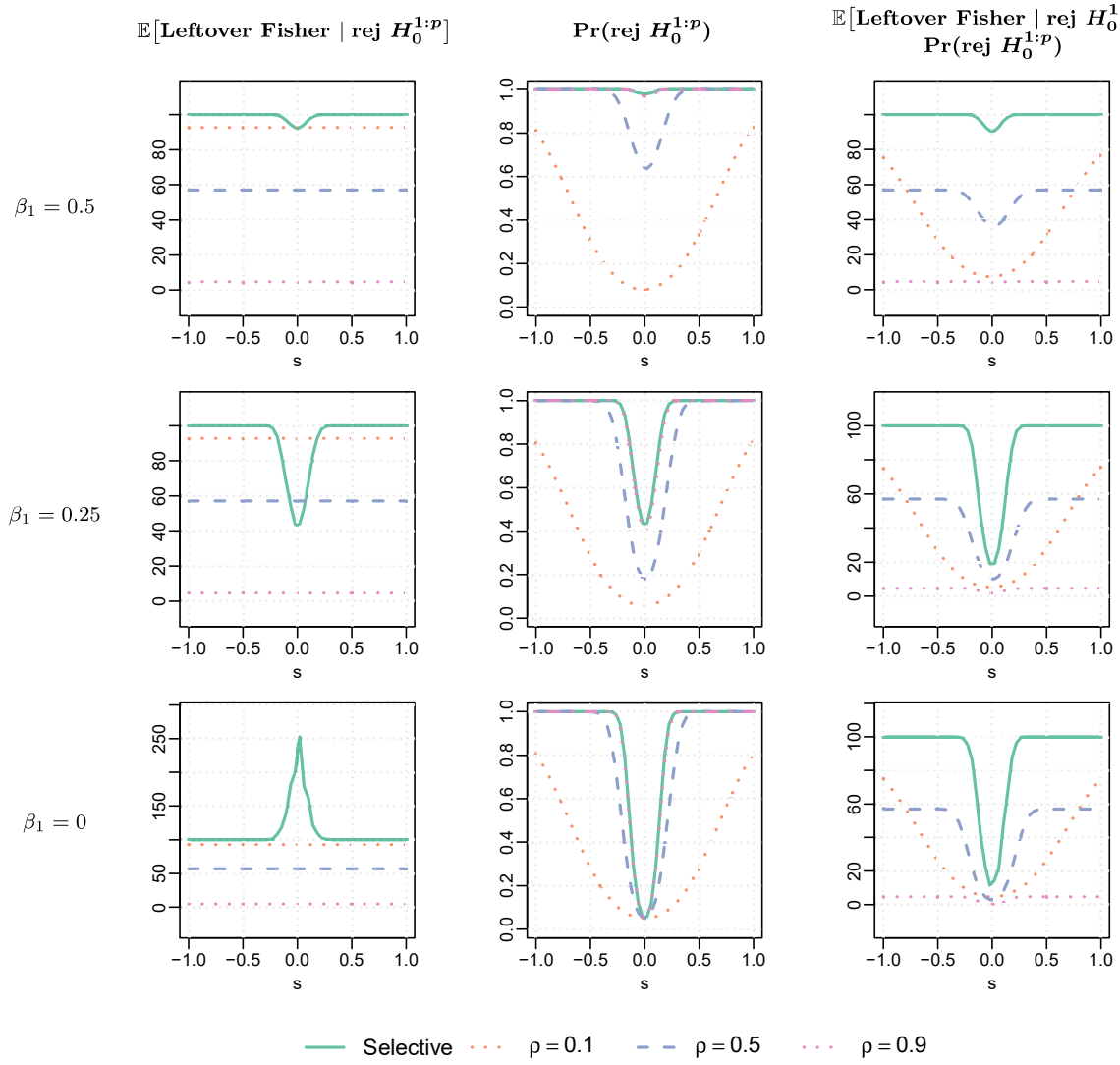


Figure 4: We construct an $n \times p$ orthogonal design matrix X with $n = 100$ and $p = 5$. Then, for $\beta_1 \in \{0.5, 0.25, 0\}$ and for a range of values of $s \in [-1, 1]$, we generate 1000 response vectors $\tilde{Y}$ according to (1) with $\sigma = 1$, $\beta_j = s$ for all $j \in \{2, \ldots, p\}$, and $\beta_0 = 0$. We then compute quantities related to the leftover Fisher information about β_1 for our selective approach, and for sample splitting with split proportions $\rho \in \{0.1, 0.5, 0.9\}$. *Left:* For each of the 1000 realizations $\tilde{y}$ of $\tilde{Y}$, we compute the leftover Fisher information $\mathcal{I}_{Y|Y \in R(\tilde{y};X)}(\beta_1; Y \in R(\tilde{y}; X))$ for our conditional selective inference procedure and display the average *over all datasets such that* $\tilde{Y} \in R_1(X)$. This approximates $\mathbb{E}\left[\mathcal{I}_{Y|Y \in R(\tilde{Y};X)}(\beta_1; Y \in R(\tilde{Y}; X)) \mid \tilde{Y} \in R_1(X)\right]$. Further, for $\rho \in \{0.1, 0.5, 0.9\}$, we display $(X^{\text{te}})_1^\top (I - P_{(X^{\text{te}})_{-1}})(X^{\text{te}})_1 / \sigma^2$. *Center:* We display $\Pr(Y \in R_1(X))$, along with $\Pr(Y_\rho^{\text{tr}} \in R_\rho(X^{\text{tr}}))$ for $\rho \in \{0.1, 0.5, 0.9\}$. *Right:* We display the product of the left-hand and center columns, i.e., the right-hand sides of (17) and (18).

3 Selective p-values in a retrospective analysis

In the applied literature, examples abound of analyses where Step 2 would not have been carried out if Step 1 had not rejected[3]. If the original data for such an analysis were available to the reader, then they could reanalyze the data using the methods in Section 2. Unfortunately, the raw data are often not available. In this section, we demostrate that one can use the usual software output of a least squares linear regression (e.g., the output of `summary(lm(y~x))` in `R`) to "correct" Step 2 of Box 1 when a standard test of H_0^M was used. We refer to this as a *retrospective* analysis, as it can be applied retrospectively to the results of a published paper, without access to the raw data.

Standard software packages for fitting linear models output the proportion of variance explained (R^2) and the residual standard error (RSE), defined as

$$\mathrm{R}^2 = 1 - \frac{y^\top (I_{n-1} - P_X) y}{y^\top y}, \quad \mathrm{RSE} = \sqrt{\frac{y^\top (I_{n-1} - P_X) y}{n-p-1}}, \tag{19}$$

as well as $F_{H_0^M}$ (4). These three quantities suffice to compute $p_{H_0^M|E}$ (12).

Proposition 3.1. *Let $F_{H_0^M}$ be defined as in* (4)*, c defined as in* (11)*, and* R^2 *and* RSE *defined as in* (19)*. Define $Z \sim \chi^2_{n-p-1}$ and $W \sim \chi^2_m$, where $Z \perp W$, and*

$$d_{\text{retro}} := \mathrm{RSE}^2 \cdot \left(\tfrac{\mathrm{R}^2 \cdot (n-p-1)}{1-\mathrm{R}^2} - F_{H_0^M} \cdot m \right), \quad a_{\text{retro}} := \tfrac{\mathrm{RSE}^2 \cdot (n-p-1)}{1-\mathrm{R}^2}.$$

The p-value in (12) *can be written as*

$$p_{H_0^M|E}(y) = \Pr\left(\frac{W}{Z} \geq F_{H_0^M} \cdot \frac{n-p-1}{m} \,\middle|\, \frac{a_{\text{retro}} \cdot W + d_{\text{retro}} \cdot Z}{(a_{\text{retro}} - d_{\text{retro}}) \cdot Z} \geq c \right). \tag{20}$$

In contrast, the selective confidence interval and point estimates described in Appendix C involve quantities that are not generally reported in software output, and thus cannot be easily obtained retrospectively.

4 The special case of ANOVA

F-screening is particularly pervasive in the context of ANOVA, where it dates back almost a century (Fisher 1935, Lindquist 1940). As a simple example, suppose that we are interested in how plant height, Y, varies between treatments by fertilizers A, B, and C. We fit the model $Y = \beta_A \cdot \boldsymbol{I}_A + \beta_B \cdot \boldsymbol{I}_B + \beta_C \cdot \boldsymbol{I}_C + \epsilon$, where $\boldsymbol{I}_A$, $\boldsymbol{I}_B$, and $\boldsymbol{I}_C$ are n-dimensional indicator variables for treatment with fertilizers A, B, and C. In the language of ANOVA, $H_0^{1:p} : \beta_A = \beta_B = \beta_C$

[3]Here are some examples of published papers that clearly state that Step 2 was carried out if and only if $H_0^{1:p}$ was rejected in Step 1: (i) "If the groups tested are different, we report the p-values of post hoc tests but omit the p-value of the omnibus test. If the groups did not test as different, only the P value of the omnibus test is reported" (Kula et al. 2024, page 13); (ii) Table 2 of Wang et al. (2024), where NA's denote hypotheses that were left untested because the overall test was not rejected; (iii) "Post hoc pairwise comparisons were only performed if the omnibus test was significant" (Tams et al. 2020, caption of Figure 5).

is often referred to as the "omnibus test." If $H_0^{1:p}$ is rejected, then one typically conducts post hoc tests (Everitt & Skrondal 2010) to see how the expected plant height differs across fertilizers; for instance, we may test $H_0 : \beta_A = \beta_B$. This post hoc test can be equivalently expressed as a test of $H_0 : \beta_{B \text{ vs. } A} = 0$ in the reparameterized model

$$\begin{aligned} Y &= \beta_A \cdot (\boldsymbol{I}_A + \boldsymbol{I}_B) + (\beta_B - \beta_A) \cdot \boldsymbol{I}_B + \beta_C \cdot \boldsymbol{I}_C + \epsilon \\ &= \beta_A \cdot (\boldsymbol{I}_A + \boldsymbol{I}_B) + \beta_{B \text{ vs. } A} \cdot \boldsymbol{I}_B + \beta_C \cdot \boldsymbol{I}_C + \epsilon. \end{aligned} \tag{21}$$

However, as we have seen in Section 2, a post hoc test of $H_0 : \beta_{B \text{ vs. } A} = 0$ that does not account for the fact that it is tested if and only if $H_0^{1:p}$ was rejected will fail to control the Type 1 error.

The machinery developed in Sections 2–3 for valid inference in Step 2 of F-screening applies directly in the setting of ANOVA. In fact, retrospective analysis, as detailed in Section 3, is even more straightforward because in the context of ANOVA, it is not necessary to have access to $F_{H_0^M}$, R^2, and RSE to conduct a retrospective analysis: instead, it suffices to know the number of samples within each group, and the mean and standard deviation of the outcome variable within each group. This is made precise in the following remark.

Remark 4. *Consider a one-way ANOVA with K groups, parameterized analogously to* (21)*. Suppose we have access to only the following quantities, for $k = 1, \ldots, K$:*

1. *n_k, the number of observations in the kth group,*
2. *$\hat{\mu}_k$, the sample mean of Y among observations in the kth group, and*
3. *$\hat{\sigma}_k = \sqrt{\frac{1}{n_k - 1} \sum_{i \in \text{ group } k} (y_i - \hat{\mu}_k)^2}$, the sample standard deviation of Y among observations in the kth group.*

Then R^2 (19)*,* RSE (19)*, $F_{H_0^{k_1 \text{ vs. } k_2}}$* (4)*, and $F_{H_0^{1:K}}$* (3) *take the form*

$$\mathrm{R}^2 = \frac{SSB}{SSB + SSW}, \quad \mathrm{RSE} = \sqrt{\frac{SSW}{n-K}}, \quad F_{H_0^{k_1 \text{ vs. } k_2}} = \frac{(n-K)(\hat{\mu}_{k_1} - \hat{\mu}_{k_2})^2}{SSW\left(\frac{1}{n_{k_1}} + \frac{1}{n_{k_2}}\right)}, \quad F_{H_0^{1:K}} = \frac{SSB/(K-1)}{SSW/(n-K)},$$

where we define $n = \sum_{k=1}^K n_k$, $\bar{y} = \frac{1}{n} \sum_{k=1}^K \hat{\mu}_k n_k$, $SSB = \sum_{k=1}^K n_k (\hat{\mu}_k - \bar{y})^2$, and $SSW = \sum_{k=1}^K (n_k - 1) \hat{\sigma}_k^2$. Thus $p_{H_0^{1:K}} = 1 - F_{K-1, n-K}(F_{H_0^{1:K}})$, and $p_{H_0^M | E}$ can be computed via (20)*.*

5 Simulation study

5.1 Type 1 error

We generate data according to (1) with $n = 100$, $p = 10$, $\sigma^2 = 1$, $X_{ij} \overset{iid}{\sim} \mathcal{N}(0, 1)$, and $\beta_1 = \ldots = \beta_{10} = 0$, so that $H_0^{1:p}$ in (2) holds. We consider testing $H_0^1 : \beta_1 = 0$ if and only if $H_0^{1:p}$ is rejected at level $\alpha_0 = 0.05$ using $F_{H_0^{1:p}}$ (3). For the subset of 100,000 datasets for which $H_0^{1:p}$ is rejected, the left-hand panel of Fig. 5 displays (i) $p_{H_0^M}$ from (5), which does not account for rejection of $H_0^{1:p}$, and (ii) $p_{H_0^M | E}$ from (12), which accounts for rejection of $H_0^{1:p}$. We see that $p_{H_0^M | E}$ controls selective Type 1 error. Because $p_{H_0^M}$ does not control the selective Type 1 error, we do not consider it in Section 5.2.

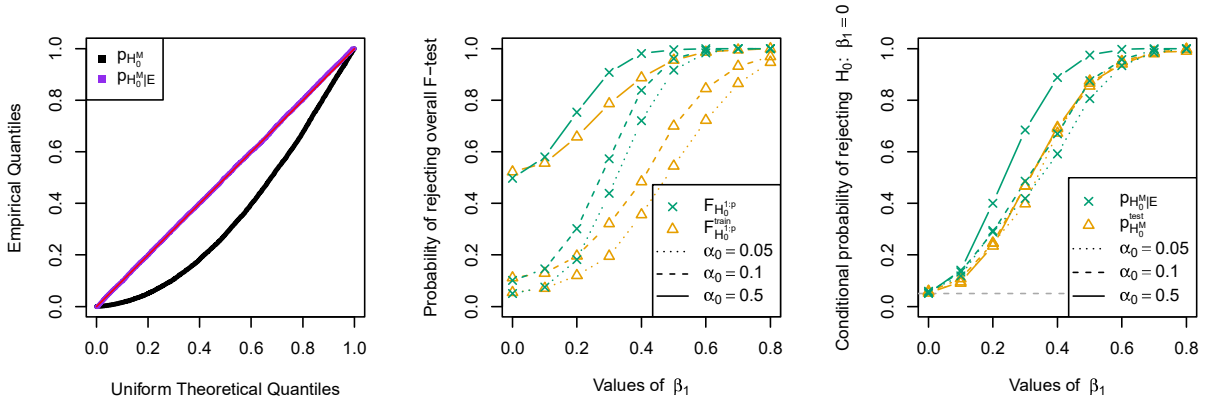


Figure 5: *Left:* For the simulation study detailed in Section 5.1, we display a Unif$(0,1)$ quantile-quantile plot of p-values for the test of $H_0^1 : \beta_1 = 0$, conditional on the event that $H_0^{1:p}$ (2) is rejected. Here, $H_0^{1:p}$ (and hence H_0^1) holds, and so a p-value for $H_0^1 : \beta_1 = 0$ that controls the selective Type 1 error should follow a Unif$(0,1)$ distribution. If the trace lies below the diagonal, the p-value is anti-conservative. For the subset of simulated datasets for which $H_0^{1:p}$ is rejected using $F_{H_0^{1:p}}$, we display $p_{H_0^M}$ and (5) $p_{H_0^M|E}$ (12). *Middle:* For the simulation setting described in Section 5.2, we plot the power to reject $H_0^{1:p}$ at three different significance levels ($\alpha_0 \in \{0.05, 0.1, 0.5\}$) with (i) the F-statistic based on all of the data, $F_{H_0^{1:p}}$ (3), and (ii) the F-statistic based on only the training set from sample splitting, $F^{\text{train}}_{H_0^{1:p}}$. *Right:* For the simulation setting described in Section 5.2, we plot the conditional probability of rejecting H_0^1 at significance level 0.05 (marked with a gray dashed line) given that $H_0^{1:p}$ was rejected at three levels of significance ($\alpha_0 \in \{0.05, 0.1, 0.5\}$). When we reject $H_0^{1:p}$ with $F_{H_0^{1:p}}$, we compute the p-value $p_{H_0^M|E}$ (12). When we reject $H_0^{1:p}$ with $F^{\text{train}}_{H_0^{1:p}}$, we compute $p^{\text{test}}_{H_0^M}$.

5.2 Power

We compare the power of our proposed approach for F-screening to an approach that relies on sample splitting (Cox 1975).

First, we formalize the sample splitting procedure. We randomly partition n independent realizations of (1) into two equally-sized training and test sets. We use the training set to test $H_0^{1:p}$ using an F-statistic (which we refer to as $F^{\text{train}}_{H_0^{1:p}}$), and, if and only if we reject $H_0^{1:p}$, we use the test set to test H_0^M using an F-statistic, which yields the p-value $p^{\text{test}}_{H_0^M}$. Since the training and test sets are independent, $p^{\text{test}}_{H_0^M}$ is uniformly distributed under H_0^M, regardless of the value of $F^{\text{train}}_{H_0^{1:p}}$.

Now we return to the simulation set-up. We generate data according to (1) with $n = 100$, $p = 10$, $\sigma^2 = 1$, $X_{ij} \overset{iid}{\sim} \mathcal{N}(0,1)$, $\beta_2 = \ldots = \beta_{10} = 0$, and with β_1 ranging from 0 to 0.8.

First, we test $H_0^{1:p}$ (2) at $\alpha_0 \in \{0.05, 0.1, 0.5\}$ using either (i) all the data, or (ii) just the training data. The middle panel of Fig. 5 displays the proportion of simulated datasets for which $H_0^{1:p}$ is rejected. Unsurprisingly, we have substantially more power to reject $H_0^{1:p}$ using all the data versus using only the training data.

Next, for the subset of simulated datasets for which we rejected $H_0^{1:p}$ in the middle panel of Fig. 5, we let $M = \{1\}$ and test the hypothesis $H_0^1 : \beta_1 = 0$. Simulated datasets for which $H_0^{1:p}$ was rejected using all the data are tested using $p_{H_0^M|E}$ (12), and those for which $H_0^{1:p}$ was rejected using the training data are tested using $p_{H_0^M}^{\text{test}}$. The right-hand panel of Fig. 5 displays the power to reject $H_0^1 : \beta_1 = 0$, conditional on having rejected $H_0^{1:p}$. When $\alpha_0 = 0.05$ or $\alpha_0 = 0.1$, the power curves are similar for $p_{H_0^M|E}$ (12) and $p_{H_0^M}^{\text{test}}$, while the power curve is substantially higher for $p_{H_0^M|E}$ when $\alpha_0 = 0.5$. The power curves for $p_{H_0^M}^{\text{test}}$ for the three values of α_0 are nearly identical, since the training data used to test $H_0^{1:p}$ is independent of the test data used to test H_0^1, and α_0 appears only in the test of $H_0^{1:p}$. By contrast, the selective p-value $p_{H_0^M|E}$ shifts substantially according to the value of α_0: when α_0 is small, the conditional probability of rejecting $H_0^1 : \beta_1 = 0$ is smaller, since relatively little signal for β_1 is available for inference after conditioning.

It is worth noting that the benefits of our selective p-value over the sample splitting p-value extend beyond the empirical results seen in Fig. 5. Depending on how sample splitting is performed, the testable hypotheses may differ between train and test sets (e.g., if rank deficiency emerges from random partitioning). Also, it may be hard in an applied setting to convince a collaborator to use only part of their data to test $H_0^{1:p}$ and to reserve the remainder for testing H_0^M. By contrast, our selective p-values enable the collaborator to use *all* of the data to test $H_0^{1:p}$, thereby yielding a result for the test of $H_0^{1:p}$ that coincides with the standard (uncorrected) F-screening procedure. Furthermore, the sample splitting p-values cannot be computed retrospectively using summary statistics, as in Section 3.

5.3 Confidence intervals

We now investigate coverage and width of the confidence intervals derived in Appendix C.

To investigate confidence interval coverage, we generate data according to (1) with $\beta_1 = \ldots = \beta_p = 0$, $n = 100$, $p = 5$, and $\sigma^2 = 1$ until we obtain 1000 datasets (X, Y) for which we reject $H_0^{1:p}$ (2) at level $\alpha_0 = 0.05$. Then, for a fixed confidence level (defined as $(1 - \alpha)$ for $\alpha \in (0, 1)$), we compute, for each of the 1000 datasets: (i) the standard confidence interval for β_1 that does not account for selection, and (ii) the selective confidence interval for β_1 (C.20).

As shown in the left panel of Fig. 6, the standard confidence intervals for β_1 that do not account for rejection of $H_0^{1:p}$ (2) exhibit severe under-coverage for all confidence levels. In contrast, the selective confidence interval achieves nominal coverage across all confidence levels.

To examine confidence interval width, we first vary $\beta_1 \in [-1, 1]$ while holding $n = 100$ constant. We set $p = 5$, $\sigma^2 = 1$, and $\beta_2 = \cdots = \beta_5 = 0$. For each fixed value of $\beta_1 \in [-1, 1]$, we generate data according to (1) until we obtain 1000 datasets where $H_0^{1:p}$ is rejected. For each dataset, we compute the selective confidence interval. (We do not depict the standard confidence interval, as we showed in the left-hand panel of Fig. 6 that it is invalid.) Finally, we compute the average confidence interval width across the 1000 datasets and plot these values as a function of β_1 in the middle panel of Fig. 6. In the right-hand panel, we repeat the analysis in the middle panel, but with $\beta_1 = 0$ and $n \in \{50, \ldots, 500\}$.

The curves associated with selective confidence intervals in the middle panel of Fig. 6 exhibit a striking shape: they are narrow both when $\beta_1 \approx 0$ and when $|\beta_1|$ is large. This

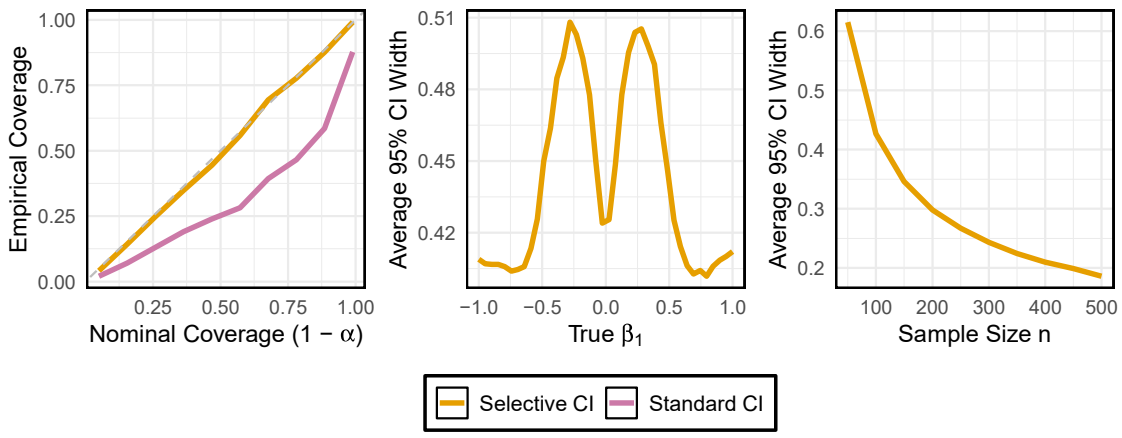


Figure 6: For the setting described in Section 5.3, we construct confidence intervals for β_1 in (1) according to Appendix C, conditional on rejection of $H_0^{1:p}$ (2) with $F_{H_0^{1:p}}$ (3). *Left:* We plot the coverage of the selective and standard confidence intervals for the datasets for which $H_0^{1:p}$ (2) is rejected, where the significance level α varies between 0 and 1, and $\beta_1 = \ldots = \beta_p = 0$. Since the standard intervals do not achieve the nominal coverage, we do not display them in the middle and right panels. *Middle:* We plot the mean width of the selective confidence intervals for the datasets for which $H_0^{1:p}$ (2) is rejected, where β_1 varies and $n = 100$. *Right:* We plot the mean width of the selective confidence intervals for which $H_0^{1:p}$ (2) is rejected for $\beta_1 = 0$ as n varies.

pattern can be understood by separately considering the cases where (i) $|\beta_1|$ is large, (ii) $|\beta_1|$ is moderate, and (iii) $|\beta_1|$ is small. When $|\beta_1|$ is large, the event E_1 (6) is highly likely, in which case conditional inference roughly coincides with standard inference, and so the intervals are not appreciably wider than the standard (non-selective) intervals. When $|\beta_1|$ is moderate, the event E_1 (6) is only moderately likely, and when it occurs, it often happens that the data falls near the boundary of the selection region, and little signal regarding β_1 remains for inference, leading to wide confidence intervals. Finally, when $|\beta_1|$ is small, the event E_1 (6) is unlikely to occur; however, when it does occur, selection is often driven by (chance) association between $X_2, \ldots, X_p$ and Y, in which case sufficient signal about β_1 remains for inference, leading to moderately narrow confidence intervals.

6 Data analysis

In this section, we demonstrate a valid data analysis involving F-screening, both with and without access to the full data. First we re-analyze the data of Xu & Wu (2018), and report selective p-values, confidence intervals, and point estimates for each coefficient. Then we conduct a retrospective analysis of Keil et al. (2023) and compute the selective p-values using only the standard ANOVA outputs because we do not have access to their data. In both analyses, we compare the selective quantities to their standard counterparts.

6.1 Prospective inference

Xu & Wu (2018) used the publicly-available National Health and Nutrition Examination Survey (NHANES) dataset from Centers for Disease Control and Prevention (CDC) (2024) to investigate possible changes in femoral neck bone mineral density (BMD)[4] across the years 2005 to 2014 in US adults over the age of 30. In the NHANES dataset, years are binned into 2005–2006, 2007–2008, 2009–2010, 2013–2014; no data from 2011–2012 is available for BMD. We restrict our analysis to the $n = 7135$ males in the dataset, and consider the model

$$\text{BMD} = \beta_{\text{AGE}} \cdot \text{AGE} + \beta_{5\text{–}6} \cdot \boldsymbol{I}_{5\text{–}6} + \beta_{7\text{–}8} \cdot \boldsymbol{I}_{7\text{–}8} + \beta_{9\text{–}10} \cdot \boldsymbol{I}_{9\text{–}10} + \beta_{13\text{–}14} \cdot \boldsymbol{I}_{13\text{–}14} + \epsilon, \tag{22}$$

where $\boldsymbol{I}_{5\text{–}6}$, $\boldsymbol{I}_{7\text{–}8}$, $\boldsymbol{I}_{9\text{–}10}$, and $\boldsymbol{I}_{13\text{–}14}$ are indicator variables for the time periods 2005–2006, 2007–2008, 2009–2010, and 2013–2014, respectively. Note that (22) contains age, as reported in the NHANES dataset, as a covariate. For convenience, we regress out age: let $U \in \mathbb{R}^{7135\times 7134}$ such that $UU^\top := I_{7135} - P_{\text{AGE}}$ and $U^\top U = I_{7135}$, and we consider the model

$$U^\top \text{BMD} = \beta_{5\text{–}6} \cdot U^\top \boldsymbol{I}_{5\text{–}6} + \beta_{7\text{–}8} \cdot U^\top \boldsymbol{I}_{7\text{–}8} + \beta_{9\text{–}10} \cdot U^\top \boldsymbol{I}_{9\text{–}10} + \beta_{13\text{–}14} \cdot U^\top \boldsymbol{I}_{13\text{–}14} + U^\top \epsilon. \tag{23}$$

In this setting, the overall null hypothesis is

$$H_0^{1:p} : \beta_{5\text{–}6} = \beta_{7\text{–}8} = \beta_{9\text{–}10} = \beta_{13\text{–}14} = 0. \tag{24}$$

In line with Xu & Wu (2018), we test whether the mean age-adjusted BMD differs between time bins (05–06 versus 07–08, 05–06 versus 09–10, 05–06 versus 13–14, 07–08 versus 09–10, 07–08 versus 13–14, 09–10 versus 13–14). To compare mean age-adjusted BMD in 2005–2006 to mean age-adjusted BMD in 2007–2008, we reparametrize (23) as

$$\begin{aligned} U^\top \text{BMD} = {} & \beta_{5\text{–}6} \cdot \left(U^\top \boldsymbol{I}_{5\text{–}6} + U^\top \boldsymbol{I}_{7\text{–}8}\right) \\ & + \beta_{7\text{–}8 \text{ vs. } 5\text{–}6} \cdot U^\top \boldsymbol{I}_{7\text{–}8} + \beta_{9\text{–}10} \cdot U^\top \boldsymbol{I}_{9\text{–}10} + \beta_{13\text{–}14} \cdot U^\top \boldsymbol{I}_{13\text{–}14} + U^\top \epsilon, \end{aligned} \tag{25}$$

and test $H_0^{7\text{–}8 \text{ vs. } 5\text{–}6} : \beta_{7\text{–}8 \text{ vs. } 5\text{–}6} = 0$. The other five pairwise tests are conducted similarly. Results are shown in Table 1. In this case, accounting for the rejection of $H_0^{1:p}$ (24) has virtually no impact—most of the selective and standard quantities are identical up three decimals.

6.2 Retrospective inference

Keil et al. (2023) explores the relationship between sleep and cognitive impairment in older adults. The authors considered four response variables: (i) Center for Epidemiologic Studies-Depression Scale (CES-D), (ii) sleep duration, (iii) Mini-Mental State Examination (MMSE), and (iv) Mattis Dementia Rating Scale (MDRS) (see references in Keil et al. (2023) for details). For each response variable corresponding to the rows in Fig. 1, Keil et al. (2023) fit a model of the form $Y = \beta_1 \cdot \boldsymbol{I}_{<65} + \beta_2 \cdot \boldsymbol{I}_{65\text{–}85} + \beta_3 \cdot \boldsymbol{I}_{>85} + \epsilon$, where $\boldsymbol{I}_{<65}$, $\boldsymbol{I}_{65\text{–}85}$, and $\boldsymbol{I}_{>85}$ denote indicator variables for the age group (in years) of $n = 826$ study participants. The second-to-last column of Fig. 1 displays the p-values for a test of $H_0^{1:p} : \beta_1 = \beta_2 = \beta_3 = 0$, and the last column reports p-values for standard tests of $H_0^j : \beta_j = 0$ for $j = 1, 2, 3$.

[4]BMD is coded as "DXXNKBMD" in the NHANES dataset.

	Standard			Selective		
Years	Estimate	95% CI	p-value	Estimate	95% CI	p-value
05–06 vs. 07–08	-0.004	(-0.013, 0.005)	0.423	-0.004	(-0.013, 0.005)	0.426
05–06 vs. 09–10	-0.003	(-0.012, 0.006)	0.473	-0.003	(-0.012, 0.006)	0.468
05–06 vs. 13–14	-0.020	(-0.029, -0.010)	0.000	-0.020	(-0.029, -0.010)	0.000
07–08 vs. 09–10	0.000	(-0.008, 0.009)	0.923	0.000	(-0.008, 0.009)	0.926
07–08 vs. 13–14	-0.016	(-0.025, -0.007)	0.000	-0.016	(-0.025, -0.007)	0.001
09–10 vs. 13–14	-0.017	(-0.025, -0.008)	0.000	-0.017	(-0.025, -0.008)	0.000

Table 1: For NHANES BMD data, we perform F-screening for the model in (22), and compare selective point estimates, confidence intervals, and p-values with standard counterparts following the rejection of the overall F-test of (24). The statistics in the "Standard" column do *not* account for rejection of (24) (see Box 1), whereas statistics in the "Selective" column do account for it.

Keil et al. (2023) are not explicit about whether they would have tested H_0^j, $j = 1, 2, 3$ had they not rejected $H_0^{1:p}$, and it is not possible to ascertain their intentions from their results, since in their Table 1 (a subset of which is displayed in our Fig. 1), all hypotheses of the form $H_0^{1:p}$ tested with ANOVA are rejected. Notably, they refer to tests of H_0^j as "post hoc tests"; this phrasing is typically reserved in the applied literature to refer to tests that are conducted *after* rejecting $H_0^{1:p}$ (Everitt & Skrondal 2010). Thus, for the sake of demonstration we shall assume that they would not have tested H_0^j for $j = 1, 2, 3$ in their Table 1 if $H_0^{1:p}$ had not been rejected.

Accordingly, we apply the methods in Section 3 and 4 to conduct a retrospective analysis, using only the summary statistics reported in Fig. 1. For each of the four outcomes in Fig. 1, we compute $p_{H_0^M|E}$ in (12) for $H_0^j : \beta_j = 0$, for $j = 1, 2, 3$, alongside the standard p-value in (5) which does not account for the fact that we would not have tested H_0^j if $H_0^{1:p}$ had not be rejected. The standard p-values that we compute match those in Fig. 1 if the Šidák correction is applied (Šidák 1967).

The results are displayed in Table 2. Generally, the values of $p_{H_0^M|E}$ (12) are equal to or larger than the corresponding values of $p_{H_0^M}$ (5). For example, the p-value for the mean difference in CES-D between the under-65 age group and the 65-85 age group is 0.003 using $p_{H_0^M}$ versus 0.118 using $p_{H_0^M|E}$, and the p-value for the mean difference in CES-D between the 65-85 age group and over-85 age group is 0.016 versus 0.100. At the significance level of 0.05 used by Keil et al. (2023), the selective p-value $p_{H_0^M|E}$ does not lead to rejection for those two post hoc tests, whereas the standard p-value $p_{H_0^M}$ does.

7 Discussion

We have shown that a common practice in applied statistics—reporting the outputs of a linear regression model if and only if an overall F-test is rejected—leads to invalid inference. We have proposed an easily-implemented approach, based on conditional selective inference, that remedies the problem. We present a toolbox for valid inference in this setting that contains the most critical linear model outputs: p-values, confidence intervals, and point

	CES-D		Sleep		MMSE		MDRS	
	$p_{H_0^M}$	$p_{H_0^M \mid E}$	$p_{H_0^M}$	$p_{H_0^M \mid E}$	$p_{H_0^M}$	$p_{H_0^M \mid E}$	$p_{H_0^M}$	$p_{H_0^M \mid E}$
< 65 vs. 65-85	0.003	0.118	0.001	0.001	0.000	0.000	0.003	0.003
< 65 vs. ≥ 85	0.397	0.397	0.000	0.003	0.000	0.000	0.000	0.000
65-85 vs. ≥ 85	0.016	0.100	0.060	0.061	0.000	0.000	0.000	0.000

Table 2: For the retrospective analysis conducted on the linear regression outputs from Keil et al. (2023) (Fig. 1), we display the selective ($p_{H_0^M \mid E}$ in (12)) and standard ($p_{H_0^M}$ in (5)) p-values from four analyses, each involving a separate outcome variable (CES-D, Sleep, MMSE, MDRS). For each of these outcome variables, we replicate the one-way ANOVA and post hoc tests conducted in Keil et al. (2023), which are reported in the last two columns of the table in Fig. 1. For the post hoc tests, we report $p_{H_0^M}$ (5) and $p_{H_0^M \mid E}$ (12). If we were to apply the Šidák correction (Šidák 1967) to the values of $p_{H_0^M}$ (5) in this table, they would match the p-values in the last column of Fig. 1.

estimates. Furthermore, our selective p-values can be computed even without access to the raw data, using only the standard outputs of a multiple linear regression.

Some natural questions that we leave for future work are as follows:

- Our results relied on an assumption that the errors are Gaussian. Can this be relaxed using ideas from e.g. Tian & Taylor (2017), Panigrahi (2023)? (We note, of course, that standard inference for a linear model also assumes Gaussianity in finite samples—thus, reliance on Gaussianity is only a very minor limitation of our method.)

- Can these results be extended beyond least squares linear regression, e.g. to generalized linear models, using ideas from Tian & Taylor (2018)?

- How should our selective p-values be adjusted to account for multiple comparisons? For instance, how can $p_{H_0^M \mid E}$ for $M \in \{M_1, \ldots, M_L\}$ be corrected in order to control family-wise selective Type 1 error rate? An extension of Scheffé's correction (Scheffé 1953) to the selective setting may be possible.

An `R` package called `lmFScreen`, implementing the methods proposed in this paper, is available at `https://github.com/mcgougho/lmFScreen`. `R` scripts to reproduce all numerical results can be found at `https://github.com/mcgougho/lmFScreen-paper`.

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A Connections to multiple testing

A.1 Family-wise error rate control does not imply selective Type 1 error rate control

One might wonder whether the problem addressed in this paper could be more simply tackled by controlling the family-wise error rate (FWER). FWER control has been studied for almost a century (Tukey 1949, Scheffé 1953, Dunn 1961, Holm 1979, Marcus et al. 1976, Hochberg & Tamhane 1987), and continues to be an area of active research: for instance, the "PoSI" proposal of Berk et al. (2013)—foundational to the field of selective inference—targets the FWER.

However, we see in the right-hand panel of Fig. A.1 that, in the context of F-screening in a setting with $p = 3$, control of the FWER does not imply control of the selective Type 1 error rate. In particular, though Bonferroni, Scheffé, and closed testing corrections are guaranteed to control FWER, they do not control the selective Type 1 error rate (Fithian et al. 2017), i.e. the probability of a Type 1 error, conditional on rejection of $H_0^{1:3}$.

Why is this the case? Consider the setting of Fig. A.1. There are three possibilities associated with the null hypotheses $H_0^{1:3} : \beta_{1:3} = 0$ and $H_0^1 : \beta_1 = 0$: (i) both are true; (ii) $H_0^{1:3}$ is false, but H_0^1 is true; or (iii) both are false. In (iii), a Type 1 error is impossible, so we consider only (i) and (ii). Let $\delta_{1:3}$ be the decision function associated with the test of $H_0^{1:3}$: it is a function of the (random) data and takes value 1 when we reject $H_0^{1:3}$ and value 0 otherwise; let δ_1 similarly denote the decision function associated with testing H_0^1. In scenario (i), the FWER is $\mathbb{E}_{H_0^{1:3}}[\max\{\delta_{1:3}, \delta_1\}]$. In scenario (ii), the FWER is $\mathbb{E}_{H_0^1}[\delta_1]$. By contrast, in scenario (i) the selective Type 1 error—which is controlled by our procedure—is $\mathbb{E}_{H_0^{1:3}}[\delta_1 \mid \delta_{1:3} = 1]$, while in scenario (ii) it is $\mathbb{E}_{H_0^1}[\delta_1 \mid \delta_{1:3} = 1]$. Clearly, the former targets are not the same as the latter targets, and control of one does not imply control of the other.

A.2 Connection to closed testing

At first glance, there appears to be a connection between the F-screening procedure described in Box 1, and so-called "closed testing procedures" (Marcus et al. 1976).

We begin by describing the closed testing procedure of Marcus et al. (1976), which controls the FWER. Let $H_1, H_2, \ldots, H_q$ denote the set of *elementary* hypotheses to be tested. In addition to the elementary hypotheses, we also consider all possible intersections $H_I = \cap_{i \in I} H_i$ for all nonempty $I \subseteq \{1, 2, \ldots, q\}$. Next, we record the results of a (preliminary) test at level α for each of the hypotheses (including the elementary *and* intersection hypotheses). We then consider an elementary hypothesis H_i to be rejected if and only if the preliminary test H_I is rejected for *all* H_I where $i \in I$.

How can we apply closed testing in our setting? First, consider elementary hypotheses $H_0^M : \beta_M = 0$ and $H_0^{-M} : \beta_{-M} = 0$. In order to conduct a closed testing procedure, the only additional hypothesis we need to consider is $H_0^{1:p} = H_0^M \cap H_0^{-M}$. In order to reject H_0^M, closed testing dictates that our preliminary tests must have rejected $H_0^{1:p}$ *and* H_0^M; of course, we can conduct the preliminary test of the former before we conduct the preliminary test of the latter. Thus, in this setting, closed testing bears a resemblance to the procedure in Box 1. The p-value associated with the test of H_0^M in this particular instance of closed-testing is

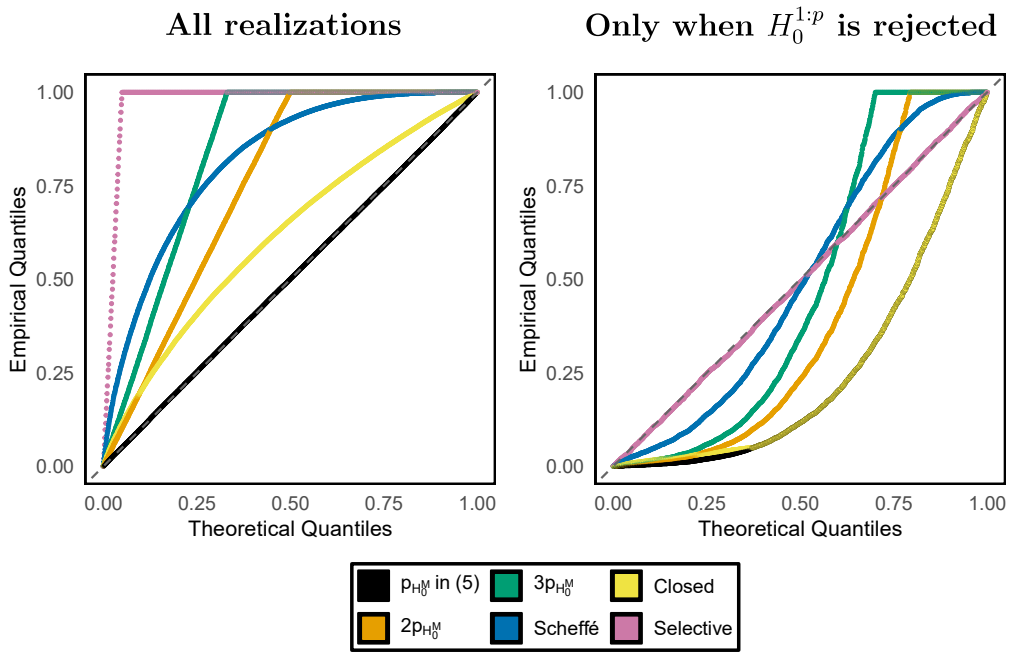


Figure A.1: We simulate data according to (1) with $n = 100$, $p = 3$, $\sigma^2 = 1$, and $\beta_1 = \beta_2 = \beta_3 = 0$, so that $H_0^{1:p}$ (2) holds. For the test of $H_0^1 : \beta_1 = 0$, we report (i) the uncorrected p-value that does not account for F-screening, i.e. $p_{H_0^M}$ in (5); (ii) $2p_{H_0^M}$, resulting from a Bonferroni correction to $p_{H_0^M}$ that accounts for the fact that two tests ($H_0^{1:p}$ and H_0^1) were conducted; (iii) $3p_{H_0^M}$, a Bonferroni correction that accounts for the fact that three post hoc tests ($H_0^j : \beta_j = 0$ for $j \in \{1, 2, 3\}$) could have been conducted; (iv) the p-value that results from using Scheffé's method; (v) the p-value associated with the closed testing procedure as described in Section A.2; and (vi) the selective p-value (12) that accounts for F-screening. *Left:* For each of 100,000 simulated datasets, the QQ-plot displays the p-values for H_0^1. The uncorrected p-value $p_{H_0^M}$ follows a Unif$(0, 1)$ distribution, while the selective p-value is extremely conservative because results for all simulated datasets are displayed, regardless of rejection of $H_0^{1:p}$. Since we are only considering a single test, the Bonferroni-corrected, Scheffé-corrected, and closed-testing p-values are also conservative. *Right:* For the 5% of the 100,000 simulated datasets for which $H_0^{1:p}$ is rejected at level $\alpha_0 = 0.05$, we display the p-values for H_0^1. The uncorrected p-value $p_{H_0^M}$ and the closed testing p-value are anti-conservative, as are the Bonferroni-corrected and Scheffé-corrected p-values when small quantiles are considered. However, the selective p-value follows a Unif$(0, 1)$ distribution.

the maximum of the two p-values associated with the preliminary tests of $H_0^{1:p}$ and H_0^M. In Fig. A.1, we plot this p-value in the case that $p = 3$ and the test of interest in Step 2 of Box 1 is $H_0^1 : \beta_1 = 0$. We see that the closed testing p-value does not control the selective Type 1 error.

How can we reconcile the FWER guarantees of closed testing with the empirical results depicted in the right-hand panel of Figure A.1? The issue hinges on how we "count" the decision regarding H_0^M in scenarios wherein we fail to reject $H_0^{1:p}$. We illustrate this point with a simple example, in which $H_0^{1:p}$ holds. Suppose that one hundred researchers independently

collect data and then perform closed testing at level α. This entails conducting preliminary tests of $H_0^{1:p}$, H_0^M, and H_0^{-M}. Suppose that five of the researchers' preliminary tests reject $H_0^{1:p}$; of these five, four find that their preliminary tests also reject H_0^M. Then, the closed testing procedure dictates that these four researchers should reject H_0^M. In this setting, the family-wise error proportion is $5/100 = 5\%$, because five of the 100 researchers falsely rejected at least one null hypothesis: this is consistent with the fact that closed testing guarantees FWER control. In contrast, the resulting selective Type 1 error proportion—defined as the proportion of researchers who rejected H_0^M among those who rejected $H_0^{1:p}$—is $4/5 = 80\%$; this is because closed testing does not control the selective Type 1 error rate.

While other work on closed testing has built upon the proposal of Marcus et al. (1976) and has considered other targets besides FWER (Goeman & Solari 2011), this literature does not target the selective Type 1 error rate that is the focus of our paper.

B Comparison to Heller et al. (2019)

In this appendix we compare our approach to the method proposed in Heller et al. (2019). We begin with a summary of their method.

For a parameter vector β and estimator $\hat{\beta} \sim \mathcal{N}_m(\beta, \Sigma)$, their proposal is as follows:

1. Conduct an aggregate test of the form $S = \hat{\beta}^\top K \hat{\beta} > S_{1-t_1}$, where K is a positive semi-definite matrix, and S_{1-t_1} is the $(1-t_1)$ quantile of the distribution of S.

2. Only if the test in Step 1 is rejected, perform inference on β conditional on this rejection.

Their Steps 1 and 2 assume that the covariance matrix Σ is known. Their Theorem 1 establishes that, for any linear contrast vector η, the statistic $\eta\hat{\beta}$ follows a truncated normal distribution conditional on $\{S > S_{1-t_1}, \vec{W}\}$, where $\vec{W} = (I_m - (\eta^\top \Sigma \eta)^{-1} \Sigma \eta \eta^\top)\hat{\beta}$. This result is then used to construct a selective p-value, which controls the selective Type 1 error (i.e., the Type 1 error conditional on rejection in Step 1).

If we attempt to apply Heller et al. (2019)'s proposal to the F-screening setting in our paper (Box 1), two challenges arise.

> Challenge 1. Step 1 of their procedure involves a χ^2-test, not an F-test. Does the distinction between F-screening and χ^2-screening in Step 1 matter? There is asymptotic equivalence between an F-test for $\beta = 0$ and a χ^2-test for $\beta = 0$ as $n \to \infty$, if a consistent estimator for σ is used in the χ^2-test. However, in finite samples the two tests will disagree.
>
> Challenge 2. Their main theoretical result, Theorem 1, requires that the covariance be known. However, in the linear model setting of our paper, σ is unknown. Their paper provides no guarantees about the validity of their procedure when the covariance is estimated.

Our proposal avoids Challenge #1 because it considers the *exact* screening step used by data analysts in practice: i.e. Step 1 of Box 1 really involves an F-test, rather than a χ^2 approximation to an F-test.

Our method also avoids Challenge #2 because we have no need to estimate σ^2. We achieve this by following the arguments in Section 4.1 of Fithian et al. (2017), and conditioning on $\|Y\|^2$, which is a sufficient statistic for σ^2 under the model $Y \sim \mathcal{N}(X\beta, \sigma^2 I_{n-1})$.

To show that Challenges #1 and #2 faced by Heller et al. (2019) are real problems in practice, we make the following points in Fig. B.1:

1. *The datasets that pass Step 1 of F-screening differ from those that pass the χ^2-screening rule proposed in Heller et al. (2019)* (Challenge #1). See the left-hand panel of Fig. B.1. Whether we use the conservative estimate of σ^2

$$\tilde{\sigma}^2 = \frac{(Y - \frac{1}{n}\sum_i Y_i)^\top (Y - \frac{1}{n}\sum_i Y_i)}{n-1}, \tag{B.1}$$

   denoted $\hat{\sigma}^2_{\text{null}}$ in Heller et al. (2019), or use the true value of σ, the F-screening and χ^2-screening events differ drastically. On the other hand, when plugging-in the standard estimate of σ^2,

$$\hat{\sigma}^2 = \frac{(Y - X\hat{\beta})^\top (Y - X\hat{\beta})}{n-p-1}, \tag{B.2}$$

   the χ^2-screening event more closely aligns with our screening event, though there are still discrepancies: 5.2% of the datasets were rejected by both the F-test and χ^2-test with the standard plug-in, while 7.2% of the datasets were rejected by the χ^2-test with the standard plug-in and *not* rejected by the F-test.

2. *When n is small, the proposal of Heller et al. (2019) that uses the standard plug-in estimator of σ, $\hat{\sigma}^2$, does not control the selective Type 1 error in Step 2 of F-screening* (Challenge #2). By contrast, our proposed selective p-value does. This is shown in the right-hand panel of Fig. B.1. If instead the oracle σ^2 or the conservative estimate $\tilde{\sigma}^2$ (B.1) is used to compute the selective p-value from Heller et al. (2019), then those p-values become very conservative.

   We note that Heller et al. (2019, Section 4.3.1) introduce a regime-switching variance estimator, which selects between (B.2) and (B.1) based on a more stringent aggregate test threshold. In the setting of Fig. B.1, this threshold is rarely exceeded, so the resulting p-values very closely match with those obtained using the conservative estimator. For this reason, we omit these from Fig. B.1.

In conclusion, our contributions relative to Heller et al. (2019) are as follows:

- whereas Heller et al. (2019) would have analysts perform Step 1 with a χ^2-test, we conduct Step 1 with an F-test, aligning with standard practice (Fisher 1935, Lindquist 1940);

- whereas Heller et al. (2019) provide no theoretical guarantees in the setting of unknown variance (i.e., the setting of a real-world data analysis), our method provides selective Type 1 error control and confidence interval coverage in finite samples;

- whereas computation of the p-value in Heller et al. (2019) requires access to the complete data (X, y), our p-value can be computed retrospectively per Section 3.

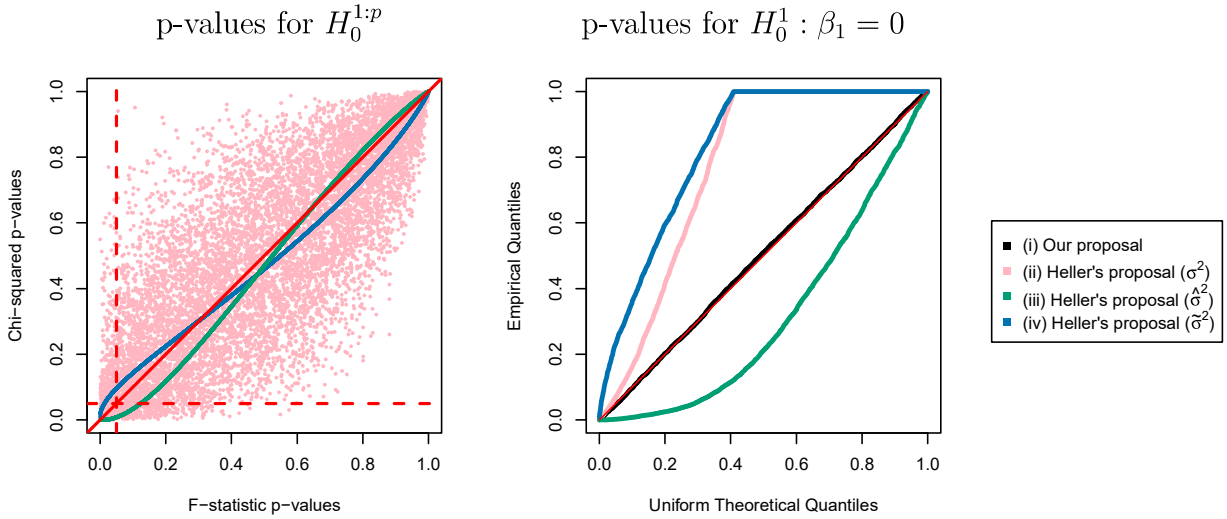


Figure B.1: *Left*: We generate 10,000 datasets (X, Y) with $n = 30$, $p = 10$, $\beta_1 = \ldots = \beta_p = 0$, and $\sigma^2 = 1$. For each dataset, we test the overall null hypothesis $H_0^{1:p} : \beta_1 = \ldots = \beta_p = 0$ at level 0.05 with (i) an F-test; (ii) a χ^2-test that uses the true value of σ^2; (iii) a χ^2-test that uses the standard plug-in estimate $\hat{\sigma}^2$ in (B.2); and (iv) a χ^2-test that uses the conservative plug-in estimate $\tilde{\sigma}^2$ in (B.1). The x-axis corresponds to the F-test p-value in (i), whereas the y-axis corresponds to the χ^2-test p-values in (ii)-(iv). The red dashed horizontal and vertical lines indicate the 0.05 cut-off for the F-statistic and χ^2-statistic p-values, and the red solid line indicates the 45 degree line. To understand why the χ^2 p-values in (iii) and (iv) are a monotone function of the F-statistic p-values in (i), note that the test statistics for (iii) after scaling by the estimated variance, (iv) after scaling by the estimated variance, and (i) are proportional to $\frac{Y^\top P_X Y}{Y^\top (I_{n-1} - P_X) Y} =: F$, $\frac{Y^\top P_X Y}{Y^\top Y} = \frac{F}{1+F}$, and F, respectively. By contrast, the χ^2 p-values in (ii) are not a monotone function of the F-statistic p-values in (i) because they are scaled by the true variance, not an estimate of the variance. *Right:* We generate datasets (X, Y) with $n = 30$, $p = 10$, $\beta_1 = \ldots = \beta_p = 0$, and $\sigma^2 = 1$ until we have 5000 for which $H_0^{1:p} : \beta_1 = \ldots = \beta_p = 0$ is rejected with an F-test at $\alpha = 0.05$. For each such dataset, for a test of $H_0^1 : \beta_1 = 0$, we compute (i) the selective p-value in (12), as well as the selective p-value based off of the truncated normal distribution in Heller et al. (2019) that uses (ii) the true value of σ, (iii) the standard plug-in estimate $\hat{\sigma}^2$ in (B.2), and (iv) the conservative plug-in estimate $\tilde{\sigma}^2$ in (B.1). A uniform QQ-plot of these p-values is displayed. Our proposal's p-values are exactly uniform under the null hypothesis, whereas those of Heller et al. (2019) are anti-conservative when the standard variance estimate $\hat{\sigma}^2$ is used, and conservative when the true value of σ^2 or the conservative estimate $\tilde{\sigma}^2$ is used. The χ^2 p-values obtained using $\hat{\sigma}^2$ are anti-conservative because conditional on rejection of $H_0^{1:p}$, the standard estimator $\hat{\sigma}^2$ is biased downwards; furthermore, the truncated normal approximation used to compute the χ^2 p-values does not hold exactly. By contrast, the χ^2 p-values obtained using the true value of σ^2 and the conservative estimate $\tilde{\sigma}^2$ are conservative because of instances where Step 1 in Box 1 is rejected but the χ^2-test would not have been (this corresponds to instances where the blue and pink points have an empirical quantile value of 1 in the right-hand panel).

C Selective confidence intervals and point estimates

In this appendix, we derive confidence intervals and point estimates for the regression coefficients that are valid conditional on rejecting $H_0^{1:p}$ (2).

C.1 Point estimates

In this section we will develop an approach to find the joint *conditional* maximum likelihood estimate (MLE) (conditional on rejecting $H_0^{1:p}$) of β_j and σ^2 under (1). First, we will arrive at a convenient expression for the conditional density. Once that is obtained, we will maximize the associated conditional likelihood with respect to β_j and σ^2.

Recall the definition of E_1 in (6). Unfortunately, we encounter an immediate challenge in finding the conditional density of Y with respect to β_j and σ^2 given $Y \in E_1$: the probability of the event E_1 depends on nuisance parameters β_{-j} through the projection $P_{X_{-j}}Y$. To eliminate this dependence, we additionally condition on the event $\tilde{E}_2(y) = \{Y : P_{X_{-j}}Y = P_{X_{-j}}y\}$ (where y is a realization of Y); this fixes the component of Y that lies in the column space of X_{-j}.

Remark C.1. *In Section 2.2, we conditioned not only on E_1 in* (6) *and $E_2(y)$ in* (8)*, but also on $E_3(y)$ in* (8)*. This is because (as discussed in Remark 1) the natural parameters are $\frac{\beta_1}{\sigma^2}, \ldots, \frac{\beta_p}{\sigma^2}, \frac{1}{2\sigma^2}$ with sufficient statistics $X_1^\top Y, \ldots, X_p^\top Y, \|Y\|$. Thus, conditioning on E_1, $X_{-j}^\top Y$, and $\|Y\|$ leaves only β_j/σ^2 in the conditional distribution. Since a test of $H_0 : \beta_j/\sigma^2 = 0$ is equivalent to a test of $H_0 : \beta_j = 0$ for any $\sigma > 0$, we were able to construct the test of interest.*

In the context of point estimation, the story is a bit different. Conditioning on E_1, $X_{-j}^\top Y$, and $\|Y\|$ again yields a conditional distribution that involves only β_j/σ^2. However, this enables estimation only of β_j/σ^2, not of β_j.

Therefore, in this section, we condition on only E_1 and $X_{-j}^\top Y$, not on $\|Y\|$. This yields a conditional distribution involving both β_j and σ^2. We jointly maximize it with respect to both parameters.

Proposition C.1. *Let $Y \sim \mathcal{N}_{n-1}(X\beta, \sigma^2 I_{n-1})$, and define*

$$D = \begin{bmatrix} I_{n-1} - P_X \\ P_X - P_{X_{-j}} \\ P_{X_{-j}} \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \\ D_3 \end{bmatrix}.$$

Define new random variables $\tilde{Z} = D_1 Y$, $\tilde{W} = D_2 Y$, and $\tilde{V} = D_3 Y$ (note that $\tilde{Z}^\top \tilde{Z}/\sigma^2 \overset{d}{=} Z$ and $\tilde{W}^\top \tilde{W}/\sigma^2 \overset{d}{=} W$, for Z and W that appear in Proposition 2.1). Let c be as defined in (11) *and E_1 as defined in* (6)*, let y denote a realization of Y, and define $\tilde{d} = P_{X_{-j}}y$. Define $\tilde{E}_2(y) = \{Y : P_{X_{-j}}Y = P_{X_{-j}}y\}$. (Note that $Y \in \tilde{E}_2(y)$ implies $Y \in E_2(y)$ in* (8)*.)*

Then, the conditional density of Y given $Y \in E_1$ and $Y \in \tilde{E}_2(y)$ is

$$f_{Y|Y\in E_1\cap\tilde{E}_2(y)}(\tilde{y}) = \frac{f_{\tilde{Z}}(D_1\tilde{y})\, f_{\tilde{W}}(D_2\tilde{y})\, \boldsymbol{I}_{\left\{\tilde{y}^\top D_2\tilde{y} - \tilde{y}^\top D_1\tilde{y} \geq -\tilde{d}^\top\tilde{d}\right\}}(\tilde{y}) \boldsymbol{I}_{\left\{D_3\tilde{y}=\tilde{d}\right\}}(\tilde{y})}{\Pr\left(\tilde{W}^\top\tilde{W} - c\tilde{Z}^\top\tilde{Z} \geq \tilde{d}^\top\tilde{d}\right)}, \tag{C.1}$$

where the marginal densities of $\tilde{Z}$ and $\tilde{W}$ are given in (C.8) *and $\boldsymbol{I}_{\{\cdot\}}(\tilde{y})$ denotes the indicator that $\tilde{y}$ satifies the condition inside the brackets.*

Proof. First, we can characterize E_1 in terms of $\left(\tilde{Z}, \tilde{W}, \tilde{V}\right)$ by observing that (almost surely)

$$\begin{aligned} Y \in E_1 &\iff \frac{Y^\top P_X Y}{Y^\top (I_{n-1} - P_X) Y} \geq c \\ &\iff \frac{Y^\top (P_X - P_{X_{-j}}) Y + Y^\top P_{X_{-j}} Y}{Y^\top (I_{n-1} - P_X) Y} \geq c \\ &\iff \frac{\tilde{W}^\top \tilde{W} + \tilde{V}^\top \tilde{V}}{\tilde{Z}^\top \tilde{Z}} \geq c \\ &\iff \tilde{W}^\top \tilde{W} - c \tilde{Z}^\top \tilde{Z} \geq -\tilde{V}^\top \tilde{V}, \end{aligned} \tag{C.2}$$

where the third equivalence exploits the idempotence and symmetry of projection matrices. We can similarly characterize $\tilde{E}_2(y)$ in terms of $\tilde{V}$ by noting that

$$\begin{aligned} Y \in \tilde{E}_2(y) &\iff P_{X_{-j}} Y = P_{X_{-j}} y \\ &\iff P_{X_{-j}} Y = \tilde{d} \\ &\iff \tilde{V} = \tilde{d}. \end{aligned} \tag{C.3}$$

Taken together, (C.2) and (C.3) enable us to characterize $E_1 \cap \tilde{E}_2(y)$ in terms of $\left(\tilde{Z}, \tilde{W}, \tilde{V}\right)$ by observing that (almost surely)

$$Y \in E_1 \cap \tilde{E}_2(y) \iff \tilde{W}^\top \tilde{W} - c \tilde{Z}^\top \tilde{Z} \geq -\tilde{d}^\top \tilde{d} \text{ and } \tilde{V} = \tilde{d}. \tag{C.4}$$

Next, note that because the transformation from Y to $\left(\tilde{Z}, \tilde{W}, \tilde{V}\right)$ is given by D, and $D^\top D = I_{n-1}$, a change of variables is straightforward:

$$f_Y(\tilde{y}) = f_{\tilde{Z}, \tilde{W}, \tilde{V}}\left(D\tilde{y}\right). \tag{C.5}$$

Then, we obtain the conditional density of Y given $Y \in \tilde{E}_2(y)$ and find

$$\begin{aligned} f_{Y | Y \in \tilde{E}_2(y)}(\tilde{y}) &= f_{Y | \tilde{V} = \tilde{d}}(\tilde{y}) \\ &= f_{\tilde{Z}, \tilde{W}, \tilde{V} | \tilde{V} = \tilde{d}}(D\tilde{y}) \\ &= f_{\tilde{Z} | \tilde{V} = \tilde{d}}\left(D_1 \tilde{y}\right) f_{\tilde{W} | \tilde{V} = \tilde{d}}\left(D_2 \tilde{y}\right) f_{\tilde{V} | \tilde{V} = \tilde{d}}\left(D_3 \tilde{y}\right) \\ &= f_{\tilde{Z}}\left(D_1 \tilde{y}\right) f_{\tilde{W}}\left(D_2 \tilde{y}\right) f_{\tilde{V} | \tilde{V} = \tilde{d}}\left(D_3 \tilde{y}\right) \\ &= f_{\tilde{Z}}\left(D_1 \tilde{y}\right) f_{\tilde{W}}\left(D_2 \tilde{y}\right) \boldsymbol{I}_{\left\{D_3 \tilde{y} = \tilde{d}\right\}}(\tilde{y}), \end{aligned} \tag{C.6}$$

where the first equality follows from (C.3), the second equality follows from (C.5), the third from mutual independence of $\tilde{Z}$, $\tilde{W}$, and $\tilde{V}$ per Lemma F.1, the fourth from the independence of $\tilde{Z}$ from $\tilde{V}$ and of $\tilde{W}$ from $\tilde{V}$, and the last equality follows because the conditional distribution of $\tilde{V} \mid \tilde{V} = \tilde{d}$ is simply an atom at $\tilde{V} = \tilde{d}$.

Next, we (additionally) condition on $Y \in E_1$ (since we had already conditioned on $Y \in \tilde{E}_2(y)$ in (C.6), we are now effectively conditioning on $Y \in E_1 \cap \tilde{E}_2(y)$). Standard

application of the definition of conditional density yields

$$\begin{aligned} f_{Y|Y\in E_1\cap\tilde{E}_2(y)}(\tilde{y}) &= f_{Y|\tilde{W}^\top\tilde{W}-c\tilde{Z}^\top\tilde{Z}\geq-\tilde{d}^\top\tilde{d},\tilde{V}=\tilde{d}}(\tilde{y}) \\ &= f_{\left\{Y|\tilde{V}=\tilde{d}\right\}|\tilde{W}^\top\tilde{W}-c\tilde{Z}^\top\tilde{Z}\geq-\tilde{d}^\top\tilde{d}}(\tilde{y}) \\ &= \frac{f_{Y|\tilde{V}=\tilde{d}}(\tilde{y})\boldsymbol{I}_{\left\{\tilde{y}\,:\,\tilde{y}^\top D_2\tilde{y}-c\tilde{y}^\top D_1\tilde{y}\geq-\tilde{d}^\top\tilde{d}\right\}}(\tilde{y})}{\Pr\left(\tilde{W}^\top\tilde{W}-c\tilde{Z}^\top\tilde{Z}\geq-\tilde{d}^\top\tilde{d}\right)} \\ &= \frac{f_{\tilde{Z}}\left(D_1\tilde{y}\right)f_{\tilde{W}}\left(D_2\tilde{y}\right)\boldsymbol{I}_{\left\{D_3\tilde{y}=\tilde{d}\right\}}(\tilde{y})\boldsymbol{I}_{\left\{\tilde{y}\,:\,\tilde{y}^\top D_2\tilde{y}-c\tilde{y}^\top D_1\tilde{y}\geq-\tilde{d}^\top\tilde{d}\right\}}(\tilde{y})}{\Pr\left(\tilde{W}^\top\tilde{W}-c\tilde{Z}^\top\tilde{Z}\geq-\tilde{d}^\top\tilde{d}\right)}, \end{aligned} \tag{C.7}$$

where the first equality follows from (C.4), the second equality follows because sequential conditioning is equivalent to simultaneous conditioning, the third equality from the standard definition of conditional density and the fact that $\tilde{W}=D_2Y$ and $\tilde{Z}=D_1Y$ from the statement of the proposition, and the fourth by substituting the result of (C.6). □

Now, we wish to view (C.7) as a likelihood, i.e., a function of β_j and σ^2 with the realization of the data y fixed. First, note that because $Y\sim\mathcal{N}_{n-1}\left(X\beta,\sigma^2I_{n-1}\right)$,

$$\begin{aligned} \tilde{Z} &= \left(I_{n-1}-P_X\right)Y\sim\mathcal{N}_{n-1}\left(0_{n-1},\sigma^2\left(I_{n-1}-P_X\right)\right), \\ \tilde{W} &= \left(P_X-P_{X_{-j}}\right)Y\sim\mathcal{N}_{n-1}\left(\beta_j\left(P_X-P_{X_{-j}}\right)X_j,\sigma^2\left(P_X-P_{X_{-j}}\right)\right). \end{aligned} \tag{C.8}$$

By inspection, we can observe that the distribution of $\tilde{Z}$ is invariant to β_j, while $\tilde{W}$ involves β_j but none of β_{-j}. When (C.7) is viewed as a conditional likelihood under the data realization y, the indicators must be satisfied (since we observed the data), and maximizing it with respect to β_j and σ^2 is equivalent to maximizing

$$L(\beta_j,\sigma^2):=\frac{f_{\tilde{Z}}(\sigma^2;\tilde{z}(y))\cdot f_{\tilde{W}}(\beta_j,\sigma^2;\tilde{w}(y))}{\Pr_{\beta_j,\sigma^2}\left(\tilde{W}^\top\tilde{W}-c\tilde{Z}^\top\tilde{Z}\geq-y^\top P_{X_{-j}}y\right)}, \tag{C.9}$$

where $\tilde{w}(y)=(P_X-P_{X_{-j}})y$, $\tilde{z}(y)=(I_{n-1}-P_X)y$, and we write $\Pr_{\beta_j,\sigma^2}$ to emphasize that the probability depends upon β_j and σ^2.

Recall that we observe y and hence can compute $\tilde{w}(y)$, $\tilde{z}(y)$, and $y^\top P_{X_{-j}}y$. For any candidate value (β_j,σ^2), the numerator can be computed exactly, and the denominator can be approximated via Monte Carlo. Therefore, to approximate the conditional MLE, we numerically jointly maximize the logarithm of the ratio in (C.9) with respect to β_j and σ^2.

We now comment on Fig. C.1. In the left-hand panel, where the overall null $H_0^{1:p}$ (2) holds and we condition on the rejection of $H_0^{1:p}$, the standard MLE has a much larger spread than the conditional MLE derived in this section. This is because the overall null is rejected when the F-statistic is sufficiently large, which tends to occur when $\hat{\beta}_1$ and/or $\hat{\beta}_2$ is large. Thus, the standard MLE for β_1 will tend to be far from zero *conditional on the rejection of the overall null.* The selective estimate, on the other hand, adjusts for the selection event, and is therefore more tightly concentrated around it. In the right-hand panel, where $\beta_1=0.5$ and $\beta_2=0$, the unconditional MLE distribution is fairly tightly concentrated, but centered around 0.8, indicating upward bias after conditioning on rejection of $H_0^{1:p}$. The conditonal MLE has a wider spread but is centered much closer to the true value.

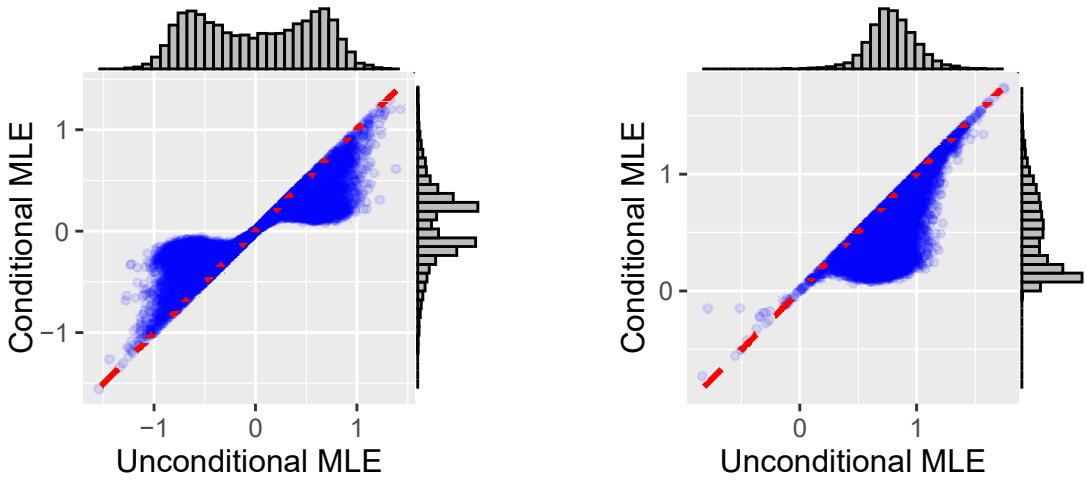


Figure C.1: We generate datasets (X, Y) with $n = 50$, $p = 2$, and $\sigma^2 = 4$ until we have 10,000 that satisfy (6). In the left-hand panel, we set $\beta_1 = \beta_2 = 0$, and in the right-hand panel we set $\beta_1 = 0.5$ and $\beta_2 = 0$. For each dataset, we compute the standard, unconditional MLE of β, $\hat{\beta}_{\text{MLE}} = (X^\top X)^{-1} X^\top Y$. For the same datasets, we compute the conditional MLE of β_1 that arises from numerically maximizing the conditional likelihood in (C.9). Each dataset corresponds to a point on the above scatterplots, with the standard point estimate on the horizontal axis and the conditional point estimate on the vertical axis.

C.2 Confidence intervals

We now consider constructing selective confidence intervals for β_j, i.e., confidence intervals that attain the nominal coverage, conditional on having rejected $H_0^{1:p}$. First, we extend our selective testing framework to accommodate more general hypotheses of the form $H_0^j(b) : \beta_j = b$. With this in hand, confidence intervals can be constructed by means of test inversion, i.e., a level $(1 - \alpha)$ confidence interval will contain all values of b for which we would *not* reject $H_0^j(b)$ at level α.

In order to construct a selective p-value for $H_0^j(b)$, we consider the transformed data vector $Y - X_j b$, as suggested by Fithian et al. (2017, Section 4.1). That is, we evaluate $F_{H_0^M}$ in (4) at $y - X_j b$ instead of y in our selective p-value. Further, instead of conditioning on $Y \in E_2(y) \cap E_3(y)$ in (8), we define

$$\tilde{E}_2(y) := \{Y : P_{X_{-j}} Y = P_{X_{-j}} y\}, \quad \tilde{E}_3^b(y) := \{Y : \|Y - X_j b\|^2 = \|y - X_j b\|^2\}, \tag{C.10}$$

and condition on

$$Y \in \tilde{E}^b(y) := E_1 \cap \tilde{E}_2(y) \cap \tilde{E}_3^b(y). \tag{C.11}$$

Thus we consider the selective p-value

$$p_{H_0^j(b)|\tilde{E}^b}(y) := \Pr_{\beta_j=b}\left(\frac{(Y-X_jb)^\top(P_X-P_{X_{-j}})(Y-X_jb)}{(Y-X_jb)^\top(I_{n-1}-P_X)(Y-X_jb)} \geq r(y-X_jb) \;\middle|\; Y\in\tilde{E}^b(y)\right), \tag{C.12}$$

where $r(\cdot)$ is defined in (11).

Observe that

$$Y\in\tilde{E}_2(y) \iff P_{X_{-j}}Y = P_{X_{-j}}y \iff P_{X_{-j}}(Y-X_jb) = P_{X_{-j}}(y-X_jb).$$

Thus (C.12) is equivalent to

$$p_{H_0^j(b)|\tilde{E}^b}(y) := \Pr_{\beta_j=b}\left(\frac{(Y-X_jb)^\top(P_X-P_{X_{-j}})(Y-X_jb)}{(Y-X_jb)^\top(I_{n-1}-P_X)(Y-X_jb)} \geq r(y-X_jb) \right.$$
$$\left. \;\middle|\; Y\in E_1,\ P_{X_{-j}}(Y-X_jb) = P_{X_{-j}}(y-X_jb),\ \|Y-X_jb\|^2 = \|y-X_jb\|^2\right). \tag{C.13}$$

In order to represent (C.13) in a way that we can approximate via Monte Carlo, we first write the relevant test statistics,

$$\frac{(Y-X_jb)^\top(P_X-P_{X_{-j}})(Y-X_jb)}{(Y-X_jb)^\top(I_{n-1}-P_X)(Y-X_jb)} \quad\text{and}\quad \frac{Y^\top P_X Y}{Y^\top(I_{n-1}-P_X)Y},$$

in terms of the following three functions of our transformed data $Y-X_jb$:

$$T_1 := \|Y-X_jb\|^2, \quad T_2 := P_{X_{-j}}(Y-X_jb), \text{ and } T_3 := X_j^\top(I_{n-1}-P_{X_{-j}})(Y-X_jb). \tag{C.14}$$

Note that

$$\mathrm{Span}(X) = \mathrm{Span}(X_{-j}) \oplus \mathrm{Span}\left((I_{n-1}-P_{X_{-j}})X_j\right).$$

Therefore, $P_X = P_{X_{-j}} + P_{(I_{n-1}-P_{X_{-j}})X_j}$, and hence $P_{(I_{n-1}-P_{X_{-j}})X_j} = P_X - P_{X_{-j}}$. It follows that

$$\begin{aligned}
&(Y-X_jb)^\top P_X(Y-X_jb)\\
&= (Y-X_jb)^\top P_{X_{-j}}(Y-X_jb) + (Y-X_jb)^\top(P_X-P_{X_{-j}})(Y-X_jb)\\
&= (Y-X_jb)^\top P_{X_{-j}}(Y-X_jb) + (Y-X_jb)^\top P_{(I_{n-1}-P_{X_{-j}})X_j}(Y-X_jb)\\
&= T_2^\top T_2 + (Y-X_jb)^\top(I_{n-1}-P_{X_{-j}})X_j(X_j^\top(I_{n-1}-P_{X_{-j}})X_j)^{-1}X_j^\top(I_{n-1}-P_{X_{-j}})(Y-X_jb)\\
&= T_2^\top T_2 + T_3^\top\lambda^{-1}T_3,
\end{aligned}$$

where $\lambda = X_j^\top (I_{n-1} - P_{X_{-j}}) X_j$. Thus the overall F-statistic can be written as

$$\begin{aligned}
F_{H_0^{1:p}} &= \frac{Y^\top P_X Y}{Y^\top (I_{n-1} - P_X) Y} \\
&= \frac{((Y - X_j b) + X_j b)^\top P_X ((Y - X_j b) + X_j b)}{((Y - X_j b) + X_j b)^\top (I_{n-1} - P_X)((Y - X_j b) + X_j b)} \\
&= \frac{(Y - X_j b)^\top P_X (Y - X_j b) + 2(Y - X_j b)^\top X_j b + b^2 X_j^\top X_j}{(Y - X_j b)^\top (I_{n-1} - P_X)(Y - X_j b)} \\
&= \frac{(Y - X_j b)^\top P_X (Y - X_j b) + 2(Y - X_j b)^\top (I_{n-1} - P_{X_{-j}}) X_j b + 2(Y - X_j b)^\top P_{X_{-j}} X_j b + b^2 X_j^\top X_j}{(Y - X_j b)^\top (Y - X_j b) - (Y - X_j b)^\top P_X (Y - X_j b)} \\
&= \frac{T_2^\top T_2 + T_3^\top \lambda^{-1} T_3 + 2 T_3^\top b + 2 T_2^\top X_j b + b^2 X_j^\top X_j}{T_1 - T_2^\top T_2 - T_3^\top \lambda^{-1} T_3}.
\end{aligned} \tag{C.15}$$

Note that $Y \in \tilde{E}_3^b(y)$ (C.10) fixes the value of T_1 (C.14), and $Y \in \tilde{E}_2(y)$ (C.10) fixes the value of T_2 (C.14). Thus—once we condition on $Y \in \tilde{E}^b(y)$ (C.11)—the only randomness in the above representation of $F_{H_0^{1:p}}$ is in the random variable T_3.

Next we write the F–statistic $F_{H_0^j(b)}$ for $H_0^j(b) : \beta_j = b$ in terms of T_1, T_2 and T_3:

$$\begin{aligned}
F_{H_0^j(b)}(Y) &= \frac{(Y - X_j b)^\top (P_X - P_{X_{-j}})(Y - X_j b)}{(Y - X_j b)^\top (I_{n-1} - P_X)(Y - X_j b)} \\
&= \frac{(Y - X_j b)^\top P_{(I_{n-1} - P_{X_{-j}}) X_j} (Y - X_j b)}{(Y - X_j b)^\top (I_{n-1} - P_X)(Y - X_j b)} \\
&= \frac{(Y - X_j b)^\top (I_{n-1} - P_{X_{-j}}) X_j (X_j^\top (I_{n-1} - P_{X_{-j}}) X_j)^{-1} X_j^\top (I_{n-1} - P_{X_{-j}})(Y - X_j b)}{\|Y - X_j b\|^2 - (Y - X_j b)^\top P_X (Y - X_j b)} \\
&= \frac{T_3^\top \lambda^{-1} T_3}{T_1 - T_2^\top T_2 - T_3^\top \lambda^{-1} T_3}.
\end{aligned} \tag{C.16}$$

With fixed T_1 and T_2, the only randomness in the above expression for $F_{H_0^M}$ is in T_3.

Finally, we characterize the conditional distribution of $T_3 \mid \{T_1,\ T_2\}$, i.e. the conditional distribution of

$$X_j^\top (I_{n-1} - P_{X_{-j}})(Y - X_j b) \ \Big|\ \left\{ \|Y - X_j b\|^2,\ P_{X_{-j}}(Y - X_j b) \right\}.$$

Under $H_0 : \beta_j = b$,

$$\begin{aligned}
(I_{n-1} - P_{X_{-j}})(Y - X_j b) &\sim \mathcal{N}\left((I_{n-1} - P_{X_{-j}})(X\beta - X_j b), \sigma^2 (I_{n-1} - P_{X_{-j}}) \right) \\
&\equiv \mathcal{N}(0, \sigma^2 (I_{n-1} - P_{X_{-j}})).
\end{aligned}$$

Let $Z \sim \mathcal{N}(0, \sigma^2 I_{n-p})$ and $Q \in \mathbb{R}^{(n-1)\times(n-p)}$ such that $QQ^\top = I_{n-1} - P_{X_{-j}}$ and $Q^\top Q = I_{n-p}$. It follows that, under $H_0 : \beta_j = b$,

$$(I_{n-1} - P_{X_{-j}})(Y - X_j b) \stackrel{d}{=} QZ.$$

Further, letting

$$\tilde{a} = \|y - X_j b\|^2 \quad \text{and} \quad \tilde{d} = P_{X_{-j}}(y - X_j b), \tag{C.17}$$

note that

$$\begin{aligned}
& X_j^\top (I_{n-1} - P_{X_{-j}})(Y - X_j b) \;\Big|\; \left\{ \|(Y - X_j b)\|^2 = \tilde{a},\ P_{X_{-j}}(Y - X_j b) = \tilde{d} \right\} \\
\overset{d}{=}\ & X_j^\top (I_{n-1} - P_{X_{-j}})(Y - X_j b) \;\Big|\; \left\{ (Y - X_j b)^\top (I_{n-1} - P_{X_{-j}})(Y - X_j b) = \tilde{a} - \tilde{d}^\top \tilde{d}, \right. \\
& \qquad\qquad \left. P_{X_{-j}}(Y - X_j b) = \tilde{d} \right\} \\
\overset{d}{=}\ & X_j^\top (I_{n-1} - P_{X_{-j}})(Y - X_j b) \;\Big|\; \left\{ (Y - X_j b)^\top (I_{n-1} - P_{X_{-j}})(Y - X_j b) = \tilde{a} - \tilde{d}^\top \tilde{d} \right\},
\end{aligned}$$

where the second equivalence follows from independence of $(I_{n-1} - P_{X_{-j}})Y$ and $P_{X_{-j}}Y$ (or, more rigorously, from the fact that if A, B, and C are random events where (A, B) is independent of C and B is independent of C, then $\Pr(A|B, C) = \Pr(A|B)$). Hence, noting that $\|QZ\| = \|Z\|$,

$$\begin{aligned}
& X_j^\top (I_{n-1} - P_{X_{-j}})(Y - X_j b) \;\Big|\; \left\{ \|Y - X_j b\|^2 = \tilde{a},\ P_{X_{-j}}(Y - X_j b) = \tilde{d} \right\} \\
\overset{d}{=}\ & X_j^\top Q Z \;\Big|\; \left\{ \|Z\| = \sqrt{\tilde{a} - \tilde{d}^\top \tilde{d}} \right\}.
\end{aligned}$$

Finally, for $\tilde{Z} \sim \mathcal{N}(0, I_{n-p})$,

$$\begin{aligned}
Z \;\Big|\; \left\{ \|Z\| = \sqrt{\tilde{a} - \tilde{d}^\top \tilde{d}} \right\} &\overset{d}{=}\ \|Z\| \cdot \frac{Z}{\|Z\|} \;\Big|\; \left\{ \|Z\| = \sqrt{\tilde{a} - \tilde{d}^\top \tilde{d}} \right\} \\
&\overset{d}{=}\ \sqrt{\tilde{a} - \tilde{d}^\top \tilde{d}} \cdot \frac{Z}{\|Z\|} \;\Big|\; \left\{ \|Z\| = \sqrt{\tilde{a} - \tilde{d}^\top \tilde{d}} \right\} \\
&\overset{d}{=}\ \sqrt{\tilde{a} - \tilde{d}^\top \tilde{d}} \cdot \frac{Z}{\|Z\|} \\
&\overset{d}{=}\ \sqrt{\tilde{a} - \tilde{d}^\top \tilde{d}} \cdot \frac{\tilde{Z}}{\|\tilde{Z}\|},
\end{aligned}$$

where the third line follows from the fact that the direction $\frac{Z}{\|Z\|}$ and magnitude $\|Z\|$ of an isotropic normal random variable are independent, and the fourth line follows from the fact that $Z \overset{d}{=} \sigma \tilde{Z}$. Hence, we can draw $\tilde{Z} \sim \mathcal{N}(0, I_{n-p})$ and obtain

$$\frac{\sqrt{\tilde{a} - \tilde{d}^\top \tilde{d}}}{\|\tilde{Z}\|} \cdot X_j^\top Q \tilde{Z} \ \overset{d}{=}\ \left[X_j^\top (I_{n-1} - P_{X_{-j}})(Y - X_j b) \;\Big|\; \|Y - X_j b\|^2 = \tilde{a},\ P_{X_{-j}}(Y - X_j b) = \tilde{d} \right], \tag{C.18}$$

under $H_0^j(b) : \beta_j = b$.

For $\tilde{Z} \sim \mathcal{N}(0, I_{n-p})$, define $G := \frac{\sqrt{\tilde{a} - \tilde{d}^\top \tilde{d}}}{\|\tilde{Z}\|} \cdot X_j^\top Q \tilde{Z}$. With $\tilde{a}$ and $\tilde{d}$ defined in (C.17), the

p-value in (C.12) for the test of $H_0^j(b) : \beta_j = b$ can be written as

$$
\begin{aligned}
p_{H_0^j(b)|\tilde{E}^b}(y) &= \Pr_{\beta_j=b}\left(\frac{(Y-X_jb)^\top(P_X-P_{X_{-j}})(Y-X_jb)}{(Y-X_jb)^\top(I_{n-1}-P_X)(Y-X_jb)} \geq r(y-X_jb) \;\middle|\; Y \in E_1 \cap \tilde{E}_2(y) \cap \tilde{E}_3^b(y)\right) \\
&= \frac{\Pr_{\beta_j=b}\left(\frac{(Y-X_jb)^\top(P_X-P_{X_{-j}})(Y-X_jb)}{(Y-X_jb)^\top(I_{n-1}-P_X)(Y-X_jb)} \geq r(y-X_jb),\; Y \in E_1 \;\middle|\; Y \in \tilde{E}_2(y) \cap \tilde{E}_3^b(y)\right)}{\Pr_{\beta_j=b}\left(Y \in E_1 \;\middle|\; \tilde{E}_2(y) \cap \tilde{E}_3^b(y)\right)} \\
&= \frac{\Pr\left(\frac{\tilde{d}^\top\tilde{d}+G^\top\lambda^{-1}G}{\tilde{a}-\tilde{d}^\top\tilde{d}-G^\top\lambda^{-1}G} \geq r(y-X_jb)\,,\; \frac{\tilde{d}^\top\tilde{d}+G^\top\lambda^{-1}G+2G^\top b+2\tilde{d}^\top X_jb+b^2X_j^\top X_j}{\tilde{a}-\tilde{d}^\top\tilde{d}-G^\top\lambda^{-1}G} \geq c\right)}{\Pr\left(\frac{\tilde{d}^\top\tilde{d}+G^\top\lambda^{-1}G+2G^\top b+2\tilde{d}^\top X_jb+b^2X_j^\top X_j}{\tilde{a}-\tilde{d}^\top\tilde{d}-G^\top\lambda^{-1}G} \geq c\right)} \\
&= \Pr\left(\frac{\tilde{d}^\top\tilde{d}+G^\top\lambda^{-1}G}{\tilde{a}-\tilde{d}^\top\tilde{d}-G^\top\lambda^{-1}G} \geq r(y-X_jb) \;\middle|\; \frac{\tilde{d}^\top\tilde{d}+G^\top\lambda^{-1}G+2G^\top b+2\tilde{d}^\top X_jb+b^2X_j^\top X_j}{\tilde{a}-\tilde{d}^\top\tilde{d}-G^\top\lambda^{-1}G} \geq c\right),
\end{aligned}
\tag{C.19}
$$

where the third equality follows from the derivations of (C.15) and (C.16). Further, (C.19) can be readily approximated via Monte Carlo. Using this p-value, we define selective confidence intervals:

$$
\mathrm{CI}_\alpha^j(y) := \left\{b \in \mathbb{R} : p_{H_0^j(b)|\tilde{E}^b}(y) \geq \alpha\right\}. \tag{C.20}
$$

The following proposition establishes that $\mathrm{CI}_\alpha^j(y)$ attains the nominal coverage, conditional on rejection of $H_0^{1:p}$. We see this empirically in the left-hand panels of Fig. 6 and Fig. D.2.

Proposition C.2. *For any $\alpha \in (0,1)$, $\mathrm{CI}_\alpha^j(y)$ in (C.20) provides nominal coverage for the parameter β_j conditional on having rejected $H_0^{1:p}$ in (2). That is,*

$$
\Pr\left(\beta_j \in \mathrm{CI}_\alpha^j(y) \mid Y \in E_1\right) = 1-\alpha \tag{C.21}
$$

for E_1 defined in (6).

Proof of Proposition C.2. First we note that under the null $H_0^j(b) : \beta_j = b$, the p-value $p_{H_0^j(b)|\tilde{E}^b}$ in (C.19) follows a $\mathrm{Unif}(0,1)$ distribution, conditional on $Y \in E_1$ (6). This follows from a very similar argument to that in the proof of Theorem 2.1. Hence,

$$
\begin{aligned}
\Pr\left(\beta_j \in \mathrm{CI}_\alpha^j(y) \mid Y \in E_1\right) &= \Pr\left(p_{H_0^j(b)|\tilde{E}^b} > \alpha \mid Y \in E_1\right) \\
&= 1 - \Pr\left(p_{H_0^j(b)|\tilde{E}^b} \leq \alpha \mid Y \in E_1\right) \\
&= 1-\alpha.
\end{aligned}
$$

□

We observe that the p-value in (C.19) depends on X_j in a way that cannot be deduced from statistics such as R^2, RSE (19), and $F_{H_0^M}$ (4). Thus confidence intervals cannot be computed retrospectively, in the sense of Section 3.

D Extended simulation results

D.1 Alternative signal regime

In the setting of Section 5, we modify the simulation so that $\beta_j = 0.1$ for $j > 1$. The results are shown in Fig. D.1 and Fig. D.2.

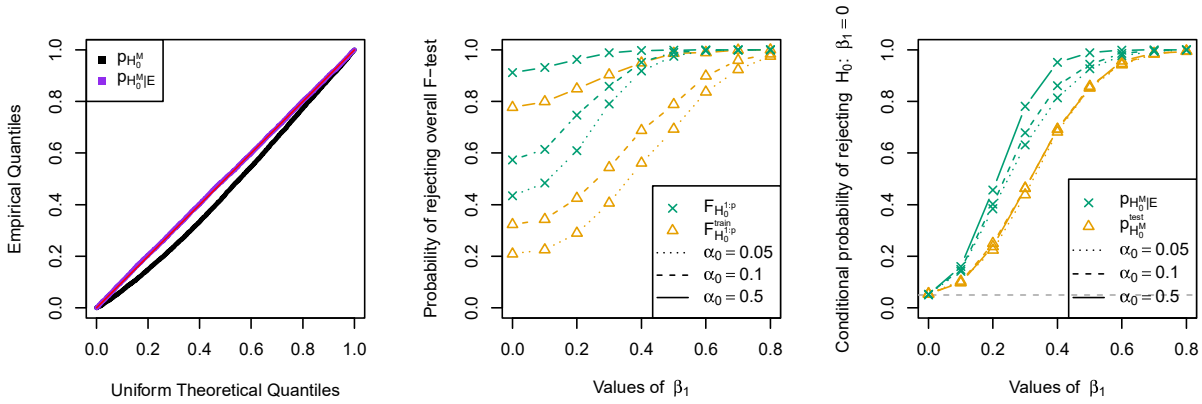


Figure D.1: Details are as in Fig. 5, but with $\beta_j = 0.1$ for $j > 1$.

Compared to the setting in Section 5 and Fig. 5, in the left panel of Fig. D.1 the standard p-value $p_{H_0^M}$ in (5) is less anti-conservative, and in the left panel of Fig. D.2 the standard confidence intervals achieve coverage closer to the nominal level, conditional on rejection of $H_0^{1:p}$ (2). These improvements can be attributed to the fact that rejection of $H_0^{1:p}$ is highly likely when $\beta_2 = \ldots = \beta_p = 0.1$, so conditioning on $Y \in E$ (9) has relatively little impact on the test of H_0^1. Thus, the performance of standard inference methods approaches the performance of selective inference methods.

The middle panel of Fig. D.1 shows that when $\beta_2 = \cdots = \beta_p = 0.1$, the overall test of $H_0^{1:p}$ based on the full dataset has higher power than a test based only on the training data; both tests have higher power to reject $H_0^{1:p}$ than in the setting of Fig. 5, in which $\beta_2 = \ldots = \beta_p = 0$. Comparing the right panel of Fig. D.1 to Fig. 5 reveals that the selective test of H_0^1, which accounts for the rejection of $H_0^{1:p}$ and uses the full dataset, has greater power when $\beta_2 = \cdots = \beta_p = 0.1$ than when $\beta_2 = \ldots = \beta_p = 0$. This is because in the former case, the rejection of $H_0^{1:p}$ is largely driven by the signal in the coefficients β_j for $j > 1$, meaning that substantial signal about β_1 remains available for inference.

The middle and right-hand plot of Fig. D.2 exhibit similar patterns to those in Fig. 6, so we do not provide further commentary.

D.2 Sensitivity to departures from assumptions

We study the behavior of our selective p-values under heteroskedasticty (Fig. D.3), and non-normal errors (Fig. D.4 and Fig. D.5) via simulation.

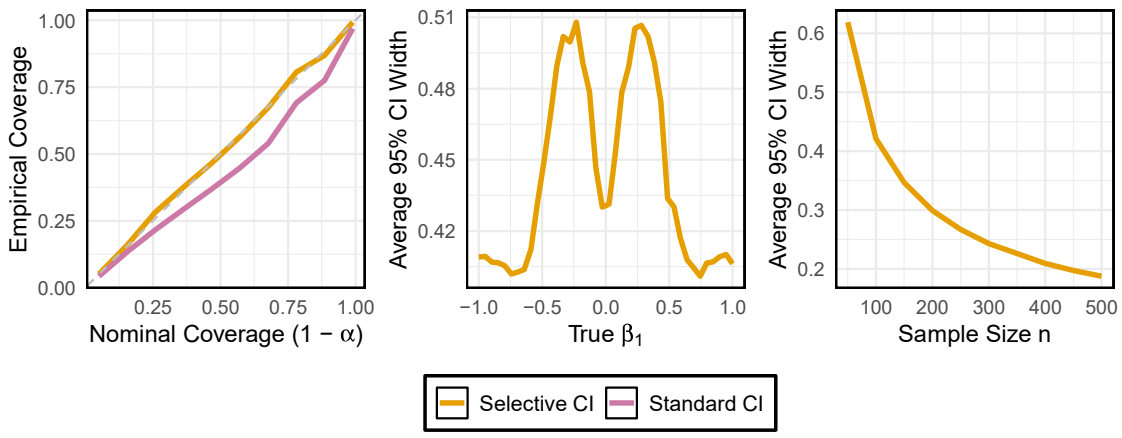


Figure D.2: Details are as in Fig. 6, but with $\beta_j = 0.1$ for $j > 1$.

In Fig. D.3, we consider a heteroskedastic ANOVA setting. We generate a design matrix X and data according to $Y = \epsilon$ (i.e., $\beta_1 = \ldots = \beta_p = 0$ and so $H_0^{1:p}$ holds) as follows:

$$X = \begin{bmatrix} 0 & 0 \\ I_{20} & 0 \\ 0 & I_{20} \end{bmatrix}, \quad \epsilon \sim \mathcal{N}_{60}\left(0, \begin{bmatrix} 1 \cdot I_{20} & 0 & 0 \\ 0 & 4 \cdot I_{20} & 0 \\ 0 & 0 & 2.25 \cdot I_{20} \end{bmatrix}\right). \tag{D.1}$$

The right-hand panel of Fig. D.3 compares the distribution of our selective p-values with the standard (unconditional) p-values in this heteroskedastic setting. While our p-value exhibits anti-conservative behavior, the deviation from the uniform is much less severe than that of the standard p-value.

In Fig. D.4 and Fig. D.5, we examine the behavior of our p-values under non-Gaussian errors. Specifically, we consider the model $Y = \epsilon$ (i.e., $\beta_1 = \ldots = \beta_p = 0$ and so $H_0^{1:p}$ holds) with error term $\epsilon \sim \mathcal{H}(0, 1)$, where $\mathcal{H}(0, 1)$ denotes some parametric distribution with mean 0 and variance 1. In particular, we consider four choices for $\mathcal{H}$: (i) a Laplace distribution with location 0 and scale $1/\sqrt{2}$; (ii) an exponential distribution with rate 1, centered to have mean 0; (iii) a beta distribution with shape parameters 2 and 1, centered and scaled to have mean 0 and variance 1; and (iv) a Rademacher distribution that takes value 1 with probability 0.5 and -1 with probability 0.5. These four distributions correspond to the four panels of Fig. D.4 and Fig. D.5 with design matrix $X_{ij} \overset{iid}{\sim} \mathcal{N}(0, 1)$ and design matrix corresponding to the ANOVA model (D.1), respectively. In Fig. D.4, our selective p-values appear uniform, while the standard p-values are anti-conservative. In Fig. D.5, the standard p-values are extremely anti-conservative, while our selective p-values only appear anti-conservative in the case of Rademacher-distributed errors. The latter setting is somewhat pathological, as $Y = \epsilon$ and ϵ Rademacher implies that $Y \in \{-1, 1\}$; in this setting it is not surprising that our proposal (with its finite-sample Gaussian guarantees) performs poorly.

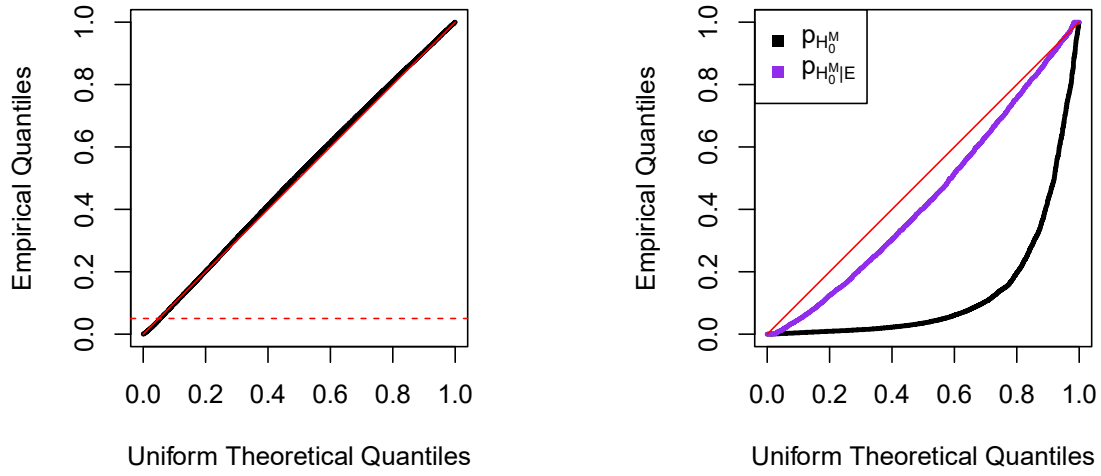


Figure D.3: For $n = 60$, we generate the $n \times 2$ design matrix X and data Y according to (D.1) until 5000 datasets are obtained that reject $H_0^{1:p}$ with an F-test at level $\alpha = 0.05$. *Left:* For all generated datasets (not just those that reject $H_0^{1:p}$), we display the p-value corresponding to the overall F-test on a uniform QQ-plot. We indicate the 45 degree line with a solid red line, and the horizontal dashed line indicates the 0.05 cutoff. Despite heteroskedasticity, these p-values appear uniform. *Right:* For each of the 5000 datasets that reject $H_0^{1:p}$, we display both the proposed selective p-value (12) and the standard (unconditional) p-value (5) for H_0^j on a uniform QQ-plot. The standard p-values are extremely anti-conservative, while the selective p-values for H_0^j are only mildly anti-conservative.

E Geometric intuition

In this section we explain the geometric view of the F-screening procedure displayed in Fig. 2 and Fig. 3. For simplicity, we assume that $\beta_0 = 0$ in the context of (1). We let $\alpha_0 = \alpha = 0.05$.

E.1 Figure 3

Let $Y = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix}$, $X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, and $\sigma^2 = 1$. In this setting, we assume that the error variance is known, and so we conduct χ^2-tests instead of F-tests.

- Step 1 of Box 1: Under $H_0^{1:p} : \beta_1 = \beta_2 = 0$, we have that

$$\frac{Y^T P_X Y}{\sigma^2} = Y^T Y = Y_1^2 + Y_2^2 \sim \chi_2^2.$$

  Thus, if the observed data (y_1, y_2) lies outside the circle defined by $Y_1^2 + Y_2^2 = F_{\chi_2^2}^{-1}(1-\alpha_0)$, then we reject this overall test, where $F_{\chi_p^2}$ is the CDF of a χ_p^2 random variable. This corresponds to the circle in both plots of Fig. 3.

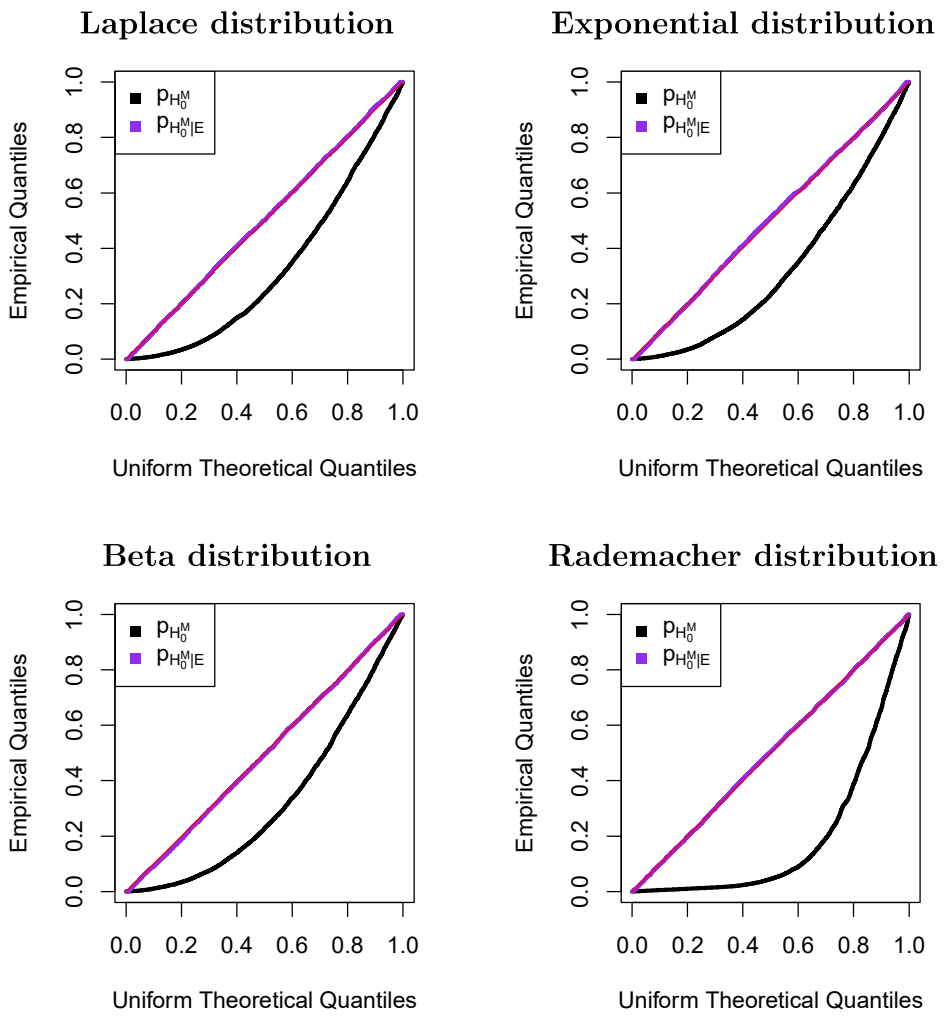


Figure D.4: For $n = 30$, we simulate an $n \times 2$ design matrix X with $X_{ij} \overset{iid}{\sim} \mathcal{N}(0,1)$, and n realizations of $Y \overset{iid}{\sim} \mathcal{H}(0,1)$, until we obtain 5000 datasets that reject $H_0^{1:p}$. Here, $\mathcal{H}$ takes four forms: (i) a Laplace distribution with location 0 and scale $1/\sqrt{2}$; (ii) an exponential distribution with rate 1, centered to have mean 0; (iii) a beta distribution with shape parameters 2 and 1, centered and scaled to have mean 0 and variance 1; and (iv) a Rademacher distribution that takes value 1 with probability 0.5 and -1 with probability 0.5. For each dataset, we display the selective p-value (12) and the standard p-value (5) for H_0^j on a uniform QQ-plot. Across all four non-Gaussian settings considered, our selective p-value for H_0^j appears uniform, while the standard p-values for H_0^j are anti-conservative because they do not account for rejection of $H_0^{1:p}$.

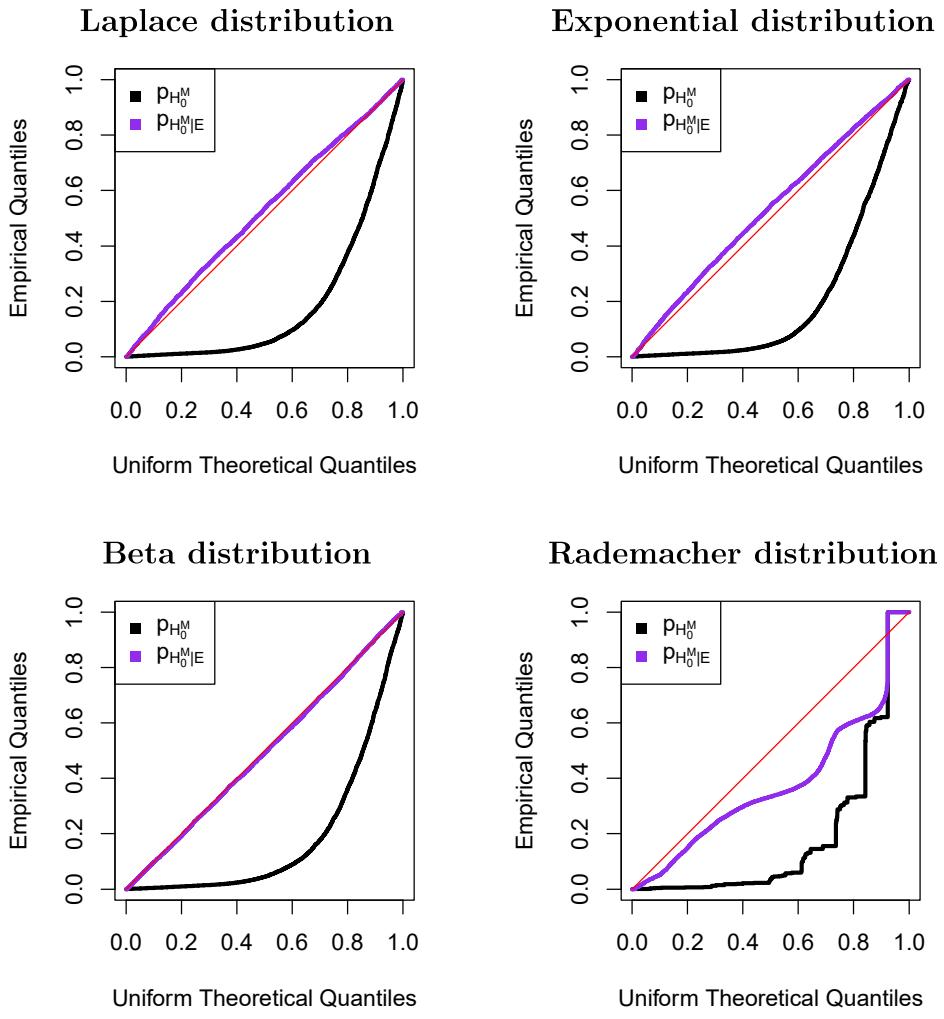


Figure D.5: For $n = 30$, we simulate an $n \times 2$ design matrix with $X = \begin{bmatrix} 0 & 0 \\ I_{10} & 0 \\ 0 & I_{10} \end{bmatrix}$, and n realizations of $Y_i \overset{iid}{\sim} \mathcal{H}(0, 1)$, until we obtain 5000 datasets that reject $H_0^{1:p}$. Here, $\mathcal{H}$ takes four forms: (i) a Laplace distribution with location 0 and scale $1/\sqrt{2}$; (ii) an exponential distribution with rate 1, centered to have mean 0; (iii) a beta distribution with shape parameters 2 and 1, centered and scaled to have mean 0 and variance 1; and (iv) a Rademacher distribution that takes value 1 with probability 0.5 and -1 with probability 0.5. For each dataset, we display the selective p-value (12) and the standard p-value (5) for H_0^j on a uniform QQ-plot. Across all four non-Gaussian settings considered, the standard p-values for H_0^j are highly anti-conservative, in part because they do not account for rejection of $H_0^{1:p}$. For Laplace and exponential errors, our selective p-values for H_0^j are slightly conservative, whereas for Beta-distributed errors they appear uniform. In contrast, under Rademacher-distributed errors — an extreme violation of our modeling assumptions since $Y = \epsilon$ with Rademacher errors implies that Y is binary — our selective p-values for H_0^j behave erratically and are anti-conservative.

- Step 2 of Box 1 (standard inference, left plot of Figure 3): Under $H_0^1 : \beta_1 = 0$, we have that

$$\begin{aligned}\frac{Y^T(P_X - P_{X_{-1}})Y}{\sigma^2} &= Y^T \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \left(\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 0 & 1 \end{bmatrix} \right) Y \\ &= Y_1^2 \sim \chi_1^2,\end{aligned}$$

unconditional on the rejection of $H_0^{1:p}$. Thus, if the observed data (y_1, y_2) satisfies $|y_1| \geq \sqrt{F_{\chi_1^2}^{-1}(1-\alpha)}$, then we reject this standard test of H_0^1. This corresponds to the vertical lines in the left plot of Fig. 3.

- Step 2 of Box 1 (selective inference, right plot in Figure 3): We construct a selective p-value for H_0^1 based on χ^2-tests in a manner similar to Proposition 2.1. We arrive at the p-value

$$\begin{aligned}p_\chi(y) &:= \Pr_{\beta_1=0} \Big(Y^\top (P_X - P_{X_{-1}})Y \geq \sigma^2 y^\top (P_X - P_{X_{-1}})y \\ &\qquad\qquad \mid Y^\top (P_X - P_{X_{-1}})Y \geq \sigma^2 F_{\chi_2^2}^{-1}(1-\alpha_0) - y^\top P_{X_{-1}} y \Big) \\ &= \Pr\left(W \geq \sigma^2 y^\top (P_X - P_{X_{-1}})y \mid W \geq \sigma^2 F_{\chi_2^2}^{-1}(1-\alpha_0) - y^\top P_{X_{-1}} y \right) \\ &= \min\left\{ 1, \frac{\Pr\left(W \geq \sigma^2 y^\top (P_X - P_{X_{-1}})y\right)}{\Pr\left(W \geq \sigma^2 F_{\chi_2^2}^{-1}(1-\alpha_0) - y^\top P_{X_{-1}} y\right)} \right\} \\ &= \min\left\{ 1, \frac{1 - F_{\chi_1^2}\left(\sigma^2 y^\top (P_X - P_{X_{-1}})y\right)}{1 - F_{\chi_1^2}\left(\sigma^2 F_{\chi_2^2}^{-1}(1-\alpha_0) - y^\top P_{X_{-1}} y\right)} \right\},\end{aligned}$$

where $W \sim \chi_1^2$. In our setting, $\sigma^2 y^\top (P_X - P_{X_{-1}})y = y_1^2$ and $y^\top P_{X_{-1}} y = y_2^2$. Thus, if the observed data (y_1, y_2) lead to

$$p_\chi(y) = \min\left\{ 1, \frac{1 - F_{\chi_1^2}\left(y_1^2\right)}{1 - F_{\chi_1^2}\left(F_{\chi_2^2}^{-1}(1-\alpha_0) - y_1^2\right)} \right\} \leq \alpha,$$

then we reject using the selective test of H_0^1. This corresponds to the purple region of the right plot in Fig. 3.

E.2 Figure 2

Now, we assume that σ^2 is unknown,

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}, \quad \text{and} \quad X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Since σ^2 is unknown, we use F-statistics to conduct the hypothesis tests.

- Step 1 of Box 1: Under $H_0^{1:p} : \beta_1 = \beta_2 = 0$,

$$\frac{Y^T P_X Y}{Y^T (I_{n-1} - P_X) Y} = \frac{Y_1^2 + Y_2^2}{Y_3^2} \sim F_{2,1}.$$

  Thus we reject $H_0^{1:p}$ if

$$\frac{y_1^2 + y_2^2}{y_3^2} \geq F_{2,1}^{-1}(1 - \alpha_0).$$

  The rejection boundary corresponds to the double cone in both panels of Fig. 2.

- Step 2 of Box 1: Under $H_0^1 : \beta_1 = 0$,

$$\frac{Y^T (P_X - P_{X_{-1}}) Y}{Y^T (I_{n-1} - P_X) Y} = \frac{Y_1^2}{Y_3^2} \sim F_{1,1},$$

  unconditional on rejection of $H_0^{1:p}$. Therefore, we reject H_0^1 if

$$\frac{y_1^2}{y_3^2} \geq F_{1,1}^{-1}(1 - \alpha) =: b.$$

  Thus the rejection boundary corresponds to the planes defined by $y_1/y_3 \geq \sqrt{b}$ and $y_1/y_3 \geq -\sqrt{b}$ in both panels of Fig. 2.

F Proofs of theoretical results in Sections 2 and 3

In this appendix, we provide proofs for all theorems and propositions from the main text. We begin by stating and proving a lemma that will be useful.

F.1 A useful lemma

Lemma F.1. *The following statements hold under the model $Y = X\beta + \epsilon$ where $X \in \mathbb{R}^{(n-1)\times p}$ and $\epsilon \sim \mathcal{N}_{n-1}(0, \sigma^2 I_{n-1})$:*

1. *Under $H_0^M : \beta_M = 0$, we have that $\frac{1}{\sigma^2} Y^\top (P_X - P_{X_{-M}}) Y \sim \chi^2_m$.*
2. *$\frac{1}{\sigma^2} Y^\top (I_{n-1} - P_X) Y \sim \chi^2_{n-p-1}$.*
3. *In general, the distribution of $Y^\top P_{X_{-M}} Y$ involves β_j for $j \notin M$.*
4. *$Y^\top (P_X - P_{X_{-M}}) Y$, $Y^\top (I_{n-1} - P_X) Y$, and $Y^\top P_{X_{-M}} Y$ are mutually independent.*

Proof of Lemma F.1. We begin by proving the fourth statement of Lemma F.1. First, note that $P_X - P_{X_{-M}}$, $I_{n-1} - P_X$ and $P_{X_{-M}}$ are projection matrices, and thus they are symmetric and idempotent. As a result, it follows that

$$Y^\top (P_X - P_{X_{-M}}) Y = Y^\top (P_X - P_{X_{-M}})^\top (P_X - P_{X_{-M}}) Y = \left\| (P_X - P_{X_{-M}}) Y \right\|_2^2.$$

Similarly,

$$Y^\top (I_{n-1} - P_X)Y = \|(I_{n-1} - P_X)Y\|_2^2 \quad \text{and} \quad Y^\top P_{X_{-M}} Y = \left\|P_{X_{-M}} Y\right\|_2^2.$$

Thus it suffices to show that $(P_X - P_{X_{-M}})Y$, $(I_{n-1} - P_X)Y$, and $P_{X_{-M}}Y$ are mutually independent. Letting

$$A = \begin{bmatrix} P_X - P_{X_{-M}} \\ I_{n-1} - P_X \\ P_{X_{-M}} \end{bmatrix},$$

observe that $AY \sim \mathcal{N}_{3(n-1)}\left(AX\beta, \sigma^2 AA^\top\right)$. Since this is a (degenerate) multivariate normal distribution, proving that $(P_X - P_{X_{-M}})Y$, $(I_{n-1} - P_X)Y$, and $P_{X_{-M}}Y$ are mutually independent is equivalent to proving that the off-diagonal blocks of the covariance matrix are all zero. The off-diagonal blocks of the covariance matrix, omitting the constant σ^2, are as follows:

$$\begin{aligned}
&(P_X - P_{X_{-M}})(I_{n-1} - P_X) = (P_X - P_{X_{-M}} - P_X + P_{X_{-M}}P_X) = P_{X_{-M}}P_X - P_{X_{-M}} \\
&(I_{n-1} - P_X)P_{X_{-M}} = P_{X_{-M}} - P_X P_{X_{-M}} \\
&(P_X - P_{X_{-M}})P_{X_{-M}} = P_X P_{X_{-M}} - P_{X_{-M}}.
\end{aligned}$$

Furthermore, noting that $R(X_{-M}) \subseteq R(X)$ where $R(\cdot)$ denotes the column space, it follows from properties of projection matrices that $P_X P_{X_{-M}} = P_{X_{-M}}$, and hence also that $P_{X_{-M}}P_X = P_{X_{-M}}$.

The fourth statement of Lemma F.1 follows.

To prove claims 1–3, we recall that if $Y \sim \mathcal{N}_{n-1}(\mu, \sigma^2 I_{n-1})$ and P is a projection matrix of rank r, then $Y^\top PY/\sigma^2 \sim \chi_r^2(\lambda)$ where $\lambda = \mu^\top P\mu/(2\sigma^2)$ is the noncentrality parameter. Using this result, to prove part 1 of Lemma F.1, observe that $\frac{1}{\sigma}Y \sim \mathcal{N}_{n-1}\left(\frac{1}{\sigma}X\beta, I_{n-1}\right)$ and $I_{n-1} - P_X$ is a projection matrix of rank $n - p - 1$. It then follows that

$$\frac{1}{\sigma^2}Y^\top (I_{n-1} - P_X)Y \sim \chi^2_{n-p-1}(0) \equiv \chi^2_{n-p-1}.$$

To prove part 2, note that

$$\begin{aligned}
(P_X - P_{X_{-M}})X\beta &= (P_X - P_{X_{-M}})X_M\beta_M + (P_X - P_{X_{-M}})X_{-M}\beta_{-M} \\
&= (P_X - P_{X_{-M}})X_M\beta_M \\
&= P_X X_M\beta_M - P_{X_{-M}}X_M\beta_M \\
&= X_M\beta_M - P_{X_{-M}}X_M\beta_M \\
&= (I_{n-1} - P_{X_{-M}})X_M\beta_M.
\end{aligned}$$

Since $(P_X - P_{X_{-M}})$ is a projection matrix of rank $|M| = m$, it follows that

$$\frac{1}{\sigma^2}Y^\top (P_X - P_{X_{-M}})Y \sim \chi^2_m\left(\frac{(X_M\beta_M)^\top (I_{n-1} - P_{X_{-M}})X_M\beta_M}{2\sigma^2}\right). \tag{F.1}$$

Therefore, under the null $H_0^M : \beta_M = 0$, we have $\frac{1}{\sigma^2}Y^\top (P_X - P_{X_{-M}})Y \sim \chi^2_m$.

Finally, to prove the third statement, note that $P_{X_{-M}}$ is a projection matrix of rank $p-m$. Hence,

$$\frac{1}{\sigma^2} Y^\top P_{X_{-M}} Y \sim \chi^2_{p-m}\left(\frac{1}{2\sigma^2}(X\beta)^\top P_{X_{-M}} X\beta\right). \tag{F.2}$$

Unless $X_M^\top X_M = 0$, the non-centrality parameter involves both β_M and β_{-M}.

□

F.2 Proof of Theorem 2.1

We begin with the following remark, which can be viewed as an extension of the probability integral transform to accommodate conditioning.

Remark F.1. *Let T be a random variable and C an event. Let*

$$F_{T|C}(t) = \Pr(T \le t \mid C),$$

i.e., the conditional CDF of T given C. Define $D = F_{T|C}(T)$. Then

$$\begin{aligned}\Pr(D \le \alpha \mid C) &= \Pr(F_{T|C}(T) \le \alpha \mid C) \\ &= \Pr(T \le F^{-1}_{T|C}(\alpha) \mid C) \\ &= F_{T|C}(F^{-1}_{T|C}(\alpha)) \\ &= \alpha.\end{aligned}$$

Thus $D \sim \text{Unif}\,(0,1)$.

Proof of Theorem 2.1. Remark F.1 implies that, for $\alpha' \in (0,1)$ and E defined in (9),

$$Pr_{\beta_M=0}(p_{H_0^M|E} \le \alpha' \mid Y \in E) = \alpha'.$$

Equivalently,

$$\mathbb{E}\left[\mathbb{1}\{p_{H_0^M|E} \le \alpha'\} \mid Y \in E\right] = \alpha'.$$

Because $Y \in E \implies Y \in E_1$ for E_1 defined in (6) and $E(y)$ defined in (9), when we apply the law of total expectation, we obtain

$$\begin{aligned}&Pr_{\beta_M=0}(p_{H_0^M|E} \le \alpha' \mid Y \in E_1) \\ &\quad= \mathbb{E}\left[\boldsymbol{I}\{p_{H_0^M|E} \le \alpha'\} \mid Y \in E_1\right] \\ &\quad= \mathbb{E}\Big[\mathbb{E}\left[\boldsymbol{I}\{p_{H_0^M|E} \le \alpha'\} \mid Y \in E_1 \cap E_2(y) \cap E_3(y)\right] \mid Y \in E_1\Big] \\ &\quad= \mathbb{E}\Big[\mathbb{E}\left[\boldsymbol{I}\{p_{H_0^M|E} \le \alpha'\} \mid Y \in E\right] \mid Y \in E_1\Big] \\ &\quad= \mathbb{E}[\alpha' \mid Y \in E_1] \\ &\quad= \alpha'.\end{aligned}$$

□

F.3 Proof of Proposition 2.1

Proof of Proposition 2.1. First, note that both $F_{H_0^M}$ and $F_{H_0^{1:p}}$ can be written in terms of the three quadratic forms $Y^\top P_{X_{-M}} Y$, $\|Y\|^2$, and $Y^\top (P_X - P_{X_{-M}})Y$. Indeed,

$$F_{H_0^M} = \tfrac{n-p-1}{m} \cdot \frac{Y^\top (P_X - P_{X_{-M}})Y}{Y^\top (I_{n-1} - P_X)Y} = \tfrac{n-p-1}{m} \cdot \frac{Y^\top (P_X - P_{X_{-M}})Y}{Y^\top Y - Y^\top P_{X_{-M}} Y - Y^\top (P_X - P_{X_{-M}})Y}$$
$$F_{H_0^{1:p}} = \tfrac{n-p-1}{p} \cdot \frac{Y^\top P_X Y}{Y^\top (I_{n-1} - P_X)Y} = \tfrac{n-p-1}{p} \cdot \frac{Y^\top P_{X_{-M}} Y + Y^\top (P_X - P_{X_{-M}})Y}{Y^\top Y - Y^\top P_{X_{-M}} Y - Y^\top (P_X - P_{X_{-M}})Y.}$$

Next, we characterize the distribution of $Y^\top (P_X - P_{X_{-M}})Y \mid \{\|Y\|^2 = a,\ \|P_{X_{-M}} Y\|^2 = d\}$, where $a \equiv a(y) = \|y\|^2$ and $d \equiv d(y) = \|P_{X_{-M}} y\|^2$ per (11). By independence of $(P_X - P_{X_{-M}})Y$ and $P_{X_{-M}} Y$ per Lemma F.1, observe that

$$\begin{aligned}
&Y^\top (P_X - P_{X_{-M}})Y \;\Big|\; \left\{\|Y\|^2 = a,\ \|P_{X_{-M}} Y\|^2 = d\right\} \\
&\overset{d}{=} Y^\top (P_X - P_{X_{-M}})Y \;\Big|\; \left\{\|(I_{n-1} - P_{X_{-M}})Y\|^2 = a - d,\ \|P_{X_{-M}} Y\|^2 = d\right\} \\
&\overset{d}{=} Y^\top (P_X - P_{X_{-M}})Y \;\Big|\; \left\{\|(I_{n-1} - P_{X_{-M}})Y\|^2 = a - d\right\} \\
&\overset{d}{=} \|(P_X - P_{X_{-M}})Y\|^2 \;\Big|\; \left\{\|(I_{n-1} - P_{X_{-M}})Y\| = \sqrt{a-d}\right\} \\
&\overset{d}{=} \|(P_X - P_{X_{-M}})(P_{X_{-M}} Y + (I_{n-1} - P_{X_{-M}})Y)\|^2 \;\Big|\; \left\{\|(I_{n-1} - P_{X_{-M}})Y\| = \sqrt{a-d}\right\} \\
&\overset{d}{=} \|(P_X - P_{X_{-M}})(I_{n-1} - P_{X_{-M}})Y\|^2 \;\Big|\; \left\{\|(I_{n-1} - P_{X_{-M}})Y\| = \sqrt{a-d}\right\}.
\end{aligned}$$

Let $R \sim \mathcal{N}\left(0, \sigma^2 I_{(n-1)-(p-m)}\right)$ and $Q \in \mathbb{R}^{(n-1)\times((n-1)-(p-m))}$ such that $QQ^\top = I_{n-1} - P_{X_{-M}}$ and $Q^\top Q = I_{(n-1)-(p-m)}$. Then, under $H_0^M : \beta_M = 0$, $(I_{n-1} - P_{X_{-M}})Y \overset{d}{=} QR$. It follows that

$$\begin{aligned}
&Y^\top (P_X - P_{X_{-M}})Y \;\Big|\; \left\{\|Y\|^2 = a,\ \|P_{X_{-M}} Y\|^2 = d\right\} \\
&\overset{d}{=} \|(P_X - P_{X_{-M}})QR\|^2 \;\Big|\; \left\{\|QR\| = \sqrt{a-d}\right\} \\
&= \|(P_X - P_{X_{-M}})QR\|^2 \;\Big|\; \left\{\|R\| = \sqrt{a-d}\right\}.
\end{aligned}$$

Further, letting $\tilde{R} \sim \mathcal{N}\left(0, I_{(n-1)-(p-m)}\right)$,

$$\begin{aligned}
&R^\top Q^\top (P_X - P_{X_{-M}})QR \;\Big|\; \left\{\|R\| = \sqrt{a-d}\right\} \\
&\overset{d}{=} \|R\|^2 \cdot \frac{R^\top}{\|R\|} Q^\top (P_X - P_{X_{-M}})Q \frac{R}{\|R\|} \;\Big|\; \left\{\|R\| = \sqrt{a-d}\right\} \\
&\overset{d}{=} (a-d) \cdot \frac{R^\top}{\|R\|} Q^\top (P_X - P_{X_{-M}})Q \frac{R}{\|R\|} \;\Big|\; \left\{\|R\| = \sqrt{a-d}\right\} \\
&\overset{d}{=} (a-d) \cdot \frac{R^\top}{\|R\|} Q^\top (P_X - P_{X_{-M}})Q \frac{R}{\|R\|} \\
&\overset{d}{=} (a-d) \cdot \frac{\tilde{R}^\top}{\|\tilde{R}\|} Q^\top (P_X - P_{X_{-M}})Q \frac{\tilde{R}}{\|\tilde{R}\|}
\end{aligned}$$

where the fourth line follows from the fact that the direction $\frac{R}{\|R\|}$ and magnitude $\|R\|$ of an isotropic normal random variable are independent, and the fifth line follows from the fact that $R \overset{d}{=} \sigma\tilde{R}$. Observing that $\text{rank}(Q^\top(P_X - P_{X_{-M}})Q) = m$, we can represent the distribution of the above ratio of quadratic forms as

$$\frac{\tilde{R}^\top(Q^\top(P_X - P_{X_{-M}})Q)\tilde{R}}{\|\tilde{R}\|^2} \overset{d}{=} \frac{W}{Z+W}$$

for $W \sim \chi^2_m$ and $Z \sim \chi^2_{n-p-1}$. We conclude that

$$Y^\top(P_X - P_{X_{-M}})Y \;\Big|\; \left\{\|Y\|^2 = a,\ \|P_{X_{-M}}Y\|^2 = d\right\} \overset{d}{=} (a-d)\cdot\frac{W}{Z+W}.$$

Finally, we can represent the selective p-value in (10) as follows:

$$\begin{aligned}
p_{H_0^M|E} &= \Pr_{\beta_M=0}\left(\frac{Y^\top(P_X - P_{X_{-M}})Y}{Y^\top(I_{n-1} - P_X)Y} \geq r(y) \;\middle|\right. \\
&\qquad \left. \|Y\|^2 = a,\ \|P_{X_{-M}}Y\|^2 = d,\ \frac{Y^\top P_X Y}{Y^\top(I - P_X)Y} \geq c\right) \\
&= \Pr_{\beta_M=0}\left(\frac{Y^\top(P_X - P_{X_{-M}})Y}{Y^\top Y - Y^\top P_{X_{-M}}Y - Y^\top(P_X - P_{X_{-M}})Y} \geq r(y) \;\middle|\right. \\
&\qquad \left. \|Y\|^2 = a,\ \|P_{X_{-M}}Y\|^2 = d,\ \frac{Y^\top P_{X_{-M}}Y + Y^\top(P_X - P_{X_{-M}})Y}{Y^\top Y - Y^\top P_{X_{-M}}Y - Y^\top(P_X - P_{X_{-M}})Y} \geq c\right) \\
&= \Pr_{\beta_M=0}\left(\frac{Y^\top(P_X - P_{X_{-M}})Y}{a - d - Y^\top(P_X - P_{X_{-M}})Y} \geq r(y) \;\middle|\right. \\
&\qquad \left. \|Y\|^2 = a,\ \|P_{X_{-M}}Y\|^2 = d,\ \frac{d + Y^\top(P_X - P_{X_{-M}})Y}{a - d - Y^\top(P_X - P_{X_{-M}})Y} \geq c\right) \\
&= \frac{\Pr_{\beta_M=0}\left(\frac{Y^\top(P_X - P_{X_{-M}})Y}{a-d-Y^\top(P_X - P_{X_{-M}})Y} \geq r(y),\ \frac{d+Y^\top(P_X - P_{X_{-M}})Y}{a-d-Y^\top(P_X - P_{X_{-M}})Y} \geq c \;\middle|\; \|Y\|^2 = a,\ \|P_{X_{-M}}Y\|^2 = d\right)}{\Pr_{\beta_M=0}\left(\frac{d+Y^\top(P_X - P_{X_{-M}})Y}{a-d-Y^\top(P_X - P_{X_{-M}})Y} \geq c \;\middle|\; \|Y\|^2 = a,\ \|P_{X_{-M}}Y\|^2 = d\right)} \\
&= \frac{\Pr\left(\frac{(a-d)\frac{W}{Z+W}}{a-d-(a-d)\frac{W}{Z+W}} \geq r(y),\ \frac{d+(a-d)\frac{W}{Z+W}}{a-d-(a-d)\frac{W}{Z+W}} \geq c\right)}{\Pr\left(\frac{d+(a-d)\frac{W}{Z+W}}{a-d-(a-d)\frac{W}{Z+W}} \geq c\right)} \\
&= \Pr\left(\frac{W}{Z} \geq r(y),\ \middle|\ \frac{aW + dZ}{(a-d)Z} \geq c\right),
\end{aligned}$$

where the last line follows from the definition of conditional probability and the fact that

$$\frac{(a-d)\frac{W}{Z+W}}{a-d-(a-d)\frac{W}{Z+W}} = \frac{(a-d)W}{(a-d)(Z+W)-(a-d)W} = \frac{W}{Z},$$

$$\frac{d + (a-d)\frac{W}{Z+W}}{a - d - (a-d)\frac{W}{Z+W}} = \frac{d(W+Z) + (a-d)W}{(a-d)(W+Z) - (a-d)W} = \frac{aW + dZ}{(a-d)Z}.$$

□

F.4 Proof of Proposition 2.2

We begin with two lemmas.

Lemma F.2. *Consider a fixed sequence of design matrices $X_n \in \mathbb{R}^{n\times p}$ and corresponding realizations of the linear model $Y_n = X_n\beta + \epsilon_n$ from (1) where $\beta \neq 0$. Suppose that $\frac{1}{n}X_n^\top X_n \to Q$ for some positive definite Q. Then $\exists \delta > 0$ and $N > 0$ such that*

$$\Pr\left(\frac{a(Y_n)W + d(Y_n)Z_n}{(a(Y_n) - d(Y_n))Z_n} \geq c_n\right) \geq \delta$$

for all $n > N$, where $W \sim \chi^2_m$, $Z_n \sim \chi^2_{n-p-1}$, $c_n = \frac{p}{n-p-1}F^{-1}_{p,n-p-1}(1-\alpha_0)$, $a(Y_n) = Y_n^\top Y_n$ and $d(Y_n) = Y_n^\top P_{(X_n)_{-M}} Y_n$.

Proof of Lemma F.2. In what follows, we suppress subscripts on X involving n to enhance readability.

To show that $\Pr\left(\frac{a(Y_n)W+d(Y_n)Z_n}{(a(Y_n)-d(Y_n))Z_n} \geq c_n\right) \geq 0$ as $n \to \infty$, first observe that

- $c_n \cdot Z_n$ converges in probability to $q := (\chi^2_p)^{-1}(1-\alpha_0)$ as $n \to \infty$,
- $d(Y_n)$ is a scaled χ^2 random variable with $p - m$ degrees of freedom and non-centrality parameter $\frac{1}{2\sigma^2}(X\beta)^\top P_{X_{-M}}(X\beta)$,
- $a(Y_n)$ is a scaled χ^2 random variable with $(n-1)$ degrees of freedom and non-centrality parameter $\frac{1}{2\sigma^2}(X\beta)^\top(X\beta)$.

Since $\beta \neq 0$, the non-centrality parameters of $d(Y_n)$ and $a(Y_n)$ are possibly non-zero. By our assumption that $\frac{1}{n}X^\top X \to Q$ deterministically for some positive definite matrix $Q \in \mathbb{R}^{p\times p}$, it follows that $\frac{1}{n}X^\top X_{-M} \to Q_{\cdot,-M}$ and $\frac{1}{n}X_{-M}^\top X_{-M} \to Q_{-M,-M}$ for $Q_{\cdot,-M} \in \mathbb{R}^{p\times(p-m)}$ and positive definite $Q_{-M,-M} \in \mathbb{R}^{(p-m)\times(p-m)}$. Hence,

$$\begin{aligned}\tfrac{1}{n}\beta^\top X^\top P_{X_{-M}} X\beta &= \beta^\top \left(\tfrac{1}{n}X^\top X_{-M}\right)\left(\tfrac{1}{n}X_{-M}^\top X_{-M}\right)^{-1}\left(\tfrac{1}{n}X_{-M}^\top X\right)\beta \\ &\to \beta^\top Q_{\cdot,-M}(Q_{-M,-M})^{-1}Q_{\cdot,-M}^\top \beta.\end{aligned}$$

and

$$\tfrac{1}{n}\beta^\top X^\top X\beta \to \beta^\top Q\beta.$$

Thus, when $\beta \neq 0$, $d(Y_n)$ and $a(Y_n)$ are each distributed as a scaled χ^2 random variable with $O(n)$ non-centrality parameters. The sequence of expectations of $d(Y_n)/n$ and $a(Y_n)/n$ converges and each has variance $O(1/n)$. Thus by Chebyshev's inequality, it follows that $\phi_n := d(Y_n)/n \xrightarrow{P} \phi^*$ and $\varphi_n := a(Y_n)/n \xrightarrow{P} \varphi^*$ for some fixed ϕ^* and φ^*, where "$\xrightarrow{P}$" denotes convergence in probability.

Therefore, for fixed $\varepsilon > 0$ and $\eta > 0$, we have that $\Pr(|\phi_n - \phi^*| \geq \varepsilon) < \eta$, $\Pr(|\varphi_n - \varphi^*| \geq \varepsilon) < \eta$ and $\Pr(|c_n Z_n - q| \geq \varepsilon) < \eta$ for sufficiently large n. For such an n, define

$$G_n(\varepsilon) = \{(\phi_n, Z_n) : |\phi_n - \phi^*| < \varepsilon,\ |\varphi_n - \varphi^*| < \varepsilon,\ |c_n Z_n - q| < \varepsilon\},$$

so that $\Pr(G_n^c(\varepsilon)) \leq 3\eta$. Note that

$$\begin{aligned}
|\varphi_n - \varphi^*| < \varepsilon &\iff \varphi_n < \varphi^* + \varepsilon \quad \text{and} \quad \varphi_n > \varphi^* - \varepsilon, \\
|\phi_n - \phi^*| < \varepsilon &\iff \phi_n < \phi^* + \varepsilon \quad \text{and} \quad \phi_n > \varphi^* - \varepsilon, \\
|c_n Z_n - q| < \varepsilon &\iff c_n Z_n < q + \varepsilon \quad \text{and} \quad c_n Z_n > q - \varepsilon.
\end{aligned}$$

Choose $\varepsilon > 0$ small enough that $\phi^* > \varepsilon$. Then

$$\begin{aligned}
&\Pr\left(\frac{a(Y_n)W + d(Y_n)Z_n}{(a(Y_n) - d(Y_n))Z_n} \geq c_n\right) \\
&= \Pr\left(\frac{n\varphi_n W + n\phi_n Z_n}{(n\varphi_n - n\phi_n)Z_n} \geq c_n\right) \\
&= \Pr\left(\frac{\varphi_n W + \phi_n Z_n}{\varphi_n - \phi_n} \geq c_n Z_n\right) \\
&\geq \Pr\left(\frac{\varphi_n W + \phi_n Z_n}{\varphi_n - \phi_n} \geq c_n Z_n,\ G_n(\varepsilon)\right) \\
&\geq \Pr\left(\frac{(\varphi^* - \varepsilon)W + (\phi^* - \varepsilon)Z_n}{(\varphi^* + \varepsilon) - (\phi^* - \varepsilon)} \geq q + \varepsilon,\ G_n(\varepsilon)\right) \\
&\geq \Pr\left(\frac{(\varphi^* - \varepsilon)W + (\phi^* - \varepsilon)Z_n}{(\varphi^* + \varepsilon) - (\phi^* - \varepsilon)} \geq q + \varepsilon\right) - \Pr(G_n^c(\varepsilon)) \\
&\geq \Pr\left(Z_n \geq \frac{(\varphi^* + \varepsilon) - (\phi^* - \varepsilon)}{\phi^* - \varepsilon}(q + \varepsilon) - \frac{\varphi^* - \varepsilon}{\phi^* - \varepsilon} W\right) - 3\eta \\
&\geq \Pr\left(Z_n \geq \frac{\varphi^* - \phi^* + 2\varepsilon}{\phi^* - \varepsilon}(q + \varepsilon)\right) - 3\eta,
\end{aligned}$$

where the last line follows from the fact that $(\varphi^* - \varepsilon)W/(\phi^* - \varepsilon)$ must be non-negative. Note that the mean of Z_n grows at rate $O(n)$, and $\frac{\varphi^* - \phi^* + 2\varepsilon}{\phi^* - \varepsilon}(q + \varepsilon)$ is a fixed finite value. Since η can be arbitrarily close to 0, we conclude that $\Pr\left(Z_n \geq \frac{\varphi^* - \phi^* + 2\varepsilon}{\phi^* - \varepsilon}(q + \varepsilon)\right) - 3\eta \to 1$ as $n \to \infty$. Thus,

$$\Pr\left(\frac{a(Y_n)W + d(Y_n)Z_n}{(a(Y_n) - d(Y_n))Z_n} \geq c_n\right)$$

is bounded away from 0 in the limit when $\beta \neq 0$.

□

Lemma F.3. *For the same assumptions on X_n and Y_n as in Lemma F.2, suppose $f : \mathbb{R}^n \to \mathbb{R}$ is a function that satisfies $\Pr(f(Y_n) > \varepsilon) \to 0$ for every $\varepsilon > 0$. Then $\Pr(f(Y_n) > \varepsilon \mid Y_n \in E_1) \to 0$ for every $\varepsilon > 0$, where E_1 is defined in (6).*

Proof of Lemma F.3. In what follows, we use the notation $\Pr_A(\cdot)$ to denote probability taken with respect to a random variable A in order to make explicit which quantity is treated as random in each probability calculation.

Note that $\Pr_{Y_n}(Y_n \in E_1) \geq \alpha_0$ where $\alpha_0 > 0$ is the level of the F-test of $H_0^{1:p}$ in Step 1 of Box 1. Thus

$$\begin{aligned}
&\Pr_{Y_n}(f(Y_n) > \varepsilon) \to 0 \\
&\quad\implies \Pr_{Y_n}(f(Y_n) > \varepsilon \mid Y_n \in E_1) \Pr_{Y_n}(Y_n \in E_1) \\
&\qquad\qquad + \Pr_{Y_n}(f(Y_n) > \varepsilon \mid Y_n \notin E_1) \Pr_{Y_n}(Y_n \notin E_1) \to 0 \\
&\quad\implies \Pr_{Y_n}(f(Y_n) > \varepsilon \mid Y_n \in E_1) \Pr_{Y_n}(Y_n \in E_1) \to 0 \\
&\quad\implies \Pr_{Y_n}(f(Y_n) > \varepsilon \mid Y_n \in E_1) \to 0.
\end{aligned}$$

□

We now continue with the proof of Proposition 2.2.

Proof of Proposition 2.2. A test is consistent if the probability of rejection under any fixed alternative converges to 1 as $n \to \infty$. Thus the test of H_0^M based on the selective p-value in (12) is consistent if and only if the selective p-value converges to 0 under any fixed alternative. That is, we want to show that if $\beta \neq 0$, then $\forall \varepsilon > 0$,

$$\begin{aligned}
&\Pr_{Y_n}\left(p_{H_0^M|E}(Y_n) > \varepsilon \mid Y_n \in E_1\right) \\
&\quad = \Pr_{Y_n}\left(\Pr_{W,Z_n}\left(\frac{W}{Z_n} \geq r(Y_n), \;\middle|\; \frac{a(Y_n)W + d(Y_n)Z_n}{(a(Y_n) - d(Y_n))Z_n} \geq c_n\right) > \varepsilon \;\middle|\; Y_n \in E_1\right) \to 0.
\end{aligned}$$

By Lemma F.3, if $\Pr_{Y_n}(p_{H_0^M|E}(Y_n) > \varepsilon) \to 0$ then $\Pr_{Y_n}(p_{H_0^M|E}(Y_n) > \varepsilon \mid Y_n \in E_1) \to 0$. Thus it is sufficient to show that whenever $\beta \neq 0$,

$$\begin{aligned}
&\Pr_{Y_n}\left(p_{H_0^M|E}(Y_n) > \varepsilon\right) \\
&\quad = \Pr_{Y_n}\left(\Pr_{W,Z_n}\left(\frac{W}{Z_n} \geq r(Y_n), \;\middle|\; \frac{a(Y_n)W + d(Y_n)Z_n}{(a(Y_n) - d(Y_n))Z_n} \geq c_n\right) > \varepsilon \;\middle|\; Y_n \in E_1\right) \to 0.
\end{aligned}$$

We begin by showing that when $\beta_M \neq 0$, it holds that

$$\lim_{n\to\infty} \Pr_{Y_n,W,Z_n}\left(\frac{W}{Z_n} > r(Y_n)\right) = \lim_{n\to\infty} \Pr_{Y_n}\left(\Pr_{W,Z_n}\left(\frac{W}{Z_n} > r(Y_n)\right)\right) = 0.$$

We have assumed that $\frac{1}{n}X^\top X \to Q$ deterministically for some positive definite matrix $Q \in \mathbb{R}^{p\times p}$, and it therefore follows that $\frac{1}{n}X_M^\top X_{-M} \to Q_{M,-M}$, $\frac{1}{n}X_{-M}^\top X_{-M} \to Q_{-M,-M}$, and $\frac{1}{n}X_M^\top X_M \to Q_{M,M}$ for $Q_{M,-M} \in \mathbb{R}^{m\times(p-m)}$ and positive definite matrices $Q_{M,M} \in \mathbb{R}^{m\times m}$ and $Q_{-M,-M} \in \mathbb{R}^{(p-m)\times(p-m)}$. Thus

$$\begin{aligned}
\tfrac{1}{n}X_M^\top(I - P_{X_{-M}})X_M &= \tfrac{1}{n}X_M^\top X_M - \tfrac{1}{n}X_M^\top P_{X_{-M}} X_M \\
&= \tfrac{1}{n}X_M^\top X_M - \tfrac{1}{n}X_M^\top X_{-M}(X_{-M}^\top X_{-M})^{-1}X_{-M}^\top X_M \\
&= \tfrac{1}{n}X_M^\top X_M - \left(\tfrac{1}{n}X_M^\top X_{-M}\right)\left(\tfrac{1}{n}X_{-M}^\top X_{-M}\right)^{-1}\left(\tfrac{1}{n}X_{-M}^\top X_M\right) \\
&\to Q_{M,M} - Q_{M,-M}(Q_{-M,-M})^{-1}Q_{M,-M}^\top.
\end{aligned}$$

It follows that $\frac{1}{2\sigma^2}(X_M\beta_M)^\top(I - P_{X_{-M}})X_M\beta_M = O(n)$ when $\beta_M \neq 0$. Hence, by (F.1), when $\beta \neq 0$, $\frac{1}{\sigma^2}Y_n^\top(P_X - P_{X_{-M}})Y_n$ is distributed as a χ^2 random variable with a non-centrality parameter that is $O(n)$. The sequence of expectations of $\frac{1}{n}Y_n^\top(P_X - P_{X_{-M}})Y_n$ converges, and the variance is $O(1/n)$. Hence, by Chebyshev's inequality, it follows that $\frac{1}{n}Y_n^\top(P_X - P_{X_{-M}})Y_n \xrightarrow{P} C$ for some constant $C > 0$. Additionally observe that $\frac{1}{n}Y_n^\top(I - P_X)Y_n \xrightarrow{P} \sigma^2$, again by Chebyshev's inequality. Hence, by the continuous mapping theorem

$$r(Y_n) = \frac{Y_n^\top(P_X - P_{X_{-M}})Y_n}{Y_n^\top(I_{n-1} - P_X)Y_n} = \frac{\frac{1}{n}Y_n^\top(P_X - P_{X_{-M}})Y_n}{\frac{1}{n}Y_n^\top(I_{n-1} - P_X)Y_n} \xrightarrow{P} \frac{C}{\sigma^2} > 0.$$

Further, $W/Z_n = \left(\frac{1}{n}W\right) / \left(\frac{1}{n}Z_n\right) \xrightarrow{P} 0/1 = 0$. That is, for any $\varepsilon > 0$, $\Pr_{W,Z_n}(W/Z_n > \varepsilon) \to 0$. We conclude that $\Pr\left(W/Z_n > r(Y_n)\right) \xrightarrow{P} 0$.

Next, note that for any $y \in \mathbb{R}^n$,

$$\begin{aligned}\Pr_{W,Z_n}\left(\frac{W}{Z_n} > r(y) \;\middle|\; \frac{a(Y_n)W + d(Y_n)Z_n}{(a(Y_n) - d(Y_n))Z_n} \geq c_n\right) &= \frac{\Pr_{W,Z_n}\left(\frac{W}{Z_n} > r(y),\ \frac{a(Y_n)W+d(Y_n)Z_n}{(a(Y_n)-d(Y_n))Z_n} \geq c_n\right)}{\Pr_{W,Z_n}\left(\frac{a(Y_n)W+d(Y_n)Z_n}{(a(Y_n)-d(Y_n))Z_n} \geq c_n\right)} \\ &\leq \frac{\Pr_{W,Z_n}\left(W/Z_n > r(y)\right)}{\Pr_{W,Z_n}\left(\frac{a(Y_n)W+d(Y_n)Z_n}{(a(Y_n)-d(Y_n))Z_n} \geq c_n\right)}.\end{aligned}$$

Thus,

$$\begin{aligned}&\Pr_{Y_n}\left(\Pr_{W,Z_n}\left(\frac{W}{Z_n} \geq r(Y_n),\ \middle|\ \frac{a(Y_n)W + d(Y_n)Z_n}{(a(Y_n) - d(Y_n))Z_n} \geq c_n\right) > \varepsilon\right) \\ &\qquad \leq \Pr_{Y_n}\left(\frac{\Pr_{W,Z_n}\left(W/Z_n > r(Y_n)\right)}{\Pr_{W,Z_n}\left(\frac{a(Y_n)W+d(Y_n)Z_n}{(a(Y_n)-d(Y_n))Z_n} \geq c_n\right)} > \varepsilon\right).\end{aligned}$$

Since the denominator $\Pr_{W,Z_n}\left(\frac{a(Y_n)W+d(Y_n)Z_n}{(a(Y_n)-d(Y_n))Z_n} \geq c_n\right)$ is bounded away from zero in the limit by Lemma F.2, we conclude that

$$\begin{aligned}\Pr_{W,Z_n}\left(W/Z_n > r(Y_n)\right) \xrightarrow{P} 0 \implies \frac{\Pr_{W,Z_n}\left(W/Z_n > r(Y_n)\right)}{\Pr_{W,Z_n}\left(\frac{a(Y_n)W+d(Y_n)Z_n}{(a(Y_n)-d(Y_n))Z_n} \geq c_n\right)} &\xrightarrow{P} 0 \\ \implies \Pr_{Y_n}\left(\frac{\Pr_{W,Z_n}\left(W/Z_n > r(Y_n)\right)}{\Pr_{W,Z_n}\left(\frac{a(Y_n)W+d(Y_n)Z_n}{(a(Y_n)-d(Y_n))Z_n} \geq c_n\right)} > \varepsilon\right) &\to 0 \\ \implies \Pr_{Y_n}\left(\Pr_{W,Z_n}\left(\frac{W}{Z_n} > r(Y_n) \;\middle|\; \frac{a(Y_n)W + d(Y_n)Z_n}{(a(Y_n) - d(Y_n))Z_n} \geq c_n\right) > \varepsilon\right) &\to 0.\end{aligned}$$

Thus we have established that $\Pr_{Y_n}\left(p_{H_0^M|E}(Y_n) > \varepsilon\right) \to 0$. By Lemma F.3, this implies that

$$\Pr\left(p_{H_0^M|E}(Y_n) > \varepsilon \mid Y \in E_1\right) \to 0,$$

and equivalently that $\Pr\left(p_{H_0^M|E}(Y_n) < \varepsilon \mid Y \in E_1\right) \to 1$. This completes the proof. □

F.5 Proof of Proposition 2.3

Proof of Proposition 2.3. Recall that for a proportion $\rho \in (0,1)$ and a realization $\tilde{y} \in \mathbb{R}^n$ we define

$$\begin{aligned} R_\rho(X) &:= \{Y \in \mathbb{R}^{(\rho n)} : Y^\top P_X Y \geq c_\chi\}, \\ R(\tilde{y}; X) &:= \{Y \in \mathbb{R}^n : Y^\top P_X Y \geq c_\chi,\ P_{X_{-1}} Y = P_{X_{-1}} \tilde{y}\}, \\ c_\chi &:= \sigma^2 (\chi^2_p)^{-1}(1-\alpha_0), \\ e_{\tilde{y}} &:= c_\chi - \tilde{y}^\top P_{X_{-1}} \tilde{y}. \end{aligned}$$

Let $(X^{\text{tr}}, Y^{\text{tr}})$ be the training dataset with ρ proportion of the observations and $(X^{\text{te}}, Y^{\text{te}})$ the test dataset with $(1-\rho)$ proportion of the observations. We begin by deriving the leftover Fisher information in Y^{te} about β_1 conditional on $Y^{\text{tr}} \in R_\rho(X^{\text{tr}})$, where we treat $\beta_2, \ldots, \beta_p$ as nuisance parameters. That is, we will find an expression for

$$\left(\left[\left[\mathcal{I}_{Y^{\text{te}} | Y^{\text{tr}} \in R_\rho(X^{\text{tr}})}(\beta; Y^{\text{tr}} \in R_\rho(X^{\text{tr}})) \right]^{-1} \right]_{1,1} \right)^{-1}, \tag{F.3}$$

where $\mathcal{I}_{Y^{\text{te}} | Y^{\text{tr}} \in R_\rho(X^{\text{tr}})}(\beta; Y^{\text{tr}} \in R_\rho(X^{\text{tr}}))$ is the Fisher information matrix for the entire parameter vector β. We consider the inverse of the $(1,1)$ element of the inverted Fisher information matrix in order to isolate the information in Y^{te} about β_1 while accounting for the fact that β_{-1} is also unknown.

Since Y^{tr} and Y^{te} are independent, the density of Y^{te} conditional on $Y^{\text{tr}} \in R_\rho(X^{\text{tr}})$ is equivalent to the unconditional density of Y^{te}, i.e.

$$f_{Y^{\text{te}} | Y^{\text{tr}} \in R_\rho(X^{\text{tr}})}(y) = f_{Y^{\text{te}}}(y) = (2\pi\sigma^2)^{-n/2} \exp\left\{ \frac{1}{2\sigma^2} (y - X^{\text{te}}\beta)^\top (y - X^{\text{te}}\beta) \right\}.$$

Since we are concerned with the derivative of this expression with respect to β, we will now consider this function as a likelihood of β with $y \in \mathbb{R}^n$ fixed. Then the Hessian of the log likelihood is

$$\frac{d^2}{d(\beta_1,\ \beta_{-1}) d(\beta_1,\ \beta_{-1})^\top} \log f_{Y^{\text{te}}}(\beta; y) = -\frac{1}{\sigma^2} \begin{bmatrix} (X^{\text{te}})_1^\top (X^{\text{te}})_1 & (X^{\text{te}})_1^\top (X^{\text{te}})_{-1} \\ (X^{\text{te}})_{-1}^\top (X^{\text{te}})_1 & (X^{\text{te}})_{-1}^\top (X^{\text{te}})_{-1} \end{bmatrix}.$$

Note that this Hessian is non-random because the test design matrix X^{te} is fixed. Treating β_{-1} as a nuisance parameter, we take the Schur complement to obtain the $(1,1)$ entry of the inverse Fisher information matrix and conclude that the leftover Fisher information is

$$\begin{aligned} &\left(\left[\left[\mathcal{I}_{Y^{\text{te}} | Y^{\text{tr}} \in R_\rho(X^{\text{tr}})}(\beta; Y^{\text{tr}} \in R_\rho(X^{\text{tr}})) \right]^{-1} \right]_{1,1} \right)^{-1} \\ &= \frac{1}{\sigma^2} \left((X^{\text{te}})_1^\top (X^{\text{te}})_1 - [(X^{\text{te}})_1^\top (X^{\text{te}})_{-1}][(X^{\text{te}})_{-1}^\top (X^{\text{te}})_{-1}]^{-1}[(X^{\text{te}})_{-1}^\top (X^{\text{te}})_1] \right) \\ &= \frac{1}{\sigma^2} \left((X^{\text{te}})_1^\top (I - P_{(X^{\text{te}})_{-1}}) (X^{\text{te}})_1 \right). \end{aligned}$$

Next, we derive an expression for $\mathcal{I}_{Y | Y \in R(\tilde{y}; X)}(\beta_1; Y \in R(\tilde{y}; X))$ for some realization $\tilde{y} \in \mathbb{R}^n$. Let

1. $U_1 \in \mathbb{R}^{n\times(n-p)}$ such that $U_1U_1^\top = I_n - P_X$ and $U_1^\top U_1 = I_{n-p}$,
2. $U_2 \in \mathbb{R}^n$ such that $U_2U_2^\top = P_X - P_{X_{-1}}$ and $U_2^\top U_2 = 1$,
3. $U_3 \in \mathbb{R}^{n\times(p-1)}$ such that $U_3U_3^\top = P_{X_{-1}}$ and $U_3^\top U_3 = I_{p-1}$.

Furthermore, define

$$\tilde{Z} = U_1^\top Y \in \mathbb{R}^{n-p}, \quad \tilde{W} = U_2^\top Y \in \mathbb{R}, \quad \tilde{V} = U_3^\top Y \in \mathbb{R}^{p-1}, \quad U = \begin{bmatrix} U_1 & U_2 & U_3 \end{bmatrix}.$$

Then note that

$$\begin{aligned}
U^\top U &= \begin{bmatrix} U_1^\top \\ U_2^\top \\ U_3^\top \end{bmatrix} \begin{bmatrix} U_1 & U_2 & U_3 \end{bmatrix} = \begin{bmatrix} U_1^\top U_1 & 0 & 0 \\ 0 & U_2^\top U_2 & 0 \\ 0 & 0 & U_3^\top U_3 \end{bmatrix} = I_n, \\
UU^\top &= \begin{bmatrix} U_1 & U_2 & U_3 \end{bmatrix} \begin{bmatrix} U_1^\top \\ U_2^\top \\ U_3^\top \end{bmatrix} = U_1U_1^\top + U_2U_2^\top + U_3U_3^\top = I_n, \\
Y &= UU^\top Y = U_1U_1^\top Y + U_2U_2^\top Y + U_3U_3^\top Y = U_1\tilde{Z} + U_2\tilde{W} + U_3\tilde{V}, \\
Y &\in R(\tilde{y}; X) \iff \tilde{W}^2 \geq e_{\tilde{y}} \text{ and } \tilde{V} = U_3^\top \tilde{y}.
\end{aligned}$$

Similarly to (C.7), this allows us to write the conditional density of Y given $Y \in R(\tilde{y}; X)$ as

$$f_{Y|Y\in R(\tilde{y};X)}(y) = \frac{f_{\tilde{Z}}(U_1^\top y) f_{\tilde{W}}(U_2^\top y) \boldsymbol{I}_{\{y\,:\,U_3^\top y = U_3^\top \tilde{y}\}}(y) \boldsymbol{I}_{\{y\,:\,(U_2^\top y)^2 \geq e_{\tilde{y}}\}}(y)}{\Pr\left(\tilde{W}^2 \geq e_{\tilde{y}}\right)}.$$

We now switch to viewing this expression as a likelihood of β for a fixed realization $y \in \mathbb{R}^n$, and thus the indicator $\boldsymbol{I}_{\{y\,:\,U_3^\top y = U_3^\top \tilde{y}\}}(y)$ must be satisfied. Note that the density $f_{\tilde{Z}}$ does not involve β. Then,

$$f_{Y|Y\in R(\tilde{y};X)}(\beta; y) \propto \frac{f_{\tilde{W}}(\beta; U_2^\top y) \boldsymbol{I}_{\{y\,:\,(U_2^\top y)^2 \geq e_{\tilde{y}}\}}(y)}{\Pr\left(\tilde{W}^2 \geq e_{\tilde{y}}\right)} = f_{\tilde{W}|\tilde{W}^2 \geq e_{\tilde{y}}}(\beta; U_2^\top y).$$

Note that $\tilde{W} = U_2^\top Y \sim N(U_2^\top X\beta, \sigma^2)$, or equivalently $\tilde{W} \sim N_1(U_2^\top X_1\beta_1, \sigma^2)$. Thus, letting $\tilde{w} = U_2^\top y$,

$$f_{\tilde{W}|\tilde{W}^2\geq e_{\tilde{y}}}(\beta; \tilde{w}) = \frac{\frac{1}{\sigma}\phi\left(\frac{\tilde{w} - U_2^\top X_1\beta_1}{\sigma}\right) \boldsymbol{I}_{\{\tilde{w}^2 \geq e_{\tilde{y}}\}}(\tilde{w})}{1 + \boldsymbol{I}_{\{e_{\tilde{y}}>0\}}\left(-\Phi\left(\frac{\sqrt{e_{\tilde{y}}} - U_2^\top X_1\beta_1}{\sigma}\right) + \Phi\left(\frac{-\sqrt{e_{\tilde{y}}} - U_2^\top X_1\beta_1}{\sigma}\right)\right)},$$

where ϕ is the density and Φ the CDF of a standard normal random variable and $\boldsymbol{I}_{\{e_{\tilde{y}}>0\}}$ is the indicator that $\tilde{y}$ is such that $e_{\tilde{y}} > 0$. Noting that the above display is free of β_{-1}, we rewrite this in terms of Y as

$$f_{Y|Y\in R(\tilde{y};X)}(\beta_1; y) \propto \frac{\frac{1}{\sigma}\phi\left(\frac{U_2^\top y - U_2^\top X_1\beta_1}{\sigma}\right) \boldsymbol{I}_{\{(U_2^\top y)^2 \geq e_{\tilde{y}}\}}(y)}{1 + \boldsymbol{I}_{\{e_{\tilde{y}}>0\}}\left(-\Phi\left(\frac{\sqrt{e_{\tilde{y}}} - U_2^\top X_1\beta_1}{\sigma}\right) + \Phi\left(\frac{-\sqrt{e_{\tilde{y}}} - U_2^\top X_1\beta_1}{\sigma}\right)\right)}.$$

Accordingly, we simply consider the second derivative of the log likelihood with respect to β_1 instead of the entire Hessian (with respect to β).

After some calculus, we find that if

$$\boldsymbol{I}_{\{y\,:\,U_3^\top y = U_3^\top \tilde{y}\}}(y) = 1 \;\text{ and }\; \boldsymbol{I}_{\{y\,:\,(U_2^\top y)^2 \geq e_{\tilde{y}}\}}(y) = 1, \tag{F.4}$$

then the second derivative has the expression

$$\begin{aligned}
&\frac{d^2}{(d\beta_1)^2} \log f_{Y|Y\in R(\tilde{y};X)}(\beta_1; y) \\
&\qquad = \frac{X_1^\top (P_X - P_{X_{-1}}) X_1}{\sigma^2} \left[-1 + \boldsymbol{I}_{\{e_{\tilde{y}} > 0\}} \left(\frac{[\phi(a) - \phi(b)]^2}{[1 - \Phi(b) + \Phi(a)]^2} + \frac{\phi(a)\cdot a - \phi(b)\cdot b}{1 - \Phi(b) + \Phi(a)} \right) \right]
\end{aligned}$$

where

$$a = \frac{-\sqrt{e_{\tilde{y}}} - U_2^\top X_1 \beta_1}{\sigma}, \quad b = \frac{\sqrt{e_{\tilde{y}}} - U_2^\top X_1 \beta_1}{\sigma}.$$

This second derivative is non-random. Additionally, the indicators in (F.4) are equivalent to $Y \in \mathbb{R}(\tilde{y}; X)$. Thus the leftover Fisher information about β_1 in Y conditional on $Y \in \mathbb{R}(\tilde{y}; X)$ is

$$\mathcal{I}_{Y|Y\in R(\tilde{y};X)}(\beta_1; Y \in R(\tilde{y}; X)) = \tfrac{X_1^\top (I - P_{X_{-1}}) X_1}{\sigma^2} \left[1 - \boldsymbol{I}_{\{e_{\tilde{y}} > 0\}} \left(\tfrac{[\phi(a) - \phi(b)]^2}{[1 - \Phi(b) + \Phi(a)]^2} + \tfrac{\phi(a)\cdot a - \phi(b)\cdot b}{1 - \Phi(b) + \Phi(a)} \right) \right].$$

□

F.6 Proof of Proposition 3.1

Proof of Proposition 3.1. When we plug in the definitions of RSE, R^2, and $F_{H_0^M}$, we find that $a_{\text{retro}} = a(y)$ and $d_{\text{retro}} = d(y)$ for $a(y)$ and $d(y)$ defined in (11) and a_{retro} and d_{retro} defined in Proposition 3.1. That is,

$$a_{\text{retro}} = \tfrac{\mathrm{RSE}^2 \cdot (n-p-1)}{1-\mathrm{R}^2} = \frac{\frac{y^\top (I_{n-1} - P_X) y}{n-p-1} \cdot (n-p-1)}{\frac{y^\top (I_{n-1} - P_X) y}{y^\top y}} = y^\top y = a(y),$$

and

$$\begin{aligned}
d_{\text{retro}} &= \mathrm{RSE}^2 \cdot \left(\tfrac{\mathrm{R}^2 \cdot (n-p-1)}{1-\mathrm{R}^2} - F_{H_0^M} \cdot m \right) \\
&= \frac{y^\top (I_{n-1} - P_X) y}{n-p-1} \cdot \left(\frac{\left(1 - \frac{y^\top (I_{n-1} - P_X) y}{y^\top y}\right) \cdot (n-p-1)}{\frac{y^\top (I_{n-1} - P_X) y}{y^\top y}} - \tfrac{n-p-1}{m} \cdot \tfrac{y^\top (P_X - P_{X_{-M}}) y}{y^\top (I_{n-1} - P_X) y} \cdot m \right) \\
&= y^\top y - y^\top (I_{n-1} - P_X) y - y^\top (P_X - P_{X_{-M}}) y \\
&= y^\top P_{X_{-M}} y \\
&= d(y).
\end{aligned}$$

Hence, beginning from the characterization of $p_{H_0^M|E}$ in (12),

$$\begin{aligned} p_{H_0^M|E}(y) &= \Pr\left(\frac{W}{Z} \geq r(y) \,\middle|\, \frac{a(y)\cdot W + d(y)\cdot Z}{(a(y)-d(y))\cdot Z} \geq c\right) \\ &= \Pr\left(\frac{W}{Z} \geq F_{H_0^M}\cdot\frac{n-p-1}{m} \,\middle|\, \frac{a_{\text{retro}}\cdot W + d_{\text{retro}}\cdot Z}{(a_{\text{retro}}-d_{\text{retro}})\cdot Z} \geq c\right), \end{aligned}$$

as claimed.

□

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